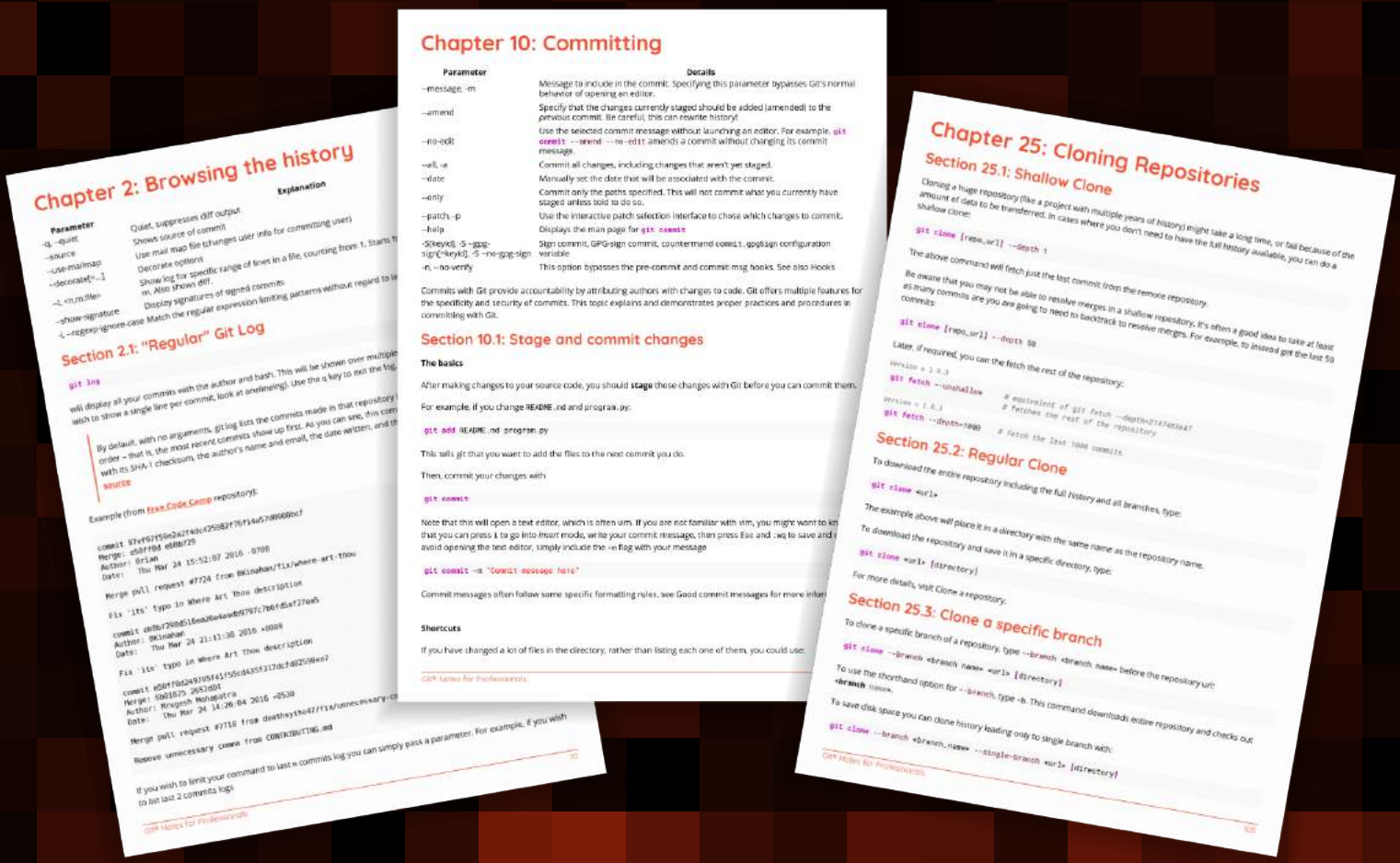
Git®

Notes for Professionals



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About

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Chapter 1: Getting started with Git

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2.6	2015-09-28
2.5	2015-07-27
2.4	2015-04-30
2.3	2015-02-05
2.2	2014-11-26
2.1	2014-08-16
2.0	2014-05-28
1.9	2014-02-14
1.8.3	2013-05-24
1.8	2012-10-21
1.7.10	2012-04-06
1.7	2010-02-13
1.6.5	2009-10-10
1.6.3	2009-05-07
1.6	2008-08-17
1.5.3	2007-09-02
1.5	2007-02-14
1.4	2006-06-10
1.3	2006-04-18
1.2	2006-02-12
1.1	2006-01-08
1.0	2005-12-21
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Section 1.1: Create your first repository, then add and commit files

At the command line, first verify that you have Git installed:

On all operating systems:

```
git --version
```

On UNIX-like operating systems:

```
which git
```

If nothing is returned, or the command is not recognized, you may have to install Git on your system by downloading and running the installer. See the Git homepage for exceptionally clear and easy installation instructions.

After installing Git, configure your username and email address. Do this *before* making a commit.

Once Git is installed, navigate to the directory you want to place under version control and create an empty Git repository:

```
git init
```

This creates a hidden folder, `.git`, which contains the plumbing needed for Git to work.

Next, check what files Git will add to your new repository; this step is worth special care:

```
git status
```

Review the resulting list of files; you can tell Git which of the files to place into version control (avoid adding files with confidential information such as passwords, or files that just clutter the repo):

```
git add <file/directory name #1> <file/directory name #2> < ... >
```

If all files in the list should be shared with everyone who has access to the repository, a single command will add everything in your current directory and its subdirectories:

```
git add .
```

This will "stage" all files to be added to version control, preparing them to be committed in your first commit.

For files that you want never under version control, create and populate a file named `.gitignore` before running the `add` command.

Commit all the files that have been added, along with a commit message:

```
git commit -m "Initial commit"
```

This creates a new commit with the given message. A commit is like a save or snapshot of your entire project. You can now push, or upload, it to a remote repository, and later you can jump back to it if necessary.
If you omit the `-m` parameter, your default editor will open and you can edit and save the commit message there.

Adding a remote

To add a new remote, use the `git remote add` command on the terminal, in the directory your repository is stored at.

The `git remote add` command takes two arguments:

1. A remote name, for example, `origin`
2. A remote URL, for example, `https://<your-git-service-address>/user/repo.git`

```
  git remote add origin https://<your-git-service-address>/owner/repository.git
```

NOTE: Before adding the remote you have to create the required repository in your git service, You'll be able to push/pull commits after adding your remote.

Section 1.2: Clone a repository

The `git clone` command is used to copy an existing Git repository from a server to the local machine.

For example, to clone a GitHub project:

```
cd <path where you would like the clone to create a directory>
git clone https://github.com/username/projectname.git
```

To clone a BitBucket project:

```
cd <path where you would like the clone to create a directory>
git clone https://yourusername@bitbucket.org/username/projectname.git
```

This creates a directory called `projectname` on the local machine, containing all the files in the remote Git repository. This includes source files for the project, as well as a `.git` sub-directory which contains the entire history and configuration for the project.

To specify a different name of the directory, e.g. `MyFolder`:

```
git clone https://github.com/username/projectname.git MyFolder
```

Or to clone in the current directory:

```
git clone https://github.com/username/projectname.git .
```

Note:

1. When cloning to a specified directory, the directory must be empty or non-existent.
2. You can also use the `ssh` version of the command:

   ```
   git clone git@github.com:username/projectname.git
   ```

The `https` version and the `ssh` version are equivalent. However, some hosting services such as GitHub recommend that you use `https` rather than `ssh`.

Section 1.3: Sharing code

To share your code you create a repository on a remote server to which you will copy your local repository.

To minimize the use of space on the remote server you create a bare repository: one which has only the `.git` objects and doesn't create a working copy in the filesystem. As a bonus you set this remote as an upstream server to easily share updates with other programmers.

On the remote server:

```
git init --bare /path/to/repo.git
```

On the local machine:

```
git remote add origin ssh://username@server:/path/to/repo.git
```

(Note that `ssh:` is just one possible way of accessing the remote repository.)

Now copy your local repository to the remote:

```
git push --set-upstream origin master
```

Adding `--set-upstream` (or `-u`) created an upstream (tracking) reference which is used by argument-less Git commands, e.g. `git pull`.

Section 1.4: Setting your user name and email

```
You need to set who you are *before* creating any commit.  That will allow commits to have the
right author name and email associated to them.
```

It has nothing to do with authentication when pushing to a remote repository (e.g. when pushing to a remote repository using your GitHub, BitBucket, or GitLab account)

To declare that identity for *all* repositories, use `git config --global`
This will store the setting in your user's `.gitconfig` file: e.g. `$HOME/.gitconfig` or for Windows, `%USERPROFILE%\.gitconfig`.

```
git config --global user.name "Your Name"
git config --global user.email mail@example.com
```

To declare an identity for a single repository, use `git config` inside a repo.
This will store the setting inside the individual repository, in the file `$GIT_DIR/config`. e.g. `/path/to/your/repo/.git/config`.

```
cd /path/to/my/repo
git config user.name "Your Login At Work"
git config user.email mail_at_work@example.com
```

Settings stored in a repository's config file will take precedence over the global config when you use that repository.

Tips: if you have different identities (one for open-source project, one at work, one for private repos, ...), and you don't want to forget to set the right one for each different repos you are working on:

- **Remove a global identity**

  ```
  git config --global --remove-section user.name
  git config --global --remove-section user.email
  ```

Version ≥ 2.8

- To force git to look for your identity only within a repository's settings, not in the global config:

  ```
  git config --global user.useConfigOnly true
  ```

That way, if you forget to set your `user.name` and `user.email` for a given repository and try to make a commit, you will see:

```
no name was given and auto-detection is disabled
no email was given and auto-detection is disabled
```

Section 1.5: Setting up the upstream remote

If you have cloned a fork (e.g. an open source project on Github) you may not have push access to the upstream repository, so you need both your fork but be able to fetch the upstream repository.

First check the remote names:

```
$ git remote -v
origin    https://github.com/myusername/repo.git (fetch)
origin    https://github.com/myusername/repo.git (push)
upstream  # this line may or may not be here
```

If `upstream` is there already (it is on *some* Git versions) you need to set the URL (currently it's empty):

```
$ git remote set-url upstream https://github.com/projectusername/repo.git
```

If the upstream is **not** there, or if you also want to add a friend/colleague's fork (currently they do not exist):

```
$ git remote add upstream https://github.com/projectusername/repo.git
$ git remote add dave https://github.com/dave/repo.git
```

Section 1.6: Learning about a command

To get more information about any git command – i.e. details about what the command does, available options and other documentation – use the `--help` option or the `help` command.

For example, to get all available information about the `git diff` command, use:

```
git diff --help
git help diff
```

Similarly, to get all available information about the `status` command, use:

```
git status --help
git help status
```

If you only want a quick help showing you the meaning of the most used command line flags, use `-h`:

```
git checkout -h
```

Section 1.7: Set up SSH for Git

If you are using **Windows** open Git Bash. If you are using **Mac** or **Linux** open your Terminal.

Before you generate an SSH key, you can check to see if you have any existing SSH keys.

List the contents of your `~/.ssh` directory:

```
$ ls -al ~/.ssh
# Lists all the files in your ~/.ssh directory
```

Check the directory listing to see if you already have a public SSH key. By default the filenames of the public keys are one of the following:

```
id_dsa.pub
id_ecdsa.pub
id_ed25519.pub
id_rsa.pub
```

If you see an existing public and private key pair listed that you would like to use on your Bitbucket, GitHub (or similar) account you can copy the contents of the `id_*.pub` file.

If not, you can create a new public and private key pair with the following command:

```
$ ssh-keygen
```

Press the Enter or Return key to accept the default location. Enter and re-enter a passphrase when prompted, or leave it empty.

Ensure your SSH key is added to the ssh-agent. Start the ssh-agent in the background if it's not already running:

```
$ eval "$(ssh-agent -s)"
```

Add you SSH key to the ssh-agent. Notice that you'll need te replace `id_rsa` in the command with the name of your **private key file**:

```
$ ssh-add ~/.ssh/id_rsa
```

If you want to change the upstream of an existing repository from HTTPS to SSH you can run the following command:

```
$ git remote set-url origin ssh://git@bitbucket.server.com:7999/projects/your_project.git
```

In order to clone a new repository over SSH you can run the following command:

```
$ git clone ssh://git@bitbucket.server.com:7999/projects/your_project.git
```

Section 1.8: Git Installation

Let's get into using some Git. First things first—you have to install it. You can get it a number of ways; the two major ones are to install it from source or to install an existing package for your platform.

Installing from Source

If you can, it's generally useful to install Git from source, because you'll get the most recent version. Each version of Git tends to include useful UI enhancements, so getting the latest version is often the best route if you feel comfortable compiling software from source. It is also the case that many Linux distributions contain very old packages; so unless you're on a very up-to-date distro or are using backports, installing from source may be the best bet.

To install Git, you need to have the following libraries that Git depends on: curl, zlib, openssl, expat, and libiconv. For example, if you're on a system that has yum (such as Fedora) or apt-get (such as a Debian based system), you can use one of these commands to install all of the dependencies:

```
$ yum install curl-devel expat-devel gettext-devel \
```

```
  openssl-devel zlib-devel

$ apt-get install libcurl4-gnutls-dev libexpat1-dev gettext \
  libz-dev libssl-dev
```

When you have all the necessary dependencies, you can go ahead and grab the latest snapshot from the Git web site:

http://git-scm.com/download Then, compile and install:

```
$ tar -zxf git-1.7.2.2.tar.gz
$ cd git-1.7.2.2
$ make prefix=/usr/local all
$ sudo make prefix=/usr/local install
```

After this is done, you can also get Git via Git itself for updates:

```
$ git clone git://git.kernel.org/pub/scm/git/git.git
```

Installing on Linux

If you want to install Git on Linux via a binary installer, you can generally do so through the basic package-management tool that comes with your distribution. If you're on Fedora, you can use yum:

```
$ yum install git
```

Or if you're on a Debian-based distribution like Ubuntu, try apt-get:

```
$ apt-get install git
```

Installing on Mac

There are three easy ways to install Git on a Mac. The easiest is to use the graphical Git installer, which you can download from the SourceForge page.

http://sourceforge.net/projects/git-osx-installer/

Figure 1-7. Git OS X installer. The other major way is to install Git via MacPorts (http://www.macports.org). If you have MacPorts installed, install Git via

```
$ sudo port install git +svn +doc +bash_completion +gitweb
```

You don't have to add all the extras, but you'll probably want to include +svn in case you ever have to use Git with Subversion repositories (see Chapter 8).

Homebrew (http://brew.sh/) is another alternative to install Git. If you have Homebrew installed, install Git via

```
$ brew install git
```

Installing on Windows

Installing Git on Windows is very easy. The msysGit project has one of the easier installation procedures. Simply download the installer exe file from the GitHub page, and run it:

```
http://msysgit.github.io
```

After it's installed, you have both a command-line version (including an SSH client that will come in handy later) and the standard GUI.

Note on Windows usage: you should use Git with the provided msysGit shell (Unix style), it allows to use the complex lines of command given in this book. If you need, for some reason, to use the native Windows shell / command line console, you have to use double quotes instead of single quotes (for parameters with spaces in them) and you must quote the parameters ending with the circumflex accent (^) if they are last on the line, as it is a continuation symbol in Windows.

Chapter 2: Browsing the history

Parameter	Explanation
-q, --quiet	Quiet, suppresses diff output
--source	Shows source of commit
--use-mailmap	Use mail map file (changes user info for committing user)
--decorate[=...]	Decorate options
--L <n,m:file>	Show log for specific range of lines in a file, counting from 1. Starts from line n, goes to line m. Also shows diff.
--show-signature	Display signatures of signed commits
-i, --regexp-ignore-case	Match the regular expression limiting patterns without regard to letter case

Section 2.1: "Regular" Git Log

```
git log
```

will display all your commits with the author and hash. This will be shown over multiple lines per commit. (If you wish to show a single line per commit, look at onelineing). Use the `q` key to exit the log.

> By default, with no arguments, git log lists the commits made in that repository in reverse chronological order – that is, the most recent commits show up first. As you can see, this command lists each commit with its SHA-1 checksum, the author's name and email, the date written, and the commit message. - **source**

Example (from **Free Code Camp** repository):

```
commit 87ef97f59e2a2f4dc425982f76f14a57d0900bcf
Merge: e50ff0d eb8b729
Author: Brian
Date:   Thu Mar 24 15:52:07 2016 -0700

Merge pull request #7724 from BKinahan/fix/where-art-thou

Fix 'its' typo in Where Art Thou description

commit eb8b7298d516ea20a4aadb9797c7b6fd5af27ea5
Author: BKinahan
Date:   Thu Mar 24 21:11:36 2016 +0000

Fix 'its' typo in Where Art Thou description

commit e50ff0d249705f41f55cd435f317dcfd02590ee7
Merge: 6b01875 2652d04
Author: Mrugesh Mohapatra
Date:   Thu Mar 24 14:26:04 2016 +0530

Merge pull request #7718 from deathsythe47/fix/unnecessary-comma

Remove unnecessary comma from CONTRIBUTING.md
```

If you wish to limit your command to last `n` commits log you can simply pass a parameter. For example, if you wish to list last 2 commits logs

```
git log -2
```

Section 2.2: Prettier log

To see the log in a prettier graph-like structure use:

```
git log --decorate --oneline --graph
```

sample output :

```
* e0c1cea (HEAD -> maint, tag: v2.9.3, origin/maint) Git 2.9.3
*   9b601ea Merge branch 'jk/difftool-in-subdir' into maint
|\
| * 32b8c58 difftool: use Git::* functions instead of passing around state
| * 98f917e difftool: avoid $GIT_DIR and $GIT_WORK_TREE
| * 9ec26e7 difftool: fix argument handling in subdirs
* |   f4fd627 Merge branch 'jk/reset-ident-time-per-commit' into maint
...
```

Since it's a pretty big command, you can assign an alias:

```
git config --global alias.lol "log --decorate --oneline --graph"
```

To use the alias version:

```
# history of current branch :
git lol

# combined history of active branch (HEAD), develop and origin/master branches :
git lol HEAD develop origin/master

# combined history of everything in your repo :
git lol --all
```

Section 2.3: Colorize Logs

```
git log --graph --pretty=format:'%C(red)%h%Creset -%C(yellow)%d%Creset %s %C(green)(%cr)
%C(yellow)<%an>%Creset'
```

The `format` option allows you to specify your own log output format:

Parameter	Details
`%C(color_name)`	option colors the output that comes after it
`%h` or `%H`	abbreviates commit hash (use %H for complete hash)
`%Creset`	resets color to default terminal color
`%d`	ref names
`%s`	subject [commit message]
`%cr`	committer date, relative to current date
`%an`	author name

Section 2.4: Oneline log

```
git log --oneline
```

will show all of your commits with only the first part of the hash and the commit message. Each commit will be in a single line, as the `oneline` flag suggests.

> The oneline option prints each commit on a single line, which is useful if you're looking at a lot of commits. - **source**

Example (from **Free Code Camp** repository, with the same section of code from the other example):

```
87ef97f Merge pull request #7724 from BKinahan/fix/where-art-thou
eb8b729 Fix 'its' typo in Where Art Thou description
e50ff0d Merge pull request #7718 from deathsythe47/fix/unnecessary-comma
2652d04 Remove unnecessary comma from CONTRIBUTING.md
6b01875 Merge pull request #7667 from zerkms/patch-1
766f088 Fixed assignment operator terminology
d1e2468 Merge pull request #7690 from BKinahan/fix/unsubscribe-crash
bed9de2 Merge pull request #7657 from Rafase282/fix/
```

If you wish to limit you command to last `n` commits log you can simply pass a parameter. For example, if you wish to list last 2 commits logs

```
git log -2 --oneline
```

Section 2.5: Log search

```
git log -S"#define SAMPLES"
```

Searches for **addition** or **removal** of specific string or the string **matching** provided REGEXP. In this case we're looking for addition/removal of the string `#define SAMPLES`. For example:

```
+#define SAMPLES  100000
```

or

```
-#define SAMPLES  100000
```

```
git log -G"#define SAMPLES"
```

Searches for **changes** in **lines containing** specific string or the string **matching** provided REGEXP. For example:

```
-#define SAMPLES  100000
+#define SAMPLES  100000000
```

Section 2.6: List all contributions grouped by author name

`git shortlog` summarizes `git log` and groups by author

If no parameters are given, a list of all commits made per committer will be shown in chronological order.

```
$ git shortlog
Committer 1 (<number_of_commits>):
    Commit Message 1
    Commit Message 2
```

```
    ...
Committer 2 (<number_of_commits>):
    Commit Message 1
    Commit Message 2
    ...
```

To simply see the number of commits and suppress the commit description, pass in the summary option:

`-s`

`--summary`

```
$ git shortlog -s
<number_of_commits> Committer 1
<number_of_commits> Committer 2
```

To sort the output by number of commits instead of alphabetically by committer name, pass in the numbered option:

`-n`

`--numbered`

To add the email of a committer, add the email option:

`-e`

`--email`

A custom format option can also be provided if you want to display information other than the commit subject:

`--format`

This can be any string accepted by the `--format` option of `git log`.

See **Colorizing Logs** above for more information on this.

Section 2.7: Searching commit string in git log

Searching git log using some string in log:

```
git log [options] --grep "search_string"
```

Example:

```
git log --all --grep "removed file"
```

Will search for `removed file` string in **all logs** in **all branches**.

Starting from git 2.4+, the search can be inverted using the `--invert-grep` option.

Example:

```
git log --grep="add file" --invert-grep
```

Will show all commits that do not contain `add file`.

Section 2.8: Log for a range of lines within a file

```
$ git log -L 1,20:index.html
commit 6a57fde739de66293231f6204cbd8b2feca3a869
Author: John Doe <john@doe.com>
Date:   Tue Mar 22 16:33:42 2016 -0500

    commit message

diff --git a/index.html b/index.html
--- a/index.html
+++ b/index.html
@@ -1,17 +1,20 @@
 <!DOCTYPE HTML>
 <html>
-       <head>
-        <meta charset="utf-8">
+
+<head>
+    <meta charset="utf-8">
     <meta http-equiv="X-UA-Compatible" content="IE=edge">
     <meta name="viewport" content="width=device-width, initial-scale=1">
```

Section 2.9: Filter logs

```
git log --after '3 days ago'
```

Specific dates work too:

```
git log --after 2016-05-01
```

As with other commands and flags that accept a date parameter, the allowed date format is as supported by GNU date (highly flexible).

An alias to `--after` is `--since`.

Flags exist for the converse too: `--before` and `--until`.

You can also filter logs by `author`. e.g.

```
git log --author=author
```

Section 2.10: Log with changes inline

To see the log with changes inline, use the `-p` or `--patch` options.

```
git log --patch
```

Example (from Trello Scientist repository)

```
ommit 8ea1452aca481a837d9504f1b2c77ad013367d25
Author: Raymond Chou <info@raychou.io>
Date:   Wed Mar 2 10:35:25 2016 -0800

    fix readme error link
```

```
diff --git a/README.md b/README.md
index 1120a00..9bef0ce 100644
--- a/README.md
+++ b/README.md
@@ -134,7 +134,7 @@ the control function threw, but *after* testing the other functions and
readying
 the logging. The criteria for matching errors is based on the constructor and
 message.

-You can find this full example at [examples/errors.js](examples/error.js).
+You can find this full example at [examples/errors.js](examples/errors.js).

 ## Asynchronous behaviors


commit d3178a22716cc35b6a2bdd679a7ec24bc8c63ffa
:
```

Section 2.11: Log showing commited files

```
git log --stat
```

Example:

```
commit 4ded994d7fc501451fa6e233361887a2365b91d1
Author: Manassés Souza <manasses.inatel@gmail.com>
Date:   Mon Jun 6 21:32:30 2016 -0300

    MercadoLibre java-sdk dependency

 mltracking-poc/.gitignore |  1 +
 mltracking-poc/pom.xml    | 14 +++++++++++--
 2 files changed, 13 insertions(+), 2 deletions(-)

commit 506fff56190f75bc051248770fb0bcd976e3f9a5
Author: Manassés Souza <manasses.inatel@gmail.com>
Date:   Sat Jun 4 12:35:16 2016 -0300

    [manasses] generated by SpringBoot initializr

 .gitignore                                                                  |  42
++++++++++++
 mltracking-poc/mvnw                                                         | 233
+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 mltracking-poc/mvnw.cmd                                                     | 145
++++++++++++++++++++++++++++++++++++++++
 mltracking-poc/pom.xml                                                      |  74
++++++++++++++++++++
 mltracking-poc/src/main/java/br/com/mls/mltracking/MltrackingPocApplication.java      |  12 ++++
 mltracking-poc/src/main/resources/application.properties                    |   0
 mltracking-poc/src/test/java/br/com/mls/mltracking/MltrackingPocApplicationTests.java |  18 +++++
 7 files changed, 524 insertions(+)
```

Section 2.12: Show the contents of a single commit

Using `git show` we can view a single commit

```
git show 48c83b3
```

```
git show 48c83b3690dfc7b0e622fd220f8f37c26a77c934
```

Example

```
commit 48c83b3690dfc7b0e622fd220f8f37c26a77c934
Author: Matt Clark <mrclark32493@gmail.com>
Date:   Wed May 4 18:26:40 2016 -0400

    The commit message will be shown here.

diff --git a/src/main/java/org/jdm/api/jenkins/BuildStatus.java
b/src/main/java/org/jdm/api/jenkins/BuildStatus.java
index 0b57e4a..fa8e6a5 100755
--- a/src/main/java/org/jdm/api/jenkins/BuildStatus.java
+++ b/src/main/java/org/jdm/api/jenkins/BuildStatus.java
@@ -50,7 +50,7 @@ public enum BuildStatus {

                        colorMap.put(BuildStatus.UNSTABLE, Color.decode( "#FFFF55" ));
-                       colorMap.put(BuildStatus.SUCCESS,  Color.decode( "#55FF55" ));
+                       colorMap.put(BuildStatus.SUCCESS,  Color.decode( "#33CC33" ));
                        colorMap.put(BuildStatus.BUILDING, Color.decode( "#5555FF" ));
```

Section 2.13: Git Log Between Two Branches

`git log master..foo` will show the commits that are on `foo` and not on `master`. Helpful for seeing what commits you've added since branching!

Section 2.14: One line showing commiter name and time since commit

```
tree = log --oneline --decorate --source --pretty=format:'"%Cblue %h %Cgreen %ar %Cblue %an
%C(yellow) %d %Creset %s"'  --all --graph
```

example

```
*    40554ac  3 months ago  Alexander Zolotov    Merge pull request #95 from
gmandnepr/external_plugins
|\
| *  e509f61  3 months ago  Ievgen Degtiarenko    Documenting new property
| *  46d4cb6  3 months ago  Ievgen Degtiarenko    Running idea with external plugins
| *  6253da4  3 months ago  Ievgen Degtiarenko    Resolve external plugin classes
| *  9fdb4e7  3 months ago  Ievgen Degtiarenko    Keep original artifact name as this may be
important for intellij
| *  22e82e4  3 months ago  Ievgen Degtiarenko    Declaring external plugin in intellij section
|/
*  bc3d2cb  3 months ago  Alexander Zolotov    Ignore DTD in plugin.xml
```

Chapter 3: Working with Remotes

Section 3.1: Deleting a Remote Branch

To delete a remote branch in Git:

```
git push [remote-name] --delete [branch-name]
```

or

```
git push [remote-name] :[branch-name]
```

Section 3.2: Changing Git Remote URL

Check existing remote

```
git remote -v
# origin https://github.com/username/repo.git (fetch)
# origin https://github.com/usernam/repo.git (push)
```

Changing repository URL

```
git remote set-url origin https://github.com/username/repo2.git
# Change the 'origin' remote's URL
```

Verify new remote URL

```
git remote -v
# origin  https://github.com/username/repo2.git (fetch)
# origin  https://github.com/username/repo2.git (push)
```

Section 3.3: List Existing Remotes

List all the existing remotes associated with this repository:

```
git remote
```

List all the existing remotes associated with this repository in detail including the `fetch` and `push` URLs:

```
git remote --verbose
```

or simply

```
git remote -v
```

Section 3.4: Removing Local Copies of Deleted Remote Branches

If a remote branch has been deleted, your local repository has to be told to prune the reference to it.

To prune deleted branches from a specific remote:

```
git fetch [remote-name] --prune
```

To prune deleted branches from *all* remotes:

```
git fetch --all --prune
```

Section 3.5: Updating from Upstream Repository

Assuming you set the upstream (as in the "setting an upstream repository")

```
git fetch remote-name
git merge remote-name/branch-name
```

The `pull` command combines a `fetch` and a `merge`.

```
git pull
```

The `pull` with `--rebase` flag command combines a `fetch` and a `rebase` instead of `merge`.

```
git pull --rebase remote-name branch-name
```

Section 3.6: ls-remote

`git ls-remote` is one unique command allowing you to query a remote repo *without having to clone/fetch it first*.

It will list refs/heads and refs/tags of said remote repo.

You will see sometimes `refs/tags/v0.1.6` *and* `refs/tags/v0.1.6^{}`: the `^{}` to list the dereferenced annotated tag (ie the commit that tag is pointing to)

Since git 2.8 (March 2016), you can avoid that double entry for a tag, and list directly those dereferenced tags with:

```
git ls-remote --ref
```

It can also help resolve the actual url used by a remote repo when you have "`url.<base>.insteadOf`" config setting. If `git remote --get-url <aremotename>` returns https://server.com/user/repo, and you have set `git config url.ssh://git@server.com:.insteadOf https://server.com/`:

```
git ls-remote --get-url <aremotename>
ssh://git@server.com:user/repo
```

Section 3.7: Adding a New Remote Repository

```
git remote add upstream git-repository-url
```

Adds remote git repository represented by `git-repository-url` as new remote named `upstream` to the git repository

Section 3.8: Set Upstream on a New Branch

You can create a new branch and switch to it using

```
git checkout -b AP-57
```

After you use git checkout to create a new branch, you will need to set that upstream origin to push to using

```
git push --set-upstream origin AP-57
```

After that, you can use git push while you are on that branch.

Section 3.9: Getting Started

Syntax for pushing to a remote branch

```
git push <remote_name> <branch_name>
```

Example

```
git push origin master
```

Section 3.10: Renaming a Remote

To rename remote, use command `git remote rename`

The `git remote rename` command takes two arguments:

- An existing remote name, for example : **origin**
- A new name for the remote, for example : **destination**

Get existing remote name

```
git remote
# origin
```

Check existing remote with URL

```
git remote -v
# origin https://github.com/username/repo.git (fetch)
# origin https://github.com/usernam/repo.git (push)
```

Rename remote

```
 git remote rename origin destination
 # Change remote name from 'origin' to 'destination'
```

Verify new name

```
git remote -v
# destination https://github.com/username/repo.git (fetch)
# destination https://github.com/usernam/repo.git (push)
```

=== Posible Errors ===

1. Could not rename config section 'remote.[old name]' to 'remote.[new name]'

   This error means that the remote you tried the old remote name (**origin**) doesn't exist.

2. Remote [new name] already exists.

   Error message is self explanatory.

Section 3.11: Show information about a Specific Remote

Output some information about a known remote: `origin`

```
git remote show origin
```

Print just the remote's URL:

```
git config --get remote.origin.url
```

With 2.7+, it is also possible to do, which is arguably better than the above one that uses the `config` command.

```
git remote get-url origin
```

Section 3.12: Set the URL for a Specific Remote

You can change the url of an existing remote by the command

```
git remote set-url remote-name url
```

Section 3.13: Get the URL for a Specific Remote

You can obtain the url for an existing remote by using the command

```
git remote get-url <name>
```

By default, this will be

```
git remote get-url origin
```

Section 3.14: Changing a Remote Repository

To change the URL of the repository you want your remote to point to, you can use the `set-url` option, like so:

```
git remote set-url <remote_name> <remote_repository_url>
```

Example:

```
git remote set-url heroku https://git.heroku.com/fictional-remote-repository.git
```

Chapter 4: Staging

Section 4.1: Staging All Changes to Files

```
git add -A
```

Version ≥ 2.0

```
git add .
```

In version 2.x, `git add .` will stage all changes to files in the current directory and all its subdirectories. However, in 1.x it will only stage new and modified files, not deleted files.

Use `git add -A`, or its equivalent command `git add --all`, to stage all changes to files in any version of git.

Section 4.2: Unstage a file that contains changes

```
git reset <filePath>
```

Section 4.3: Add changes by hunk

You can see what "hunks" of work would be staged for commit using the patch flag:

```
git add -p
```

or

```
git add --patch
```

This opens an interactive prompt that allows you to look at the diffs and let you decide whether you want to include them or not.

```
Stage this hunk [y,n,q,a,d,/,s,e,?]?
```

- y stage this hunk for the next commit
- n do not stage this hunk for the next commit
- q quit; do not stage this hunk or any of the remaining hunks
- a stage this hunk and all later hunks in the file
- d do not stage this hunk or any of the later hunks in the file
- g select a hunk to go to
- / search for a hunk matching the given regex
- j leave this hunk undecided, see next undecided hunk
- J leave this hunk undecided, see next hunk
- k leave this hunk undecided, see previous undecided hunk
- K leave this hunk undecided, see previous hunk
- s split the current hunk into smaller hunks
- e manually edit the current hunk
- ? print hunk help

This makes it easy to catch changes which you do not want to commit.

You can also open this via `git add --interactive` and selecting `p`.

Section 4.4: Interactive add

`git add -i` (or `--interactive`) will give you an interactive interface where you can edit the index, to prepare what you want to have in the next commit. You can add and remove changes to whole files, add untracked files and remove files from being tracked, but also select subsection of changes to put in the index, by selecting chunks of changes to be added, splitting those chunks, or even editing the diff. Many graphical commit tools for Git (like e.g. `git gui`) include such feature; this might be easier to use than the command line version.

It is very useful (1) if you have entangled changes in the working directory that you want to put in separate commits, and not all in one single commit (2) if you are in the middle of an interactive rebase and want to split too large commit.

```
$ git add -i
           staged     unstaged path
  1:    unchanged        +4/-4 index.js
  2:        +1/-0      nothing package.json

*** Commands ***
  1: status       2: update       3: revert       4: add untracked
  5: patch        6: diff         7: quit         8: help
What now>
```

The top half of this output shows the current state of the index broken up into staged and unstaged columns:

1. `index.js` has had 4 lines added and 4 lines removed. It is currently not staged, as the current status reports "unchanged." When this file becomes staged, the `+4/-4` bit will be transferred to the staged column and the unstaged column will read "nothing."
2. `package.json` has had one line added and has been staged. There are no further changes since it has been staged as indicated by the "nothing" line under the unstaged column.

The bottom half shows what you can do. Either enter a number (1-8) or a letter (s, u, r, a, p, d, q, h).

`status` shows output identical to the top part of the output above.

`update` allows you to make further changes to the staged commits with additional syntax.

`revert` will revert the staged commit information back to HEAD.

`add untracked` allows you to add filepaths previously untracked by version control.

`patch` allows for one path to be selected out of an output similar to `status` for further analysis.

`diff` displays what will be committed.

`quit` exits the command.

`help` presents further help on using this command.

Section 4.5: Show Staged Changes

To display the hunks that are staged for commit:

```
git diff --cached
```

Section 4.6: Staging A Single File

To stage a file for committing, run

```
git add <filename>
```

Section 4.7: Stage deleted files

```
git rm filename
```

To delete the file from git without removing it from disk, use the `--cached` flag

```
git rm --cached filename
```

Chapter 5: Ignoring Files and Folders

This topic illustrates how to avoid adding unwanted files (or file changes) in a Git repo. There are several ways (global or local `.gitignore`, `.git/exclude`, `git update-index --assume-unchanged`, and `git update-index --skip-tree`), but keep in mind Git is managing *content*, which means: ignoring actually ignores a folder *content* (i.e. files). An empty folder would be ignored by default, since it cannot be added anyway.

Section 5.1: Ignoring files and directories with a .gitignore file

You can make Git ignore certain files and directories — that is, exclude them from being tracked by Git — by creating one or more `.gitignore` files in your repository.

In software projects, `.gitignore` typically contains a listing of files and/or directories that are generated during the build process or at runtime. Entries in the `.gitignore` file may include names or paths pointing to:

1. temporary resources e.g. caches, log files, compiled code, etc.
2. local configuration files that should not be shared with other developers
3. files containing secret information, such as login passwords, keys and credentials

When created in the top level directory, the rules will apply recursively to all files and sub-directories throughout the entire repository. When created in a sub-directory, the rules will apply to that specific directory and its sub-directories.

When a file or directory is ignored, it will not be:

1. tracked by Git
2. reported by commands such as `git status` or `git diff`
3. staged with commands such as `git add -A`

In the unusual case that you need to ignore tracked files, special care should be taken. See: Ignore files that have already been committed to a Git repository.

Examples

Here are some generic examples of rules in a `.gitignore` file, based on glob file patterns:

```
# Lines starting with `#` are comments.

# Ignore files called 'file.ext'
file.ext

# Comments can't be on the same line as rules!
# The following line ignores files called 'file.ext # not a comment'
file.ext # not a comment

# Ignoring files with full path.
# This matches files in the root directory and subdirectories too.
# i.e. otherfile.ext will be ignored anywhere on the tree.
dir/otherdir/file.ext
otherfile.ext

# Ignoring directories
# Both the directory itself and its contents will be ignored.
bin/
gen/
```

```
# Glob pattern can also be used here to ignore paths with certain characters.
# For example, the below rule will match both build/ and Build/
[bB]uild/

# Without the trailing slash, the rule will match a file and/or
# a directory, so the following would ignore both a file named `gen`
# and a directory named `gen`, as well as any contents of that directory
bin
gen

# Ignoring files by extension
# All files with these extensions will be ignored in
# this directory and all its sub-directories.
*.apk
*.class

# It's possible to combine both forms to ignore files with certain
# extensions in certain directories. The following rules would be
# redundant with generic rules defined above.
java/*.apk
gen/*.class

# To ignore files only at the top level directory, but not in its
# subdirectories, prefix the rule with a `/`
/*.apk
/*.class

# To ignore any directories named DirectoryA
# in any depth use ** before DirectoryA
# Do not forget the last /,
# Otherwise it will ignore all files named DirectoryA, rather than directories
**/DirectoryA/
# This would ignore
# DirectoryA/
# DirectoryB/DirectoryA/
# DirectoryC/DirectoryB/DirectoryA/
# It would not ignore a file named DirectoryA, at any level

# To ignore any directory named DirectoryB within a
# directory named DirectoryA with any number of
# directories in between, use ** between the directories
DirectoryA/**/DirectoryB/
# This would ignore
# DirectoryA/DirectoryB/
# DirectoryA/DirectoryQ/DirectoryB/
# DirectoryA/DirectoryQ/DirectoryW/DirectoryB/

# To ignore a set of files, wildcards can be used, as can be seen above.
# A sole '*' will ignore everything in your folder, including your .gitignore file.
# To exclude specific files when using wildcards, negate them.
# So they are excluded from the ignore list:
!.gitignore

# Use the backslash as escape character to ignore files with a hash (#)
# (supported since 1.6.2.1)
\#*#
```

Most `.gitignore` files are standard across various languages, so to get started, here is set of sample `.gitignore` files listed by language from which to clone or copy/modify into your project. Alternatively, for a fresh project you may consider auto-generating a starter file using an online tool.

Other forms of .gitignore

`.gitignore` files are intended to be committed as part of the repository. If you want to ignore certain files without committing the ignore rules, here are some options:

- Edit the `.git/info/exclude` file (using the same syntax as `.gitignore`). The rules will be global in the scope of the repository;
- Set up a global gitignore file that will apply ignore rules to all your local repositories:

Furthermore, you can ignore local changes to tracked files without changing the global git configuration with:

- `git update-index --skip-worktree [<file>...]`: for minor local modifications
- `git update-index --assume-unchanged [<file>...]`: for production ready, non-changing files upstream

See more details on differences between the latter flags and the `git update-index` documentation for further options.

Cleaning up ignored files

You can use `git clean -X` to cleanup ignored files:

```
git clean -Xn #display a list of ignored files
git clean -Xf #remove the previously displayed files
```

Note: `-X` (caps) cleans up *only* ignored files. Use `-x` (no caps) to also remove untracked files.

See the `git clean` documentation for more details.

See the Git manual for more details.

Section 5.2: Checking if a file is ignored

The `git check-ignore` command reports on files ignored by Git.

You can pass filenames on the command line, and `git check-ignore` will list the filenames that are ignored. For example:

```
$ cat .gitignore
*.o
$ git check-ignore example.o Readme.md
example.o
```

Here, only *.o files are defined in .gitignore, so Readme.md is not listed in the output of `git check-ignore`.

If you want to see line of which .gitignore is responsible for ignoring a file, add -v to the git check-ignore command:

```
$ git check-ignore -v example.o Readme.md
.gitignore:1:*.o        example.o
```

From Git 1.7.6 onwards you can also use `git status --ignored` in order to see ignored files. You can find more info on this in the official documentation or in Finding files ignored by .gitignore.

Section 5.3: Exceptions in a .gitignore file

If you ignore files by using a pattern but have exceptions, prefix an exclamation mark(!) to the exception. For example:

```
*.txt
!important.txt
```

The above example instructs Git to ignore all files with the `.txt` extension except for files named `important.txt`.

If the file is in an ignored folder, you can **NOT** re-include it so easily:

```
folder/
!folder/*.txt
```

In this example all .txt files in the folder would remain ignored.

The right way is re-include the folder itself on a separate line, then ignore all files in `folder` by `*`, finally re-include the `*.txt` in `folder`, as the following:

```
!folder/
folder/*
!folder/*.txt
```

Note: For file names beginning with an exclamation mark, add two exclamation marks or escape with the `\` character:

```
!!includethis
\!excludethis
```

Section 5.4: A global .gitignore file

To have Git ignore certain files across all repositories you can create a global .gitignore with the following command in your terminal or command prompt:

```
$ git config --global core.excludesfile <Path_To_Global_gitignore_file>
```

Git will now use this in addition to each repository's own .gitignore file. Rules for this are:

- If the local `.gitignore` file explicitly includes a file while the global `.gitignore` ignores it, the local `.gitignore` takes priority (the file will be included)
- If the repository is cloned on multiple machines, then the global `.gigignore` must be loaded on all machines or at least include it, as the ignored files will be pushed up to the repo while the PC with the global `.gitignore` wouldn't update it. This is why a repo specific `.gitignore` is a better idea than a global one if the project is worked on by a team

This file is a good place to keep platform, machine or user specific ignores, e.g. OSX `.DS_Store`, Windows `Thumbs.db` or Vim `*.ext~` and `*.ext.swp` ignores if you don't want to keep those in the repository. So one team member working on OS X can add all `.DS_STORE` and `_MACOSX` (which is actually useless), while another team member on Windows can ignore all `thumbs.bd`

Section 5.5: Ignore files that have already been committed to

a Git repository

If you have already added a file to your Git repository and now want to **stop tracking it** (so that it won't be present in future commits), you can remove it from the index:

```
git rm --cached <file>
```

This will remove the file from the repository and prevent further changes from being tracked by Git. The `--cached` option will make sure that the file is not physically deleted.

Note that previously added contents of the file will still be visible via the Git history.

Keep in mind that if anyone else pulls from the repository after you removed the file from the index, **their copy will be physically deleted**.

You can make Git pretend that the working directory version of the file is up to date and read the index version instead (thus ignoring changes in it) with "skip worktree" bit:

```
git update-index --skip-worktree <file>
```

Writing is not affected by this bit, content safety is still first priority. You will never lose your precious ignored changes; on the other hand this bit conflicts with stashing: to remove this bit, use

```
git update-index --no-skip-worktree <file>
```

It is sometimes ***wrongly*** recommended to lie to Git and have it assume that file is unchanged without examining it. It looks at first glance as ignoring any further changes to the file, without removing it from its index:

```
git update-index --assume-unchanged <file>
```

This will force git to ignore any change made in the file (keep in mind that if you pull any changes to this file, or you stash it, **your ignored changes will be lost**)

If you want git to "care" about this file again, run the following command:

```
git update-index --no-assume-unchanged <file>
```

Section 5.6: Ignore files locally without committing ignore rules

`.gitignore` ignores files locally, but it is intended to be committed to the repository and shared with other contributors and users. You can set a global `.gitignore`, but then all your repositories would share those settings.

If you want to ignore certain files in a repository locally and not make the file part of any repository, edit `.git/info/exclude` inside your repository.

For example:

```
# these files are only ignored on this repo
# these rules are not shared with anyone
# as they are personal
gtk_tests.py
gui/gtk/tests/*
localhost
```

```
pushReports.py
server/
```

Section 5.7: Ignoring subsequent changes to a file (without removing it)

Sometimes you want to have a file held in Git but ignore subsequent changes.

Tell Git to ignore changes to a file or directory using `update-index`:

```
git update-index --assume-unchanged my-file.txt
```

The above command instructs Git to assume `my-file.txt` hasn't been changed, and not to check or report changes. The file is still present in the repository.

This can be useful for providing defaults and allowing local environment overrides, e.g.:

```
# create a file with some values in
cat <<EOF
MYSQL_USER=app
MYSQL_PASSWORD=FIXME_SECRET_PASSWORD
EOF > .env

# commit to Git
git add .env
git commit -m "Adding .env template"

# ignore future changes to .env
git update-index --assume-unchanged .env

# update your password
vi .env

# no changes!
git status
```

Section 5.8: Ignoring a file in any directory

To ignore a file `foo.txt` in **any** directory you should just write its name:

```
foo.txt # matches all files 'foo.txt' in any directory
```

If you want to ignore the file only in part of the tree, you can specify the subdirectories of a specific directory with `**` pattern:

```
bar/**/foo.txt # matches all files 'foo.txt' in 'bar' and all subdirectories
```

Or you can create a `.gitignore` file in the `bar/` directory. Equivalent to the previous example would be creating file `bar/.gitignore` with these contents:

```
foo.txt # matches all files 'foo.txt' in any directory under bar/
```

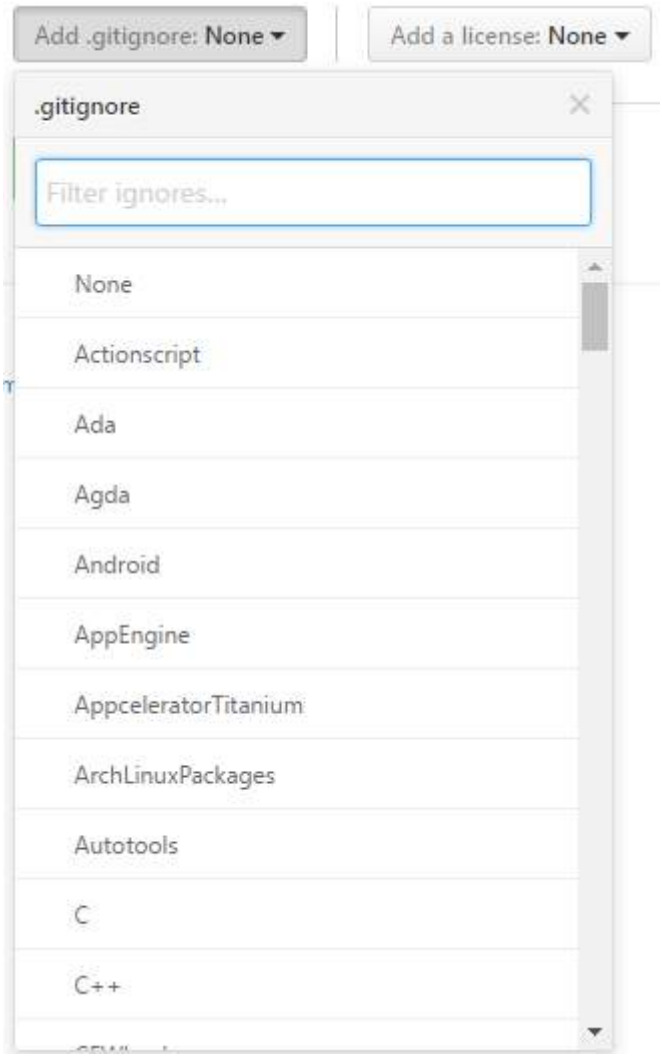
Section 5.9: Prefilled .gitignore Templates

If you are unsure which rules to list in your `.gitignore` file, or you just want to add generally accepted exceptions

to your project, you can choose or generate a `.gitignore` file:

- https://www.gitignore.io/
- https://github.com/github/gitignore

Many hosting services such as GitHub and BitBucket offer the ability to generate `.gitignore` files based upon the programming languages and IDEs you may be using:



Section 5.10: Ignoring files in subfolders (Multiple gitignore files)

Suppose you have a repository structure like this:

```
examples/
    output.log
src/
    <files not shown>
    output.log
README.md
```

`output.log` in the examples directory is valid and required for the project to gather an understanding while the one beneath `src/` is created while debugging and should not be in the history or part of the repository.

There are two ways to ignore this file. You can place an absolute path into the `.gitignore` file at the root of the working directory:

```
# /.gitignore
src/output.log
```

Alternatively, you can create a `.gitignore` file in the `src/` directory and ignore the file that is relative to this `.gitignore`:

```
# /src/.gitignore
output.log
```

Section 5.11: Create an Empty Folder

It is not possible to add and commit an empty folder in Git due to the fact that Git manages *files* and attaches their directory to them, which slims down commits and improves speed. To get around this, there are two methods:

Method one: `.gitkeep`

One hack to get around this is to use a `.gitkeep` file to register the folder for Git. To do this, just create the required directory and add a `.gitkeep` file to the folder. This file is blank and doesn't serve any purpose other than to just register the folder. To do this in Windows (which has awkward file naming conventions) just open git bash in the directory and run the command:

> $ touch .gitkeep

This command just makes a blank `.gitkeep` file in the current directory

Method two: `dummy.txt`

Another hack for this is very similar to the above and the same steps can be followed, but instead of a `.gitkeep`, just use a `dummy.txt` instead. This has the added bonus of being able to easily create it in Windows using the context menu. And you get to leave funny messages in them too.You can also use `.gitkeep` file to track the empty directory. `.gitkeep` normally is an empty file that is added to track the empty directory.

Section 5.12: Finding files ignored by .gitignore

You can list all files ignored by git in current directory with command:

```
git status --ignored
```

So if we have repository structure like this:

```
.git
.gitignore
./example_1
./dir/example_2
./example_2
```

...and .gitignore file containing:

```
example_2
```

...than result of the command will be:

```
$ git status --ignored
```

```
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)

.gitignore
.example_1

Ignored files:
  (use "git add -f <file>..." to include in what will be committed)

dir/
example_2
```

If you want to list recursively ignored files in directories, you have to use additional parameter - `--untracked-files=all`

Result will look like this:

```
$ git status --ignored --untracked-files=all
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)

.gitignore
example_1

Ignored files:
  (use "git add -f <file>..." to include in what will be committed)

dir/example_2
example_2
```

Section 5.13: Ignoring only part of a file [stub]

Sometimes you may want to have local changes in a file you don't want to commit or publish. Ideally local settings should be concentrated in a separate file that can be placed into `.gitignore`, but sometimes as a short-term solution it can be helpful to have something local in a checked-in file.

You can make Git "unsee" those lines using clean filter. They won't even show up in diffs.

Suppose here is snippet from file `file1.c`:

```
struct settings s;
s.host = "localhost";
s.port = 5653;
s.auth = 1;
s.port = 15653; // NOCOMMIT
s.debug = 1; // NOCOMMIT
s.auth = 0; // NOCOMMIT
```

You don't want to publish `NOCOMMIT` lines anywhere.

Create "nocommit" filter by adding this to Git config file like `.git/config`:

```
[filter "nocommit"]
    clean=grep -v NOCOMMIT
```

Add (or create) this to `.git/info/attributes` or `.gitmodules`:

```
file1.c filter=nocommit
```

And your NOCOMMIT lines are hidden from Git.

Caveats:

- Using clean filter slows down processing of files, especially on Windows.
- The ignored line may disappear from file when Git updates it. It can be counteracted with a smudge filter, but it is trickier.
- Not tested on Windows

Section 5.14: Ignoring changes in tracked files. [stub]

.gitignore and `.git/info/exclude` work only for untracked files.

To set ignore flag on a tracked file, use the command update-index:

```
git update-index --skip-worktree myfile.c
```

To revert this, use:

```
git update-index --no-skip-worktree myfile.c
```

You can add this snippet to your global git config to have more convenient `git hide`, `git unhide` and `git hidden` commands:

```
[alias]
    hide   = update-index --skip-worktree
    unhide = update-index --no-skip-worktree
    hidden  = "!git ls-files -v | grep ^[hsS] | cut -c 3-"
```

You can also use the option --assume-unchanged with the update-index function

```
git update-index --assume-unchanged <file>
```

If you want to watch this file again for the changes, use

```
git update-index --no-assume-unchanged <file>
```

When --assume-unchanged flag is specified, the user promises not to change the file and allows Git to assume that the working tree file matches what is recorded in the index.Git will fail in case it needs to modify this file in the index e.g. when merging in a commit; thus, in case the assumed-untracked file is changed upstream, you will need to handle the situation manually.The focus lies on performance in this case.

While --skip-worktree flag is useful when you instruct git not to touch a specific file ever because the file is going to be changed locally and you don't want to accidentally commit the changes (i.e configuration/properties file configured for a particular environment). Skip-worktree takes precedence over assume-unchanged when both are set.

Section 5.15: Clear already committed files, but included in .gitignore

Sometimes it happens that a file was being tracked by git, but in a later point in time was added to .gitignore, in order to stop tracking it. It's a very common scenario to forget to clean up such files before its addition to .gitignore. In this case, the old file will still be hanging around in the repository.

To fix this problem, one could perform a "dry-run" removal of everything in the repository, followed by re-adding all the files back. As long as you don't have pending changes and the `--cached` parameter is passed, this command is fairly safe to run:

```
# Remove everything from the index (the files will stay in the file system)
$ git rm -r --cached .

# Re-add everything (they'll be added in the current state, changes included)
$ git add .

# Commit, if anything changed. You should see only deletions
$ git commit -m 'Remove all files that are in the .gitignore'

# Update the remote
$ git push origin master
```

Chapter 6: Git Diff

Parameter	Details
-p, -u, --patch	Generate patch
-s, --no-patch	Suppress diff output. Useful for commands like `git show` that show the patch by default, or to cancel the effect of `--patch`
--raw	Generate the diff in raw format
--diff-algorithm=	Choose a diff algorithm. The variants are as follows: `myers`, `minimal`, `patience`, `histogram`
--summary	Output a condensed summary of extended header information such as creations, renames and mode changes
--name-only	Show only names of changed files
--name-status	Show names and statuses of changed files The most common statuses are M (Modified), A (Added), and D (Deleted)
--check	Warn if changes introduce conflict markers or whitespace errors. What are considered whitespace errors is controlled by `core.whitespace` configuration. By default, trailing whitespaces (including lines that solely consist of whitespaces) and a space character that is immediately followed by a tab character inside the initial indent of the line are considered whitespace errors. Exits with non-zero status if problems are found. Not compatible with --exit-code
--full-index	Instead of the first handful of characters, show the full pre- and post-image blob object names on the "index" line when generating patch format output
--binary	In addition to `--full-index`, output a binary diff that can be applied with `git apply`
-a, --text	Treat all files as text.
--color	Set the color mode; i.e. use `--color=always` if you would like to pipe a diff to less and keep git's coloring

Section 6.1: Show differences in working branch

```
git diff
```

This will show the *unstaged* changes on the current branch from the commit before it. It will only show changes relative to the index, meaning it shows what you *could* add to the next commit, but haven't. To add (stage) these changes, you can use `git add`.

If a file is staged, but was modified after it was staged, `git diff` will show the differences between the current file and the staged version.

Section 6.2: Show changes between two commits

```
git diff 1234abc..6789def    # old   new
```

E.g.: Show the changes made in the last 3 commits:

```
git diff @~3..@    # HEAD -3   HEAD
```

Note: the two dots (..) is optional, but adds clarity.

This will show the textual difference between the commits, regardless of where they are in the tree.

Section 6.3: Show differences for staged files

```
git diff --staged
```

This will show the changes between the previous commit and the currently staged files.

NOTE: You can also use the following commands to accomplish the same thing:

```
git diff --cached
```

Which is just a synonym for `--staged` or

```
git status -v
```

Which will trigger the verbose settings of the `status` command.

Section 6.4: Comparing branches

Show the changes between the tip of **`new`** and the tip of **`original`**:

```
git diff original new     # equivalent to original..new
```

Show all changes on **`new`** since it branched from **`original`**:

```
git diff original...new     # equivalent to $(git merge-base original new)..new
```

Using only one parameter such as

git diff original

is equivalent to

git diff original..HEAD

Section 6.5: Show both staged and unstaged changes

To show all staged *and* unstaged changes, use:

```
git diff HEAD
```

NOTE: You can also use the following command:

```
git status -vv
```

The difference being that the output of the latter will actually tell you which changes are staged for commit and which are not.

Section 6.6: Show differences for a specific file or directory

```
git diff myfile.txt
```

Shows the changes between the previous commit of the specified file (`myfile.txt`) and the locally-modified version that has not yet been staged.

This also works for directories:

```
git diff documentation
```

The above shows the changes between the previous commit of all files in the specified directory (`documentation/`) and the locally-modified versions of these files, that have not yet been staged.

To show the difference between some version of a file in a given commit and the local `HEAD` version you can specify the commit you want to compare against:

```
git diff 27fa75e myfile.txt
```

Or if you want to see the version between two separate commits:

```
git diff 27fa75e ada9b57 myfile.txt
```

To show the difference between the version specified by the hash `ada9b57` and the latest commit on the branch `my_branchname` for only the relative directory called `my_changed_directory/` you can do this:

```
git diff ada9b57 my_branchname my_changed_directory/
```

Section 6.7: Viewing a word-diff for long lines

```
git diff [HEAD|--staged...] --word-diff
```

Rather than displaying lines changed, this will display differences within lines. For example, rather than:

```
-Hello world
+Hello world!
```

Where the whole line is marked as changed, `word-diff` alters the output to:

```
Hello [-world-]{+world!+}
```

You can omit the markers `[-`, `-]`, `{+`, `+}` by specifying `--word-diff=color` or `--color-words`. This will only use color coding to mark the difference:

```
@@ -1 +1 @@
Hello worldworld!
```

Section 6.8: Show differences between current version and last version

```
git diff HEAD^ HEAD
```

This will show the changes between the previous commit and the current commit.

Section 6.9: Produce a patch-compatible diff

Sometimes you just need a diff to apply using patch. The regular `git --diff` does not work. Try this instead:

```
git diff --no-prefix > some_file.patch
```

Then somewhere else you can reverse it:

```
patch -p0 < some_file.patch
```

Section 6.10: difference between two commit or branch

To view difference between two branch

```
git diff <branch1>..<branch2>
```

To view difference between two branch

```
git diff <commitId1>..<commitId2>
```

To view diff with current branch

```
git diff <branch/commitId>
```

To view summary of changes

```
git diff --stat <branch/commitId>
```

To view files that changed after a certain commit

```
git diff --name-only <commitId>
```

To view files that are different than a branch

```
git diff --name-only <branchName>
```

To view files that changed in a folder after a certain commit

```
git diff --name-only <commitId> <folder_path>
```

Section 6.11: Using meld to see all modifications in the working directory

```
git difftool -t meld --dir-diff
```

will show the working directory changes. Alternatively,

```
git difftool -t meld --dir-diff  [COMMIT_A] [COMMIT_B]
```

will show the differences between 2 specific commits.

Section 6.12: Diff UTF-16 encoded text and binary plist files

You can diff UTF-16 encoded files (localization strings file os iOS and macOS are examples) by specifying how git should diff these files.

Add the following to your `~/.gitconfig` file.

```
[diff "utf16"]
textconv = "iconv -f utf-16 -t utf-8"
```

`iconv` is a program to convert different encodings.

Then edit or create a `.gitattributes` file in the root of the repository where you want to use it. Or just edit `~/.gitattributes`.

```
*.strings diff=utf16
```

This will convert all files ending in `.strings` before git diffs.

You can do similar things for other files, that can be converted to text.

For binary plist files you edit `.gitconfig`

```
[diff "plist"]
textconv = plutil -convert xml1 -o -
```

and `.gitattributes`

```
*.plist diff=plist
```

Chapter 7: Undoing

Section 7.1: Return to a previous commit

To jump back to a previous commit, first find the commit's hash using `git log`.

To temporarily jump back to that commit, detach your head with:

```
git checkout 789abcd
```

This places you at commit `789abcd`. You can now make new commits on top of this old commit without affecting the branch your head is on. Any changes can be made into a proper branch using either `branch` or `checkout -b`.

To roll back to a previous commit while keeping the changes:

```
git reset --soft 789abcd
```

To roll back the ***last*** commit:

```
git reset --soft HEAD~
```

To permanently discard any changes made after a specific commit, use:

```
git reset --hard 789abcd
```

To permanently discard any changes made after the ***last*** commit:

```
git reset --hard HEAD~
```

Beware: While you can recover the discarded commits using `reflog` and `reset`, uncommitted changes cannot be recovered. Use `git stash`; `git reset` instead of `git reset --hard` to be safe.

Section 7.2: Undoing changes

Undo changes to a file or directory in the **working copy**.

```
git checkout -- file.txt
```

Used over all file paths, recursively from the current directory, it will undo all changes in the working copy.

```
git checkout -- .
```

To only undo parts of the changes use `--patch`. You will be asked, for each change, if it should be undone or not.

```
git checkout --patch -- dir
```

To undo changes added to the **index**.

```
git reset --hard
```

Without the `--hard` flag this will do a soft reset.

With local commits that you have yet to push to a remote you can also do a soft reset. You can thus rework the files

and then the commits.

```
git reset HEAD~2
```

The above example would unwind your last two commits and return the files to your working copy. You could then make further changes and new commits.

Beware: All of these operations, apart from soft resets, will permanently delete your changes. For a safer option, use `git stash -p` or `git stash`, respectively. You can later undo with `stash pop` or delete forever with `stash drop`.

Section 7.3: Using reflog

If you screw up a rebase, one option to start again is to go back to the commit (pre rebase). You can do this using `reflog` (which has the history of everything you've done for the last 90 days - this can be configured):

```
$ git reflog
4a5cbb3 HEAD@{0}: rebase finished: returning to refs/heads/foo
4a5cbb3 HEAD@{1}: rebase: fixed such and such
904f7f0 HEAD@{2}: rebase: checkout upstream/master
3cbe20a HEAD@{3}: commit: fixed such and such
...
```

You can see the commit before the rebase was `HEAD@{3}` (you can also checkout the hash):

```
git checkout HEAD@{3}
```

Now you create a new branch / delete the old one / try the rebase again.

You can also reset directly back to a point in your `reflog`, but only do this if you're 100% sure it's what you want to do:

```
git reset --hard HEAD@{3}
```

This will set your current git tree to match how it was at that point (See Undoing Changes).

This can be used if you're temporarily seeing how well a branch works when rebased on another branch, but you don't want to keep the results.

Section 7.4: Undoing merges

Undoing a merge not yet pushed to a remote

If you haven't yet pushed your merge to the remote repository then you can follow the same procedure as in undo the commit although there are some subtle differences.

A reset is the simplest option as it will undo both the merge commit and any commits added from the branch. However, you will need to know what SHA to reset back to, this can be tricky as your `git log` will now show commits from both branches. If you reset to the wrong commit (e.g. one on the other branch) **it can destroy committed work.**

```
> git reset --hard <last commit from the branch you are on>
```

Or, assuming the merge was your most recent commit.

```
> git reset HEAD~
```

A revert is safer, in that it won't destroy committed work, but involves more work as you have to revert the revert before you can merge the branch back in again (see the next section).

Undoing a merge pushed to a remote

Assume you merge in a new feature (add-gremlins)

```
> git merge feature/add-gremlins
...
   #Resolve any merge conflicts
> git commit #commit the merge
...
> git push
...
   501b75d..17a51fd  master -> master
```

Afterwards you discover that the feature you just merged in broke the system for other developers, it must be undone right away, and fixing the feature itself will take too long so you simply want to undo the merge.

```
> git revert -m 1 17a51fd
...
> git push
...
   17a51fd..e443799  master -> master
```

At this point the gremlins are out of the system and your fellow developers have stopped yelling at you. However, we are not finished just yet. Once you fix the problem with the add-gremlins feature you will need to undo this revert before you can merge back in.

```
> git checkout feature/add-gremlins
...
   #Various commits to fix the bug.
> git checkout master
...
> git revert e443799
...
> git merge feature/add-gremlins
...
   #Fix any merge conflicts introduced by the bug fix
> git commit #commit the merge
...
> git push
```

At this point your feature is now successfully added. However, given that bugs of this type are often introduced by merge conflicts a slightly different workflow is sometimes more helpful as it lets you fix the merge conflict on your branch.

```
> git checkout feature/add-gremlins
...
   #Merge in master and revert the revert right away.  This puts your branch in
   #the same broken state that master was in before.
> git merge master
...
> git revert e443799
...
   #Now go ahead and fix the bug (various commits go here)
```

```
> git checkout master
...
   #Don't need to revert the revert at this point since it was done earlier
> git merge feature/add-gremlins
...
   #Fix any merge conflicts introduced by the bug fix
> git commit #commit the merge
...
> git push
```

Section 7.5: Revert some existing commits

Use git revert to revert existing commits, especially when those commits have been pushed to a remote repository. It records some new commits to reverse the effect of some earlier commits, which you can push safely without rewriting history.

Don't use `git push --force` unless you wish to bring down the opprobrium of all other users of that repository. Never rewrite public history.

If, for example, you've just pushed up a commit that contains a bug and you need to back it out, do the following:

```
git revert HEAD~1
git push
```

Now you are free to revert the revert commit locally, fix your code, and push the good code:

```
git revert HEAD~1
work .. work .. work ..
git add -A .
git commit -m "Update error code"
git push
```

If the commit you want to revert is already further back in the history, you can simply pass the commit hash. Git will create a counter-commit undoing your original commit, which you can push to your remote safely.

```
git revert 912aaf0228338d0c8fb8cca0a064b0161a451fdc
git push
```

Section 7.6: Undo / Redo a series of commits

Assume you want to undo a dozen of commits and you want only some of them.

```
git rebase -i <earlier SHA>
```

-i puts rebase in "interactive mode". It starts off like the rebase discussed above, but before replaying any commits, it pauses and allows you to gently modify each commit as it's replayed.`rebase -i` will open in your default text editor, with a list of commits being applied, like this:

```
1  pick 84c4823 Early work on feature (we now know this was wrong)
2  pick 0835fe2 More work (this is okay)
3  pick 1e6e80f Still more work (also wrong)
4  pick 31dba49 Yet more work (yet also wrong)
5  pick 6943e85 Getting there now (starting a better path)
6  pick 38f5e4e Even better (finally working out well)
7  pick af67f82 Ooops, this belongs with 'Even better'
8
9  # Rebase 311731b..af67f82 onto 311731b (       7 TODO item(s))
```

To drop a commit, just delete that line in your editor. If you no longer want the bad commits in your project, you can delete lines 1 and 3-4 above.If you want to combine two commits together, you can use the `squash` or `fixup` commands

```
1  pick 0835fe2 More work (this is okay)
2  squash 6943e85 Getting there now (starting a better path)
3  pick 38f5e4e Even better (finally working out well)
4  fixup af67f82 Ooops, this belongs with an earlier commit
5
6  # Rebase 311731b..af67f82 onto 311731b (       7 TODO item(s))
```

Chapter 8: Merging

Parameter	Details
`-m`	Message to be included in the merge commit
`-v`	Show verbose output
`--abort`	Attempt to revert all files back to their state
`--ff-only`	Aborts instantly when a merge-commit would be required
`--no-ff`	Forces creation of a merge-commit, even if it wasn't mandatory
`--no-commit`	Pretends the merge failed to allow inspection and tweaking of the result
`--stat`	Show a diffstat after merge completion
`-n`/`--no-stat`	Don't show the diffstat
`--squash`	Allows for a single commit on the current branch with the merged changes

Section 8.1: Automatic Merging

When the commits on two branches don't conflict, Git can automatically merge them:

```
~/Stack Overflow(branch:master) » git merge another_branch
Auto-merging file_a
Merge made by the 'recursive' strategy.
 file_a | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Section 8.2: Finding all branches with no merged changes

Sometimes you might have branches lying around that have already had their changes merged into master. This finds all branches that are not `master` that have no unique commits as compared to `master`. This is very useful for finding branches that were not deleted after the PR was merged into master.

```
 for branch in $(git branch -r) ; do
   [ "${branch}" != "origin/master" ] && [ $(git diff master...${branch} | wc -l) -eq 0 ] && echo -
e `git show --pretty=format:"%ci %cr" $branch | head -n 1`\\t$branch
 done | sort -r
```

Section 8.3: Aborting a merge

After starting a merge, you might want to stop the merge and return everything to its pre-merge state. Use `--abort`:

```
git merge --abort
```

Section 8.4: Merge with a commit

Default behaviour is when the merge resolves as a fast-forward, only update the branch pointer, without creating a merge commit. Use `--no-ff` to resolve.

```
git merge <branch_name> --no-ff -m "<commit message>"
```

Section 8.5: Keep changes from only one side of a merge

During a merge, you can pass `--ours` or `--theirs` to `git checkout` to take all changes for a file from one side or the other of a merge.

```
$ git checkout --ours   -- file1.txt # Use our version of file1, delete all their changes
$ git checkout --theirs -- file2.txt # Use their version of file2, delete all our changes
```

Section 8.6: Merge one branch into another

```
git merge incomingBranch
```

This merges the branch `incomingBranch` into the branch you are currently in. For example, if you are currently in `master`, then `incomingBranch` will be merged into `master`.

Merging can create conflicts in some cases. If this happens, you will see the message `Automatic merge failed; fix conflicts and then commit the result.` You will need to manually edit the conflicted files, or to undo your merge attempt, run:

```
git merge --abort
```

Chapter 9: Submodules

Section 9.1: Cloning a Git repository having submodules

When you clone a repository that uses submodules, you'll need to initialize and update them.

```
$ git clone --recursive https://github.com/username/repo.git
```

This will clone the referenced submodules and place them in the appropriate folders (including submodules within submodules). This is equivalent to running `git submodule update --init --recursive` immediately after the clone is finished.

Section 9.2: Updating a Submodule

A submodule references a specific commit in another repository. To check out the exact state that is referenced for all submodules, run

```
git submodule update --recursive
```

Sometimes instead of using the state that is referenced you want to update to your local checkout to the latest state of that submodule on a remote. To check out all submodules to the latest state on the remote with a single command, you can use

```
git submodule foreach git pull <remote> <branch>
```

or use the default `git pull` arguments

```
git submodule foreach git pull
```

Note that this will just update your local working copy. Running `git status` will list the submodule directory as dirty if it changed because of this command. To update your repository to reference the new state instead, you have to commit the changes:

```
git add <submodule_directory>
git commit
```

There might be some changes you have that can have merge conflict if you use `git pull` so you can use `git pull --rebase` to rewind your changes to top, most of the time it decreases the chances of conflict. Also it pulls all the branches to local.

```
git submodule foreach git pull --rebase
```

To checkout the latest state of a specific submodule, you can use :

```
git submodule update --remote <submodule_directory>
```

Section 9.3: Adding a submodule

You can include another Git repository as a folder within your project, tracked by Git:

```
$ git submodule add https://github.com/jquery/jquery.git
```

You should add and commit the new `.gitmodules` file; this tells Git what submodules should be cloned when `git submodule update` is run.

Section 9.4: Setting a submodule to follow a branch

A submodule is always checked out at a specific commit SHA1 (the "gitlink", special entry in the index of the parent repo)

But one can request to update that submodule to the latest commit of a branch of the submodule remote repo.

Rather than going in each submodule, doing a `git checkout abranch --track origin/abranch`, `git pull`, you can simply do (from the parent repo) a:

```
git submodule update --remote --recursive
```

Since the SHA1 of the submodule would change, you would still need to follow that with:

```
git add .
git commit -m "update submodules"
```

That supposes the submodules were:

- either added with a branch to follow:

  ```
  git submodule -b abranch -- /url/of/submodule/repo
  ```

- or configured (for an existing submodule) to follow a branch:

  ```
  cd /path/to/parent/repo
  git config -f .gitmodules submodule.asubmodule.branch abranch
  ```

Section 9.5: Moving a submodule

Version > 1.8

Run:

$ `git mv` /path/to/module *new/path/to/module*

Version ≤ 1.8

1. Edit `.gitmodules` and change the path of the submodule appropriately, and put it in the index with `git add .gitmodules`.
2. If needed, create the parent directory of the new location of the submodule (`mkdir -p` /path/to).
3. Move all content from the old to the new directory (`mv -vi` /path/to/module *new/path/to/submodule*).
4. Make sure Git tracks this directory (`git add` /path/to).
5. Remove the old directory with `git rm --cached` /path/to/module.
6. Move the directory `.git/modules/`/path/to/module with all its content to `.git/modules/`/path/to/module.
7. Edit the `.git/modules/`/path/to/config file, make sure that worktree item points to the new locations, so in this example it should be `worktree = ../../../../../`/path/to/module. Typically there should be two more `..` then directories in the direct path in that place. . Edit the file /path/to/module/.git, make sure that the path

in it points to the correct new location inside the main project `.git` folder, so in this example `gitdir`: `../../../.git/modules/`/path/to/module.

`git status` output looks like this afterwards:

```
# On branch master
# Changes to be committed:
#   (use "git reset HEAD <file>..." to unstage)
#
#       modified:   .gitmodules
#       renamed:    old/path/to/submodule -> new/path/to/submodule
#
```

8. Finally, commit the changes.

This example from Stack Overflow, by Axel Beckert

Section 9.6: Removing a submodule

Version > 1.8

You can remove a submodule (e.g. `the_submodule`) by calling:

```
$ git submodule deinit the_submodule
$ git rm the_submodule
```

- `git submodule deinit the_submodule` deletes `the_submodules`' entry from .git/config. This excludes the_submodule from `git submodule update`, `git submodule sync` and `git submodule foreach` calls and deletes its local content (source). Also, this will not be shown as change in your parent repository. `git submodule init` and `git submodule update` will restore the submodule, again without commitable changes in your parent repository.
- `git rm the_submodule` will remove the submodule from the work tree. The files will be gone as well as the submodules' entry in the `.gitmodules` file (source). If only `git rm the_submodule` (without prior `git submodule deinit the_submodule` is run, however, the submodules' entry in your .git/config file will remain.

Version < 1.8

Taken from here:

1. Delete the relevant section from the `.gitmodules` file.
2. Stage the `.gitmodules` changes `git add .gitmodules`
3. Delete the relevant section from `.git/config`.
4. Run `git rm --cached path_to_submodule` (no trailing slash).
5. Run `rm -rf .git/modules/path_to_submodule`
6. Commit `git commit -m "Removed submodule <name>"`
7. Delete the now untracked submodule files
8. `rm -rf path_to_submodule`

Chapter 10: Committing

Parameter	Details
--message, -m	Message to include in the commit. Specifying this parameter bypasses Git's normal behavior of opening an editor.
--amend	Specify that the changes currently staged should be added (amended) to the *previous* commit. Be careful, this can rewrite history!
--no-edit	Use the selected commit message without launching an editor. For example, `git commit --amend --no-edit` amends a commit without changing its commit message.
--all, -a	Commit all changes, including changes that aren't yet staged.
--date	Manually set the date that will be associated with the commit.
--only	Commit only the paths specified. This will not commit what you currently have staged unless told to do so.
--patch, -p	Use the interactive patch selection interface to chose which changes to commit.
--help	Displays the man page for `git commit`
-S[keyid], -S --gpg-sign[=keyid], -S --no-gpg-sign	Sign commit, GPG-sign commit, countermand `commit.gpgSign` configuration variable
-n, --no-verify	This option bypasses the pre-commit and commit-msg hooks. See also Hooks

Commits with Git provide accountability by attributing authors with changes to code. Git offers multiple features for the specificity and security of commits. This topic explains and demonstrates proper practices and procedures in committing with Git.

Section 10.1: Stage and commit changes

The basics

After making changes to your source code, you should **stage** those changes with Git before you can commit them.

For example, if you change `README.md` and `program.py`:

```
git add README.md program.py
```

This tells git that you want to add the files to the next commit you do.

Then, commit your changes with

```
git commit
```

Note that this will open a text editor, which is often vim. If you are not familiar with vim, you might want to know that you can press `i` to go into *insert* mode, write your commit message, then press `Esc` and `:wq` to save and quit. To avoid opening the text editor, simply include the `-m` flag with your message

```
git commit -m "Commit message here"
```

Commit messages often follow some specific formatting rules, see Good commit messages for more information.

Shortcuts

If you have changed a lot of files in the directory, rather than listing each one of them, you could use:

```
git add --all        # equivalent to "git add -a"
```

Or to add all changes, *not including files that have been deleted*, from the top-level directory and subdirectories:

```
git add .
```

Or to only add files which are currently tracked ("update"):

```
git add -u
```

If desired, review the staged changes:

```
git status           # display a list of changed files
git diff --cached    # shows staged changes inside staged files
```

Finally, commit the changes:

```
git commit -m "Commit message here"
```

Alternately, if you have only modified existing files or deleted files, and have not created any new ones, you can combine the actions of `git add` and `git commit` in a single command:

```
git commit -am "Commit message here"
```

Note that this will stage **all** modified files in the same way as `git add --all`.

Sensitive data

You should never commit any sensitive data, such as passwords or even private keys. If this case happens and the changes are already pushed to a central server, consider any sensitive data as compromised. Otherwise, it is possible to remove such data afterwards. A fast and easy solution is the usage of the "BFG Repo-Cleaner": https://rtyley.github.io/bfg-repo-cleaner/.

The command `bfg --replace-text passwords.txt my-repo.git` reads passwords out of the `passwords.txt` file and replaces these with `***REMOVED***`. This operation considers all previous commits of the entire repository.

Section 10.2: Good commit messages

It is important for someone traversing through the `git log` to easily understand what each commit was all about. Good commit messages usually include a number of a task or an issue in a tracker and a concise description of what has been done and why, and sometimes also how it has been done.

Better messages may look like:

```
TASK-123: Implement login through OAuth
TASK-124: Add auto minification of JS/CSS files
TASK-125: Fix minifier error when name > 200 chars
```

Whereas the following messages would not be quite as useful:

```
fix                          // What has been fixed?
```

```
just a bit of a change      // What has changed?
TASK-371                    // No description at all, reader will need to look at the tracker
themselves for an explanation
Implemented IFoo in IBar    // Why it was needed?
```

A way to test if a commit message is written in the correct mood is to replace the blank with the message and see if it makes sense:

If I add this commit, I will ___ to my repository.

The seven rules of a great git commit message

1. Separate the subject line from body with a blank line
2. Limit the subject line to 50 characters
3. Capitalize the subject line
4. Do not end the subject line with a period
5. Use the imperative mood in the subject line
6. Manually wrap each line of the body at 72 characters
7. Use the body to explain *what* and *why* instead of *how*

7 rules from Chris Beam's blog.

Section 10.3: Amending a commit

If your **latest commit is not published yet** (not pushed to an upstream repository) then you can amend your commit.

```
git commit --amend
```

This will put the currently staged changes onto the previous commit.

Note: This can also be used to edit an incorrect commit message. It will bring up the default editor (usually `vi` / `vim` / `emacs`) and allow you to change the prior message.

To specify the commit message inline:

```
git commit --amend -m "New commit message"
```

Or to use the previous commit message without changing it:

```
git commit --amend --no-edit
```

Amending updates the commit date but leaves the author date untouched. You can tell git to refresh the information.

```
git commit --amend --reset-author
```

You can also change the author of the commit with:

```
git commit --amend --author "New Author <email@address.com>"
```

Note: Be aware that amending the most recent commit replaces it entirely and the previous commit is removed from the branch's history. This should be kept in mind when working with public repositories and on branches with other collaborators.

This means that if the earlier commit had already been pushed, after amending it you will have to push `--force`.

Section 10.4: Committing without opening an editor

Git will usually open an editor (like `vim` or `emacs`) when you run `git commit`. Pass the `-m` option to specify a message from the command line:

```
git commit -m "Commit message here"
```

Your commit message can go over multiple lines:

```
git commit -m "Commit 'subject line' message here

More detailed description follows here (after a blank line)."
```

Alternatively, you can pass in multiple `-m` arguments:

```
git commit -m "Commit summary" -m "More detailed description follows here"
```

See How to Write a Git Commit Message.

Udacity Git Commit Message Style Guide

Section 10.5: Committing changes directly

Usually, you have to use `git add` or `git rm` to add changes to the index before you can `git commit` them. Pass the `-a` or `--all` option to automatically add every change (to tracked files) to the index, including removals:

```
git commit -a
```

If you would like to also add a commit message you would do:

```
git commit -a -m "your commit message goes here"
```

Also, you can join two flags:

```
git commit -am "your commit message goes here"
```

You don't necessarily need to commit all files at once. Omit the `-a` or `--all` flag and specify which file you want to commit directly:

```
git commit path/to/a/file -m "your commit message goes here"
```

For directly committing more than one specific file, you can specify one or multiple files, directories and patterns as well:

```
git commit path/to/a/file path/to/a/folder/* path/to/b/file -m "your commit message goes here"
```

Section 10.6: Selecting which lines should be staged for committing

Suppose you have many changes in one or more files but from each file you only want to commit some of the changes, you can select the desired changes using:

```
git add -p
```

or

```
git add -p [file]
```

Each of your changes will be displayed individually, and for each change you will be prompted to choose one of he following options:

```
y - Yes, add this hunk

n - No, don't add this hunk

d - No, don't add this hunk, or any other remaining hunks for this file.
Useful if you've already added what you want to, and want to skip over the rest.

s - Split the hunk into smaller hunks, if possible

e - Manually edit the hunk.  This is probably the most powerful option.
It will open the hunk in a text editor and you can edit it as needed.
```

This will stage the parts of the files you choose. Then you can commit all the staged changes like this:

```
git commit -m 'Commit Message'
```

The changes that were not staged or committed will still appear in your working files, and can be committed later if required. Or if the remaining changes are unwanted, they can be discarded with:

```
git reset --hard
```

Apart from breaking up a big change into smaller commits, this approach is also useful for *reviewing* what you are about to commit. By individually confirming each change, you have an opportunity to check what you wrote, and can avoid accidentally staging unwanted code such as println/logging statements.

Section 10.7: Creating an empty commit

Generally speaking, empty commits (or commits with state that is identical to the parent) is an error.

However, when testing build hooks, CI systems, and other systems that trigger off a commit, it's handy to be able to easily create commits without having to edit/touch a dummy file.

The `--allow-empty` commit will bypass the check.

```
git commit -m "This is a blank commit" --allow-empty
```

Section 10.8: Committing on behalf of someone else

If someone else wrote the code you are committing, you can give them credit with the `--author` option:

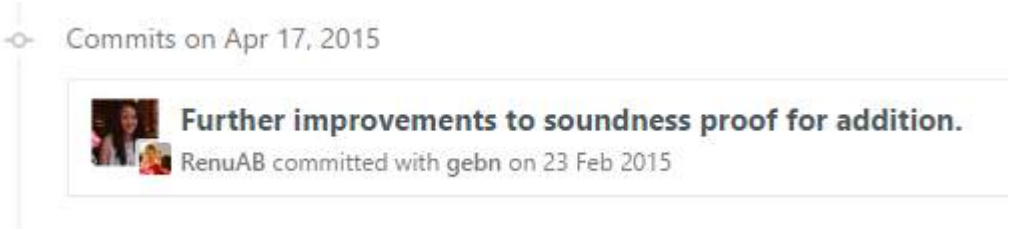
```
git commit -m "msg" --author "John Smith <johnsmith@example.com>"
```

You can also provide a pattern, which Git will use to search for previous authors:

```
git commit -m "msg" --author "John"
```

In this case, the author information from the most recent commit with an author containing "John" will be used.

On GitHub, commits made in either of the above ways will show a large author's thumbnail, with the committer's smaller and in front:



Section 10.9: GPG signing commits

1. Determine your key ID

```
gpg --list-secret-keys --keyid-format LONG

/Users/davidcondrey/.gnupg/secring.gpg
--------------------------------------
sec   2048R/YOUR-16-DIGIT-KEY-ID YYYY-MM-DD [expires: YYYY-MM-DD]
```

   Your ID is a alphanumeric 16-digit code following the first forward-slash.

2. Define your key ID in your git config

```
git config --global user.signingkey YOUR-16-DIGIT-KEY-ID
```

3. As of version 1.7.9, git commit accepts the -S option to attach a signature to your commits. Using this option will prompt for your GPG passphrase and will add your signature to the commit log.

```
git commit -S -m "Your commit message"
```

Section 10.10: Commiting changes in specific files

You can commit changes made to specific files and skip staging them using `git add`:

```
git commit file1.c file2.h
```

Or you can first stage the files:

```
git add file1.c file2.h
```

and commit them later:

```
git commit
```

Section 10.11: Committing at a specific date

```
git commit -m 'Fix UI bug' --date 2016-07-01
```

The `--date` parameter sets the *author date*. This date will appear in the standard output of `git log`, for example.

To force the *commit date* too:

```
GIT_COMMITTER_DATE=2016-07-01 git commit -m 'Fix UI bug' --date 2016-07-01
```

The date parameter accepts the flexible formats as supported by GNU date, for example:

```
git commit -m 'Fix UI bug' --date yesterday
git commit -m 'Fix UI bug' --date '3 days ago'
git commit -m 'Fix UI bug' --date '3 hours ago'
```

When the date doesn't specify time, the current time will be used and only the date will be overridden.

Section 10.12: Amending the time of a commit

You cam amend the time of a commit using

```
git commit --amend --date="Thu Jul 28 11:30 2016 -0400"
```

or even

```
git commit --amend --date="now"
```

Section 10.13: Amending the author of a commit

If you make a commit as the wrong author, you can change it, and then amend

```
git config user.name "Full Name"
git config user.email "email@example.com"

git commit --amend --reset-author
```

Chapter 11: Aliases

Section 11.1: Simple aliases

There are two ways of creating aliases in Git:

- with the `~/.gitconfig` file:

```
[alias]
    ci = commit
    st = status
    co = checkout
```

- with the command line:

```
git config --global alias.ci "commit"
git config --global alias.st "status"
git config --global alias.co "checkout"
```

After the alias is created - type:

- `git ci` instead of `git commit`,
- `git st` instead of `git status`,
- `git co` instead of `git checkout`.

As with regular git commands, aliases can be used beside arguments. For example:

```
git ci -m "Commit message..."
git co -b feature-42
```

Section 11.2: List / search existing aliases

You can list existing git aliases using `--get-regexp`:

```
$ git config --get-regexp '^alias\.'
```

Searching aliases

To search aliases, add the following to your `.gitconfig` under `[alias]`:

```
aliases = !git config --list | grep ^alias\\. | cut -c 7- | grep -Ei --color \"$1\" "#"
```

Then you can:

- `git aliases` - show ALL aliases
- `git aliases commit` - only aliases containing "commit"

Section 11.3: Advanced Aliases

Git lets you use non-git commands and full sh shell syntax in your aliases if you prefix them with `!`.

In your `~/.gitconfig` file:

```
[alias]
```

```
    temp = !git add -A && git commit -m "Temp"
```

The fact that full shell syntax is available in these prefixed aliases also means you can use shell functions to construct more complex aliases, such as ones which utilize command line arguments:

```
[alias]
    ignore = "!f() { echo $1 >> .gitignore; }; f"
```

The above alias defines the `f` function, then runs it with any arguments you pass to the alias. So running `git ignore .tmp/` would add `.tmp/` to your `.gitignore` file.

In fact, this pattern is so useful that Git defines $1, $2, etc. variables for you, so you don't even have to define a special function for it. (But keep in mind that Git will also append the arguments anyway, even if you access it via these variables, so you may want to add a dummy command at the end.)

Note that aliases prefixed with `!` in this way are run from the root directory of your git checkout, even if your current directory is deeper in the tree. This can be a useful way to run a command from the root without having to `cd` there explicitly.

```
[alias]
    ignore = "! echo $1 >> .gitignore"
```

Section 11.4: Temporarily ignore tracked files

To temporarily mark a file as ignored (pass file as parameter to alias) - type:

```
unwatch = update-index --assume-unchanged
```

To start tracking file again - type:

```
watch = update-index --no-assume-unchanged
```

To list all files that has been temporarily ignored - type:

```
unwatched = "!git ls-files -v | grep '^[[:lower:]]'"
```

To clear the unwatched list - type:

```
watchall = "!git unwatched | xargs -L 1 -I % sh -c 'git watch `echo % | cut -c 2-`'"
```

Example of using the aliases:

```
git unwatch my_file.txt
git watch my_file.txt
git unwatched
git watchall
```

Section 11.5: Show pretty log with branch graph

```
[alias]
  logp=log --pretty=format:'%h %ad | %s%d [%an]' --graph --date=short

  lg =  log --graph --date-order --first-parent \
     --pretty=format:'%C(auto)%h%Creset %C(auto)%d%Creset %s %C(green)(%ad) %C(bold
```

```
cyan)<%an>%Creset'
  lgb = log --graph --date-order --branches --first-parent \
     --pretty=format:'%C(auto)%h%Creset %C(auto)%d%Creset %s %C(green)(%ad) %C(bold
cyan)<%an>%Creset'
 lga = log --graph --date-order --all \
   --pretty=format:'%C(auto)%h%Creset %C(auto)%d%Creset %s %C(green)(%ad) %C(bold
cyan)<%an>%Creset'
```

Here an explanation of the options and placeholder used in the `--pretty` format (exhaustive list are available with `git help log`)

--graph - draw the commit tree

--date-order - use commit timestamp order when possible

--first-parent - follow only the first parent on merge node.

--branches - show all local branches (by default, only current branch is shown)

--all - show all local and remotes branches

%h - hash value for commit (abbreviated)

%ad - Date stamp (author)

%an - Author username

%an - Commit username

%C(auto) - to use colors defined in [color] section

%Creset - to reset color

%d - --decorate (branch & tag names)

%s - commit message

%ad - author date (will follow --date directive) (and not commiter date)

%an - author name (can be %cn for commiter name)

Section 11.6: See which files are being ignored by your .gitignore configuration

```
[ alias ]

    ignored = ! git ls-files --others --ignored --exclude-standard --directory \
            && git ls-files --others -i --exclude-standard
```

Shows one line per file, so you can grep (only directories):

```
$ git ignored | grep '/$'
.yardoc/
doc/
```

Or count:

```
~$ git ignored | wc -l
199811                  # oops, my home directory is getting crowded
```

Section 11.7: Updating code while keeping a linear history

Sometimes you need to keep a linear (non-branching) history of your code commits. If you are working on a branch for a while, this can be tricky if you have to do a regular `git pull` since that will record a merge with upstream.

```
[alias]
  up = pull --rebase
```

This will update with your upstream source, then reapply any work you have not pushed on top of whatever you pulled down.

To use:

```
git up
```

Section 11.8: Unstage staged files

Normally, to remove files that are staged to be committed using the `git reset` commit, `reset` has a lot of functions depending on the arguments provided to it. To completely unstage all files staged, we can make use of git aliases to create a new alias that uses `reset` but now we do not need to remember to provide the correct arguments to `reset`.

```
git config --global alias.unstage "reset --"
```

Now, any time you want to **unstage** stages files, type `git unstage` and you are good to go.

Chapter 12: Rebasing

Parameter	Details
--continue	Restart the rebasing process after having resolved a merge conflict.
--abort	Abort the rebase operation and reset HEAD to the original branch. If branch was provided when the rebase operation was started, then HEAD will be reset to branch. Otherwise HEAD will be reset to where it was when the rebase operation was started.
--keep-empty	Keep the commits that do not change anything from its parents in the result.
--skip	Restart the rebasing process by skipping the current patch.
-m, --merge	Use merging strategies to rebase. When the recursive (default) merge strategy is used, this allows rebase to be aware of renames on the upstream side. Note that a rebase merge works by replaying each commit from the working branch on top of the upstream branch. Because of this, when a merge conflict happens, the side reported as ours is the so-far rebased series, starting with upstream, and theirs is the working branch. In other words, the sides are swapped.
--stat	Show a diffstat of what changed upstream since the last rebase. The diffstat is also controlled by the configuration option rebase.stat.
-x, --exec `command`	Perform interactive rebase, stopping between each commit and executing `command`

Section 12.1: Local Branch Rebasing

Rebasing reapplies a series of commits on top of another commit.

To `rebase` a branch, checkout the branch and then `rebase` it on top of another branch.

```
git checkout topic
git rebase master  # rebase current branch onto master branch
```

This would cause:

```
      A---B---C topic
/
D---E---F---G master
```

To turn into:

```
              A'--B'--C' topic
/
D---E---F---G master
```

These operations can be combined into a single command that checks out the branch and immediately rebases it:

```
git rebase master topic   # rebase topic branch onto master branch
```

Important: After the rebase, the applied commits will have a different hash. You should not rebase commits you have already pushed to a remote host. A consequence may be an inability to `git push` your local rebased branch to a remote host, leaving your only option to `git push --force`.

Section 12.2: Rebase: ours and theirs, local and remote

A rebase switches the meaning of "ours" and "theirs":

```
git checkout topic
git rebase   master    # rebase topic branch on top of master branch
```

Whatever HEAD's pointing to is "ours"

The first thing a rebase does is resetting the `HEAD` to `master`; before cherry-picking commits from the old branch `topic` to a new one (every commit in the former `topic` branch will be rewritten and will be identified by a different hash).

With respect to terminologies used by merge tools (not to be confused with local ref or remote ref)

```
=> local is master ("ours"),
=> remote is topic ("theirs")
```

That means a merge/diff tool will present the upstream branch as **`local`** (`master`: the branch on top of which you are rebasing), and the working branch as `remote` (`topic`: the branch being rebased)

```
+-----------------------------------------+
| LOCAL:master |    BASE    | REMOTE:topic |
+-----------------------------------------+
|             MERGED                      |
+-----------------------------------------+
```

Inversion illustrated
On a merge:

```
c--c--x--x--x(*) <- current branch topic ('*'=HEAD)
\
\
\--y--y--y <- other branch to merge
```

We don't change the current branch `topic`, so what we have is still what we were working on (and we merge from another branch)

```
c--c--x--x--x---------o(*)  MERGE, still on branch topic
\       ^        /
\     ours     /
\            /
--y--y--y--/
^
theirs
```

On a rebase:

But **on a rebase** we switch sides because the first thing a rebase does is to checkout the upstream branch to replay the current commits on top of it!

```
c--c--x--x--x(*) <- current branch topic ('*'=HEAD)
\
\
\--y--y--y <- upstream branch
```

A `git rebase upstream` will first set `HEAD` to the upstream branch, hence the switch of 'ours' and 'theirs' compared to the previous "current" working branch.

```
c--c--x--x--x <- former "current" branch, new "theirs"
\
\
\--y--y--y(*) <- set HEAD to this commit, to replay x's on it
^       this will be the new "ours"
|
upstream
```

The rebase will then replay 'their' commits on the new 'our' `topic` branch:

```
c--c..x..x..x <- old "theirs" commits, now "ghosts", available through "reflogs"
\
\
\--y--y--y--x'--x'--x'(*) <- topic  once all x's are replayed,
^                       point branch topic to this commit
|
upstream branch
```

Section 12.3: Interactive Rebase

This example aims to describe how one can utilize `git rebase` in interactive mode. It is expected that one has a basic understanding of what `git rebase` is and what it does.

Interactive rebase is initiated using following command:

```
git rebase -i
```

The `-i` option refers to *interactive mode*. Using interactive rebase, the user can change commit messages, as well as reorder, split, and/or squash (combine to one) commits.

Say you want to rearrange your last three commits. To do this you can run:

```
git rebase -i HEAD~3
```

After executing the above instruction, a file will be opened in your text editor where you will be able to select how your commits will be rebased. For the purpose of this example, just change the order of your commits, save the file, and close the editor. This will initiate a rebase with the order you've applied. If you check `git log` you will see your commits in the new order you specified.

Rewording commit messages

Now, you've decided that one of the commit messages is vague and you want it to be more descriptive. Let's examine the last three commits using the same command.

```
git rebase -i HEAD~3
```

Instead of rearranging the order the commits will be rebased, this time we will change `pick`, the default, to `reword` on a commit where you would like to change the message.

When you close the editor, the rebase will initiate and it will stop at the specific commit message that you wanted to

reword. This will let you change the commit message to whichever you desire. After you've changed the message, simply close the editor to proceed.

Changing the content of a commit

Besides changing the commit message you can also adapt the changes done by the commit. To do so just change `pick` to `edit` for one commit. Git will stop when it arrives at that commit and provide the original changes of the commit in the staging area. You can now adapt those changes by unstaging them or adding new changes.

As soon as the staging area contains all changes you want in that commit, commit the changes. The old commit message will be shown and can be adapted to reflect the new commit.

Splitting a single commit into multiple

Say you've made a commit but decided at a later point this commit could be split into two or more commits instead. Using the same command as before, replace `pick` with `edit` instead and hit enter.

Now, git will stop at the commit you have marked for editing and place all of its content into the staging area. From that point you can run `git reset HEAD^` to place the commit into your working directory. Then, you can add and commit your files in a different sequence - ultimately splitting a single commit into *n* commits instead.

Squashing multiple commits into one

Say you have done some work and have multiple commits which you think could be a single commit instead. For that you can carry out `git rebase -i HEAD~3`, replacing `3` with an appropriate amount of commits.

This time replace `pick` with `squash` instead. During the rebase, the commit which you've instructed to be squashed will be squashed on top of the previous commit; turning them into a single commit instead.

Section 12.4: Rebase down to the initial commit

Since Git 1.7.12 it is possible to rebase down to the root commit. The root commit is the first commit ever made in a repository, and normally cannot be edited. Use the following command:

```
git rebase -i --root
```

Section 12.5: Configuring autostash

Autostash is a very useful configuration option when using rebase for local changes. Oftentimes, you may need to bring in commits from the upstream branch, but are not ready to commit just yet.

However, Git does not allow a rebase to start if the working directory is not clean. Autostash to the rescue:

```
git config --global rebase.autostash    # one time configuration
git rebase @{u}                         # example rebase on upstream branch
```

The autostash will be applied whenever the rebase is finished. It does not matter whether the rebase finishes successfully, or if it is aborted. Either way, the autostash will be applied. If the rebase was successful, and the base commit therefore changed, then there may be a conflict between the autostash and the new commits. In this case, you will have to resolve the conflicts before committing. This is no different than if you would have manually stashed, and then applied, so there is no downside to doing it automatically.

Section 12.6: Testing all commits during rebase

Before making a pull request, it is useful to make sure that compile is successful and tests are passing for each commit in the branch. We can do that automatically using `-x` parameter.

For example:

```
git rebase -i -x make
```

will perform the interactive rebase and stop after each commit to execute `make`. In case `make` fails, git will stop to give you an opportunity to fix the issues and amend the commit before proceeding with picking the next one.

Section 12.7: Rebasing before a code review

Summary

This goal is to reorganize all of your scattered commits into more meaningful commits for easier code reviews. If there are too many layers of changes across too many files at once, it is harder to do a code review. If you can reorganize your chronologically created commits into topical commits, then the code review process is easier (and possibly less bugs slip through the code review process).

This overly-simplified example is not the only strategy for using git to do better code reviews. It is the way I do it, and it's something to inspire others to consider how to make code reviews and git history easier/better.

This also pedagogically demonstrates the power of rebase in general.

This example assumes you know about interactive rebasing.

Assuming:

- you're working on a feature branch off of master
- your feature has three main layers: front-end, back-end, DB
- you have made a lot of commits while working on a feature branch. Each commit touches multiple layers at once
- you want (in the end) only three commits in your branch
    - one containing all front end changes
    - one containing all back end changes
    - one containing all DB changes

Strategy:

- we are going to change our chronological commits into "topical" commits.
- first, split all commits into multiple, smaller commits -- each containing only one topic at a time (in our example, the topics are front end, back end, DB changes)
- Then reorder our topical commits together and 'squash' them into single topical commits

Example:

```
$ git log --oneline master..
975430b db adding works: db.sql logic.rb
3702650 trying to allow adding todo items: page.html logic.rb
43b075a first draft: page.html and db.sql
$ git rebase -i master
```

This will be shown in text editor:

```
pick 43b075a first draft: page.html and db.sql
pick 3702650 trying to allow adding todo items: page.html logic.rb
pick 975430b db adding works: db.sql logic.rb
```

Change it to this:

```
e 43b075a first draft: page.html and db.sql
e 3702650 trying to allow adding todo items: page.html logic.rb
e 975430b db adding works: db.sql logic.rb
```

Then git will apply one commit at a time. After each commit, it will show a prompt, and then you can do the following:

```
Stopped at 43b075a92a952faf999e76c4e4d7fa0f44576579... first draft: page.html and db.sql
You can amend the commit now, with

        git commit --amend

Once you are satisfied with your changes, run

        git rebase --continue

$ git status
rebase in progress; onto 4975ae9
You are currently editing a commit while rebasing branch 'feature' on '4975ae9'.
  (use "git commit --amend" to amend the current commit)
  (use "git rebase --continue" once you are satisfied with your changes)

nothing to commit, working directory clean
$ git reset HEAD^ #This 'uncommits' all the changes in this commit.
$ git status -s
 M db.sql
 M page.html
$ git add db.sql  #now we will create the smaller topical commits
$ git commit -m "first draft: db.sql"
$ git add page.html
$ git commit -m "first draft: page.html"
$ git rebase --continue
```

Then you will repeat those steps for every commit. In the end, you have this:

```
$ git log --oneline
0309336 db adding works: logic.rb
06f81c9 db adding works: db.sql
3264de2 adding todo items: page.html
675a02b adding todo items: logic.rb
272c674 first draft: page.html
08c275d first draft: db.sql
```

Now we run rebase one more time to reorder and squash:

```
$ git rebase -i master
```

This will be shown in text editor:

```
pick 08c275d first draft: db.sql
pick 272c674 first draft: page.html
pick 675a02b adding todo items: logic.rb
```

```
pick 3264de2 adding todo items: page.html
pick 06f81c9 db adding works: db.sql
pick 0309336 db adding works: logic.rb
```

Change it to this:

```
pick 08c275d first draft: db.sql
s 06f81c9 db adding works: db.sql
pick 675a02b adding todo items: logic.rb
s 0309336 db adding works: logic.rb
pick 272c674 first draft: page.html
s  3264de2 adding todo items: page.html
```

NOTICE: make sure that you tell git rebase to apply/squash the smaller topical commits *in the order they were chronologically commited*. Otherwise you might have false, needless merge conflicts to deal with.

When that interactive rebase is all said and done, you get this:

```
$ git log --oneline master..
74bdd5f adding todos: GUI layer
e8d8f7e adding todos: business logic layer
121c578 adding todos: DB layer
```

Recap

You have now rebased your chronological commits into topical commits. In real life, you may not need to do this every single time, but when you do want or need to do this, now you can. Plus, hopefully you learned more about git rebase.

Section 12.8: Aborting an Interactive Rebase

You have started an interactive rebase. In the editor where you pick your commits, you decide that something is going wrong (for example a commit is missing, or you chose the wrong rebase destination), and you want to abort the rebase.

To do this, simply delete all commits and actions (i.e. all lines not starting with the `#` sign) and the rebase will be aborted!

The help text in the editor actually provides this hint:

```
# Rebase 36d15de..612f2f7 onto 36d15de (3 command(s))
#
# Commands:
# p, pick = use commit
# r, reword = use commit, but edit the commit message
# e, edit = use commit, but stop for amending
# s, squash = use commit, but meld into previous commit
# f, fixup = like "squash", but discard this commit's log message
# x, exec = run command (the rest of the line) using shell
#
# These lines can be re-ordered; they are executed from top to bottom.
#
# If you remove a line here THAT COMMIT WILL BE LOST.
#
# However, if you remove everything, the rebase will be aborted.
#          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
# Note that empty commits are commented out
```

Section 12.9: Setup git-pull for automatically perform a rebase instead of a merge

If your team is following a rebase-based workflow, it may be a advantageous to setup git so that each newly created branch will perform a rebase operation, instead of a merge operation, during a `git pull`.

To setup every *new* branch to automatically rebase, add the following to your `.gitconfig` or `.git/config`:

```
[branch]
autosetuprebase = always
```

Command line: `git config [--global] branch.autosetuprebase always`

Alternatively, you can setup the `git pull` command to always behave as if the option `--rebase` was passed:

```
[pull]
rebase = true
```

Command line: `git config [--global] pull.rebase true`

Section 12.10: Pushing after a rebase

Sometimes you need rewrite history with a rebase, but `git push` complains about doing so because you rewrote history.

This can be solved with a `git push --force`, but consider `git push --force-with-lease`, indicating that you want the push to fail if the local remote-tracking branch differs from the branch on the remote, e.g., someone else pushed to the remote after the last fetch. This avoids inadvertently overwriting someone else's recent push.

Note: `git push --force` - and even `--force-with-lease` for that matter - can be a dangerous command because it rewrites the history of the branch. If another person had pulled the branch before the forced push, his/her `git pull` or `git fetch` will have errors because the local history and the remote history are diverged. This may cause the person to have unexpected errors. With enough looking at the reflogs the other user's work can be recovered, but it can lead to a lot of wasted time. If you must do a forced push to a branch with other contributors, try to coordinate with them so that they do not have to deal with errors.

Chapter 13: Configuration

Parameter	Details
`--system`	Edits the system-wide configuration file, which is used for every user (on Linux, this file is located at `$(prefix)/etc/gitconfig`)
`--global`	Edits the global configuration file, which is used for every repository you work on (on Linux, this file is located at `~/.gitconfig`
`--local`	Edits the respository-specific configuration file, which is located at `.git/config` in your repository; this is the default setting

Section 13.1: Setting which editor to use

There are several ways to set which editor to use for committing, rebasing, etc.

- Change the `core.editor` configuration setting.

  ```
  $ git config --global core.editor nano
  ```

- Set the `GIT_EDITOR` environment variable.

  For one command:

  ```
  $ GIT_EDITOR=nano git commit
  ```

  Or for all commands run in a terminal. **Note:** This only applies until you close the terminal.

  ```
  $ export GIT_EDITOR=nano
  ```

- To change the editor for *all* terminal programs, not just Git, set the `VISUAL` or `EDITOR` environment variable. (See VISUAL vs EDITOR.)

  ```
  $ export EDITOR=nano
  ```

  Note: As above, this only applies to the current terminal; your shell will usually have a configuration file to allow you to set it permanently. (On `bash`, for example, add the above line to your `~/.bashrc` or `~/.bash_profile`.)

Some text editors (mostly GUI ones) will only run one instance at a time, and generally quit if you already have an instance of them open. If this is the case for your text editor, Git will print the message `Aborting commit due to empty commit message.` without allowing you to edit the commit message first. If this happens to you, consult your text editor's documentation to see if it has a `--wait` flag (or similar) that will make it pause until the document is closed.

Section 13.2: Auto correct typos

```
git config --global help.autocorrect 17
```

This enables autocorrect in git and will forgive you for your minor mistakes (e.g. `git stats` instead of `git status`). The parameter you supply to `help.autocorrect` determines how long the system should wait, in tenths of a second, before automatically applying the autocorrected command. In the command above, 17 means that git

should wait 1.7 seconds before applying the autocorrected command.

However, bigger mistakes will be considered as missing commands, so typing something like `git testingit` would result in `testingit is not a git command.`

Section 13.3: List and edit the current configuration

Git config allows you to customize how git works. It is commonly used to set your name and email or favorite editor or how merges should be done.

To see the current configuration.

```
$ git config --list
...
core.editor=vim
credential.helper=osxkeychain
...
```

To edit the config:

```
$ git config <key> <value>
$ git config core.ignorecase true
```

If you intend the change to be true for all your repositories, use `--global`

```
$ git config --global user.name "Your Name"
$ git config --global user.email "Your Email"
$ git config --global core.editor vi
```

You can list again to see your changes.

Section 13.4: Username and email address

Right after you install Git, the first thing you should do is set your username and email address. From a shell, type:

```
git config --global user.name "Mr. Bean"
git config --global user.email mrbean@example.com
```

- `git config` is the command to get or set options
- `--global` means that the configuration file specific to your user account will be edited
- `user.name` and `user.email` are the keys for the configuration variables; `user` is the section of the configuration file. `name` and `email` are the names of the variables.
- `"Mr. Bean"` and `mrbean@example.com` are the values that you're storing in the two variables. Note the quotes around `"Mr. Bean"`, which are required because the value you are storing contains a space.

Section 13.5: Multiple usernames and email address

Since Git 2.13, multiple usernames and email addresses could be configured by using a folder filter.

Example for Windows:
.gitconfig

Edit: `git config --global -e`

Add:

```
[includeIf "gitdir:D:/work"]
  path = .gitconfig-work.config

[includeIf "gitdir:D:/opensource/"]
  path = .gitconfig-opensource.config
```

Notes

- The order is depended, the last one who matches "wins".
- the `/` at the end is needed - e.g. `"gitdir:D:/work"` won't work.
- the `gitdir:` prefix is required.

.gitconfig-work.config

File in the same directory as *.gitconfig*

```
[user]
  name = Money
  email = work@somewhere.com
```

.gitconfig-opensource.config

File in the same directory as *.gitconfig*

```
[user]
  name = Nice
  email = cool@opensource.stuff
```

Example for Linux

```
[includeIf "gitdir:~/work/"]
  path = .gitconfig-work
[includeIf "gitdir:~/opensource/"]
  path = .gitconfig-opensource
```

The file content and notes under section Windows.

Section 13.6: Multiple git configurations

You have up to 5 sources for git configuration:

- 6 files:
  - **`%ALLUSERSPROFILE%\Git\Config`** (Windows only)
  - (system) **`<git>/etc/gitconfig`**, with **`<git>`** being the git installation path.
    (on Windows, it is **`<git>\mingw64\etc\gitconfig`**)
  - (system) **`$XDG_CONFIG_HOME/git/config`** (Linux/Mac only)
  - (global) `~/.gitconfig` (Windows: `%USERPROFILE%\.gitconfig`)
  - (local) `.git/config` (within a git repo `$GIT_DIR`)
  - a **dedicated file** (with `git config -f`), used for instance to modify the config of submodules: `git config -f .gitmodules ...`
- **the command line with `git -c`**: `git -c core.autocrlf=false fetch` would override *any* other `core.autocrlf` to `false`, *just* for that `fetch` command.

The order is important: any config set in one source can be overridden by a source listed below it.

`git config --system/global/local` is the command to list 3 of those sources, but only git config -l would list *all resolved* configs.
"resolved" means it lists only the final overridden config value.

Since git 2.8, if you want to see which config comes from which file, you type:

```
git config --list --show-origin
```

Section 13.7: Configuring line endings

Description

When working with a team who uses different operating systems (OS) across the project, sometimes you may run into trouble when dealing with line endings.

Microsoft Windows

When working on Microsoft Windows operating system (OS), the line endings are normally of form - carriage return + line feed (CR+LF). Opening a file which has been edited using Unix machine such as Linux or OSX may cause trouble, making it seem that text has no line endings at all. This is due to the fact that Unix systems apply different line-endings of form line feeds (LF) only.

In order to fix this you can run following instruction

```
git config --global core.autocrlf=true
```

On **checkout**, This instruction will ensure line-endings are configured in accordance with Microsoft Windows OS (LF -> CR+LF)

Unix Based (Linux/OSX)

Similarly, there might be issues when the user on Unix based OS tries to read files which have been edited on Microsoft Windows OS. In order to prevent any unexpected issues run

```
git config --global core.autocrlf=input
```

On **commit**, this will change line-endings from CR+LF -> +LF

Section 13.8: configuration for one command only

you can use `-c <name>=<value>` to add a configuration only for one command.

To commit as an other user without having to change your settings in .gitconfig :

```
git -c user.email = mail@example commit -m "some message"
```

Note: for that example you don't need to precise both `user.name` and `user.email`, git will complete the missing information from the previous commits.

Section 13.9: Setup a proxy

If you are behind a proxy, you have to tell git about it:

```
git config --global http.proxy http://my.proxy.com:portnumber
```

If you are no more behind a proxy:

```
git config --global --unset http.proxy
```

Chapter 14: Branching

Parameter	Details
-d, --delete	Delete a branch. The branch must be fully merged in its upstream branch, or in `HEAD` if no upstream was set with `--track` or `--set-upstream`
-D	Shortcut for `--delete --force`
-m, --move	Move/rename a branch and the corresponding reflog
-M	Shortcut for `--move --force`
-r, --remotes	List or delete (if used with -d) the remote-tracking branches
-a, --all	List both remote-tracking branches and local branches
--list	Activate the list mode. `git branch <pattern>` would try to create a branch, use `git branch --list <pattern>` to list matching branches
--set-upstream	If specified branch does not exist yet or if `--force` has been given, acts exactly like `--track`. Otherwise sets up configuration like --track would when creating the branch, except that where branch points to is not changed

Section 14.1: Creating and checking out new branches

To create a new branch, while staying on the current branch, use:

```
git branch <name>
```

Generally, the branch name must not contain spaces and is subject to other specifications listed here. To switch to an existing branch :

```
git checkout <name>
```

To create a new branch and switch to it:

```
git checkout -b <name>
```

To create a branch at a point other than the last commit of the current branch (also known as HEAD), use either of these commands:

```
git branch <name> [<start-point>]
git checkout -b <name> [<start-point>]
```

The **`<start-point>`** can be any revision known to git (e.g. another branch name, commit SHA, or a symbolic reference such as HEAD or a tag name):

```
git checkout -b <name> some_other_branch
git checkout -b <name> af295
git checkout -b <name> HEAD~5
git checkout -b <name> v1.0.5
```

To create a branch from a remote branch (the default **`<remote_name>`** is origin):

```
git branch <name> <remote_name>/<branch_name>
git checkout -b <name> <remote_name>/<branch_name>
```

If a given branch name is only found on one remote, you can simply use

```
git checkout -b <branch_name>
```

which is equivalent to

```
git checkout -b <branch_name> <remote_name>/<branch_name>
```

Sometimes you may need to move several of your recent commits to a new branch. This can be achieved by branching and "rolling back", like so:

```
git branch <new_name>
git reset --hard HEAD~2 # Go back 2 commits, you will lose uncommitted work.
git checkout <new_name>
```

Here is an illustrative explanation of this technique:

```
 Initial state       After git branch <new_name>    After git reset --hard HEAD~2
                              newBranch                          newBranch
                                  ↓                                  ↓
A-B-C-D-E (HEAD)         A-B-C-D-E (HEAD)                 A-B-C-D-E (HEAD)
        ↑                         ↑                                ↑
      master                    master                           master
```

Section 14.2: Listing branches

Git provides multiple commands for listing branches. All commands use the function of `git branch`, which will provide a list of a certain branches, depending on which options are put on the command line. Git will if possible, indicate the currently selected branch with a star next to it.

Goal	Command
List local branches	`git branch`
List local branches verbose	`git branch -v`
List remote and local branches	`git branch -a` OR `git branch --all`
List remote and local branches (verbose)	`git branch -av`
List remote branches	`git branch -r`
List remote branches with latest commit	`git branch -rv`
List merged branches	`git branch --merged`
List unmerged branches	`git branch --no-merged`
List branches containing commit	`git branch --contains [<commit>]`

Notes:

- Adding an additional `v` to `-v` e.g. `$ git branch -avv` or `$ git branch -vv` will print the name of the upstream branch as well.
- Branches shown in red color are remote branches

Section 14.3: Delete a remote branch

To delete a branch on the `origin` remote repository, you can use for Git version 1.5.0 and newer

```
git push origin :<branchName>
```

and as of Git version 1.7.0, you can delete a remote branch using

```
git push origin --delete <branchName>
```

To delete a local remote-tracking branch:

```
git branch --delete --remotes <remote>/<branch>
git branch -dr <remote>/<branch> # Shorter

git fetch <remote> --prune # Delete multiple obsolete tracking branches
git fetch <remote> -p      # Shorter
```

To delete a branch locally. Note that this will not delete the branch if it has any unmerged changes:

```
git branch -d <branchName>
```

To delete a branch, even if it has unmerged changes:

```
git branch -D <branchName>
```

Section 14.4: Quick switch to the previous branch

You can quickly switch to the previous branch using

```
git checkout -
```

Section 14.5: Check out a new branch tracking a remote branch

There are three ways of creating a new branch `feature` which tracks the remote branch `origin/feature`:

- `git checkout --track -b feature origin/feature`,
- `git checkout -t origin/feature`,
- `git checkout feature` - assuming that there is no local `feature` branch and there is only one remote with the `feature` branch.

To set upstream to track the remote branch - type:

- `git branch --set-upstream-to=<remote>/<branch> <branch>`
- `git branch -u <remote>/<branch> <branch>`

where:

- **`<remote>`** can be: `origin`, `develop` or the one created by user,
- **`<branch>`** is user's branch to track on remote.

To verify which remote branches your local branches are tracking:

- `git branch -vv`

Section 14.6: Delete a branch locally

```
$ git branch -d dev
```

Deletes the branch named `dev` *if* its changes are merged with another branch and will not be lost. If the `dev` branch does contain changes that have not yet been merged that would be lost, `git branch -d` will fail:

```
$ git branch -d dev
error: The branch 'dev' is not fully merged.
If you are sure you want to delete it, run 'git branch -D dev'.
```

Per the warning message, you can force delete the branch (and lose any unmerged changes in that branch) by using the `-D` flag:

```
$ git branch -D dev
```

Section 14.7: Create an orphan branch (i.e. branch with no parent commit)

```
git checkout --orphan new-orphan-branch
```

> The first commit made on this new branch will have no parents and it will be the root of a new history totally disconnected from all the other branches and commits.

source

Section 14.8: Rename a branch

Rename the branch you have checked out:

```
git branch -m new_branch_name
```

Rename another branch:

```
git branch -m branch_you_want_to_rename new_branch_name
```

Section 14.9: Searching in branches

To list local branches that contain a specific commit or tag

```
git branch --contains <commit>
```

To list local and remote branches that contain a specific commit or tag

```
git branch -a --contains <commit>
```

Section 14.10: Push branch to remote

Use to push commits made on your local branch to a remote repository.

The `git push` command takes two arguments:

- A remote name, for example, `origin`
- A branch name, for example, `master`

For example:

```
git push  <REMOTENAME> <BRANCHNAME>
```

As an example, you usually run `git push origin master` to push your local changes to your online repository.

Using `-u` (short for `--set-upstream`) will set up the tracking information during the push.

```
git push -u <REMOTENAME> <BRANCHNAME>
```

By default, `git` pushes the local branch to a remote branch with the same name. For example, if you have a local called `new-feature`, if you push the local branch it will create a remote branch `new-feature` as well. If you want to use a different name for the remote branch, append the remote name after the local branch name, separated by `:`:

```
git push <REMOTENAME> <LOCALBRANCHNAME>:<REMOTEBRANCHNAME>
```

Section 14.11: Move current branch HEAD to an arbitrary commit

A branch is just a pointer to a commit, so you can freely move it around. To make it so that the branch is referring to the commit `aabbcc`, issue the command

```
git reset --hard aabbcc
```

Please note that this will overwrite your branch's current commit, and as so, its entire history. You might loose some work by issuing this command. If that's the case, you can use the reflog to recover the lost commits. It can be advised to perform this command on a new branch instead of your current one.

However, this command can be particularly useful when rebasing or doing such other large history modifications.

Chapter 15: Rev-List

Parameter	Details
--oneline	Display commits as a single line with their title.

Section 15.1: List Commits in master but not in origin/master

```
git rev-list --oneline master ^origin/master
```

Git rev-list will list commits in one branch that are not in another branch. It is a great tool when you're trying to figure out if code has been merged into a branch or not.

- Using the `--oneline` option will display the title of each commit.
- The `^` operator excludes commits in the specified branch from the list.
- You can pass more than two branches if you want. For example, `git rev-list foo bar ^baz` lists commits in foo and bar, but not baz.

Chapter 16: Squashing

Section 16.1: Squash Recent Commits Without Rebasing

If you want to squash the previous `x` commits into a single one, you can use the following commands:

```
git reset --soft HEAD~x
git commit
```

Replacing `x` with the number of previous commits you want to be included in the squashed commit.

Mind that this will create a *new* commit, essentially forgetting information about the previous `x` commits including their author, message and date. You probably want to *first* copy-paste an existing commit message.

Section 16.2: Squashing Commit During Merge

You can use `git merge --squash` to squash changes introduced by a branch into a single commit. No actual commit will be created.

```
git merge --squash <branch>
git commit
```

This is more or less equivalent to using `git reset`, but is more convenient when changes being incorporated have a symbolic name. Compare:

```
git checkout <branch>
git reset --soft $(git merge-base master <branch>)
git commit
```

Section 16.3: Squashing Commits During a Rebase

Commits can be squashed during a `git rebase`. It is recommended that you understand rebasing before attempting to squash commits in this fashion.

1. Determine which commit you would like to rebase from, and note its commit hash.
2. Run `git rebase -i [commit hash]`.

   Alternatively, you can type `HEAD~4` instead of a commit hash, to view the latest commit and 4 more commits before the latest one.

3. In the editor that opens when running this command, determine which commits you want to squash. Replace `pick` at the beginning of those lines with `squash` to squash them into the previous commit.
4. After selecting which commits you would like to squash, you will be prompted to write a commit message.

Logging Commits to determine where to rebase

```
> git log --oneline
612f2f7 This commit should not be squashed
d84b05d This commit should be squashed
ac60234 Yet another commit
36d15de Rebase from here
```

```
17692d1 Did some more stuff
e647334 Another Commit
2e30df6 Initial commit

> git rebase -i 36d15de
```

At this point your editor of choice pops up where you can describe what you want to do with the commits. Git provides help in the comments. If you leave it as is then nothing will happen because every commit will be kept and their order will be the same as they were before the rebase. In this example we apply the following commands:

```
pick ac60234 Yet another commit
squash d84b05d This commit should be squashed
pick 612f2f7 This commit should not be squashed

# Rebase 36d15de..612f2f7 onto 36d15de (3 command(s))
#
# Commands:
# p, pick = use commit
# r, reword = use commit, but edit the commit message
# e, edit = use commit, but stop for amending
# s, squash = use commit, but meld into previous commit
# f, fixup = like "squash", but discard this commit's log message
# x, exec = run command (the rest of the line) using shell
#
# These lines can be re-ordered; they are executed from top to bottom.
#
# If you remove a line here THAT COMMIT WILL BE LOST.
#
# However, if you remove everything, the rebase will be aborted.
#
# Note that empty commits are commented out
```

Git log after writing commit message

```
> git log --oneline
77393eb This commit should not be squashed
e090a8c Yet another commit
36d15de Rebase from here
17692d1 Did some more stuff
e647334 Another Commit
2e30df6 Initial commit
```

Section 16.4: Autosquashing and fixups

When committing changes it is possible to specify that the commit will in future be squashed to another commit and this can be done like so,

```
git commit --squash=[commit hash of commit to which this commit will be squashed to]
```

One might also use, `--fixup=[commit hash]` alternatively to fixup.

It is also possible to use words from the commit message instead of the commit hash, like so,

```
git commit --squash :/things
```

where the most recent commit with the word 'things' would be used.

These commits' message would begin with `'fixup!'` or `'squash!'` followed by the rest of the commit message to which these commits will be squashed to.

When rebasing `--autosquash` flag should be used to use the autosquash/fixup feature.

Section 16.5: Autosquash: Committing code you want to squash during a rebase

Given the following history, imagine you make a change that you want to squash into the commit `bbb2222 A second commit`:

```
$ git log --oneline --decorate
ccc3333 (HEAD -> master) A third commit
bbb2222 A second commit
aaa1111 A first commit
9999999 Initial commit
```

Once you've made your changes, you can add them to the index as usual, then commit them using the `--fixup` argument with a reference to the commit you want to squash into:

```
$ git add .
$ git commit --fixup bbb2222
[my-feature-branch ddd4444] fixup! A second commit
```

This will create a new commit with a commit message that Git can recognize during an interactive rebase:

```
$ git log --oneline --decorate
ddd4444 (HEAD -> master) fixup! A second commit
ccc3333 A third commit
bbb2222 A second commit
aaa1111 A first commit
9999999 Initial commit
```

Next, do an interactive rebase with the `--autosquash` argument:

```
$ git rebase --autosquash --interactive HEAD~4
```

Git will propose you to squash the commit you made with the `commit --fixup` into the correct position:

```
pick aaa1111 A first commit
pick bbb2222 A second commit
fixup ddd4444 fixup! A second commit
pick ccc3333 A third commit
```

To avoid having to type `--autosquash` on every rebase, you can enable this option by default:

```
$ git config --global rebase.autosquash true
```

Chapter 17: Cherry Picking

Parameters	Details
-e, --edit	With this option, `git cherry-pick` will let you edit the commit message prior to committing.
-x	When recording the commit, append a line that says "(cherry picked from commit …)" to the original commit message in order to indicate which commit this change was cherry-picked from. This is done only for cherry picks without conflicts.
--ff	If the current HEAD is the same as the parent of the cherry-pick'ed commit, then a fast forward to this commit will be performed.
--continue	Continue the operation in progress using the information in .git/sequencer. Can be used to continue after resolving conflicts in a failed cherry-pick or revert.
--quit	Forget about the current operation in progress. Can be used to clear the sequencer state after a failed cherry-pick or revert.
--abort	Cancel the operation and return to the pre-sequence state.

A cherry-pick takes the patch that was introduced in a commit and tries to reapply it on the branch you're currently on.

Source: Git SCM Book

Section 17.1: Copying a commit from one branch to another

`git cherry-pick <commit-hash>` will apply the changes made in an existing commit to another branch, while recording a new commit. Essentially, you can copy commits from branch to branch.

Given the following tree (Source)

```
dd2e86 - 946992 - 9143a9 - a6fd86 - 5a6057 [master]
            \
             76cada - 62ecb3 - b886a0 [feature]
```

Let's say we want to copy `b886a0` to master (on top of `5a6057`).

We can run

```
git checkout master
git cherry-pick b886a0
```

Now our tree will look something like:

```
dd2e86 - 946992 - 9143a9 - a6fd86 - 5a6057 - a66b23 [master]
            \
             76cada - 62ecb3 - b886a0 [feature]
```

Where the new commit `a66b23` has the same content (source diff, commit message) as `b886a0` (but a different parent). Note that cherry-picking will only pick up changes on that commit(`b886a0` in this case) not all the changes in feature branch (for this you will have to either use rebasing or merging).

Section 17.2: Copying a range of commits from one branch to another

`git cherry-pick <commit-A>..<commit-B>` will place every commit *after* A and up to and including B on top of the currently checked-out branch.

`git cherry-pick <commit-A>^..<commit-B>` will place commit A and every commit up to and including B on top of the currently checked-out branch.

Section 17.3: Checking if a cherry-pick is required

Before you start the cherry-pick process, you can check if the commit you want to cherry-pick already exists in the target branch, in which case you don't have to do anything.

`git branch --contains <commit>` lists local branches that contain the specified commit.

`git branch -r --contains <commit>` also includes remote tracking branches in the list.

Section 17.4: Find commits yet to be applied to upstream

Command `git cherry` shows the changes which haven't yet been cherry-picked.

Example:

```
git checkout master
git cherry development
```

... and see output a bit like this:

```
+ 492508acab7b454eee8b805f8ba906056eede0ff
- 5ceb5a9077ddb9e78b1e8f24bfc70e674c627949
+ b4459544c000f4d51d1ec23f279d9cdb19c1d32b
+ b6ce3b78e938644a293b2dd2a15b2fecb1b54cd9
```

The commits that being with `+` will be the ones that haven't yet cherry-picked into `development`.

Syntax:

`git cherry [-v] [<upstream> [<head> [<limit>]]]`

Options:

-v Show the commit subjects next to the SHA1s.

< upstream > Upstream branch to search for equivalent commits. Defaults to the upstream branch of HEAD.

< head > Working branch; defaults to HEAD.

< limit > Do not report commits up to (and including) limit.

Check git-cherry documentation for more info.

Chapter 18: Recovering

Section 18.1: Recovering from a reset

With Git, you can (almost) always turn the clock back

Don't be afraid to experiment with commands that rewrite history*. Git doesn't delete your commits for 90 days by default, and during that time you can easily recover them from the reflog:

```
$ git reset @~3   # go back 3 commits
$ git reflog
c4f708b HEAD@{0}: reset: moving to @~3
2c52489 HEAD@{1}: commit: more changes
4a5246d HEAD@{2}: commit: make important changes
e8571e4 HEAD@{3}: commit: make some changes
... earlier commits ...
$ git reset 2c52489
... and you're back where you started
```

** Watch out for options like `--hard` and `--force` though — they can discard data.*
** Also, avoid rewriting history on any branches you're collaborating on.*

Section 18.2: Recover from git stash

To get your most recent stash after running git stash, use

```
git stash apply
```

To see a list of your stashes, use

```
git stash list
```

You will get a list that looks something like this

```
stash@{0}: WIP on master: 67a4e01 Merge tests into develop
stash@{1}: WIP on master: 70f0d95 Add user role to localStorage on user login
```

Choose a different git stash to restore with the number that shows up for the stash you want

```
git stash apply stash@{2}
```

You can also choose 'git stash pop', it works same as 'git stash apply' like..

```
 git stash pop
```

or

```
 git stash pop stash@{2}
```

Difference in git stash apply and git stash pop...

git stash pop: stash data will be remove from stack of stash list.

Ex:

```
git stash list
```

You will get a list that looks something like this

```
stash@{0}: WIP on master: 67a4e01 Merge tests into develop
stash@{1}: WIP on master: 70f0d95 Add user role to localStorage on user login
```

Now pop stash data using command

```
git stash pop
```

Again Check for stash list

```
git stash list
```

You will get a list that looks something like this

```
stash@{0}: WIP on master: 70f0d95 Add user role to localStorage on user login
```

You can see one stash data is removed (popped) from stash list and stash@{1} became stash@{0}.

Section 18.3: Recovering from a lost commit

In case you have reverted back to a past commit and lost a newer commit you can recover the lost commit by running

```
git reflog
```

Then find your lost commit, and reset back to it by doing

```
git reset HEAD --hard <sha1-of-commit>
```

Section 18.4: Restore a deleted file after a commit

In case you have accidentally commited a delete on a file and later realized that you need it back.

First find the commit id of the commit that deleted your file.

```
git log --diff-filter=D --summary
```

Will give you a sorted summary of commits which deleted files.

Then proceed to restore the file by

```
git checkout 81eeccf~1 <your-lost-file-name>
```

(Replace 81eeccf with your own commit id)

Section 18.5: Restore file to a previous version

To restore a file to a previous version you can use `reset`.

```
git reset <sha1-of-commit> <file-name>
```

If you have already made local changes to the file (that you do not require!) you can also use the `--hard` option

Section 18.6: Recover a deleted branch

To recover a deleted branch you need to find the commit which was the head of your deleted branch by running

```
git reflog
```

You can then recreate the branch by running

```
git checkout -b <branch-name> <sha1-of-commit>
```

You will not be able to recover deleted branches if git's garbage collector deleted dangling commits - those without refs. Always have a backup of your repository, especially when you work in a small team / proprietary project

Chapter 19: Git Clean

Parameter	Details
-d	Remove untracked directories in addition to untracked files. If an untracked directory is managed by a different Git repository, it is not removed by default. Use -f option twice if you really want to remove such a directory.
-f, --force	If the Git configuration variable clean. `requireForce` is not set to false, git clean will refuse to delete files or directories unless given -f, -n or -i. Git will refuse to delete directories with .git sub directory or file unless a second -f is given.
-i, --interactive	Interactively prompts the removal of each file.
-n, --dry-run	Only displays a list of files to be removed, without actually removing them.
-q,--quiet	Only display errors, not the list of successfully removed files.

Section 19.1: Clean Interactively

```
git clean -i
```

Will print out items to be removed and ask for a confirmation via commands like the follow:

```
Would remove the following items:
folder/file1.py
folder/file2.py
*** Commands ***
1: clean        2: filter by pattern        3: select by numbers        4: ask each
5: quit        6: help
What now>
```

Interactive option `i` can be added along with other options like `X`, `d`, etc.

Section 19.2: Forcefully remove untracked files

```
git clean -f
```

Will remove all untracked files.

Section 19.3: Clean Ignored Files

```
git clean -fX
```

Will remove all ignored files from the current directory and all subdirectories.

```
git clean -Xn
```

Will preview all files that will be cleaned.

Section 19.4: Clean All Untracked Directories

```
git clean -fd
```

Will remove all untracked directories and the files within them. It will start at the current working directory and will iterate through all subdirectories.

```
git clean -dn
```

Will preview all directories that will be cleaned.

Chapter 20: Using a .gitattributes file

Section 20.1: Automatic Line Ending Normalization

Create a `.gitattributes` file in the project root containing:

```
* text=auto
```

This will result in all text files (as identified by Git) being committed with LF, but checked out according to the host operating system default.

This is equivalent to the recommended `core.autocrlf` defaults of:

- `input` on Linux/macOS
- `true` on Windows

Section 20.2: Identify Binary Files

Git is pretty good at identifying binary files, but you can explicitly specify which files are binary. Create a `.gitattributes` file in the project root containing:

```
*.png binary
```

`binary` is a built-in macro attribute equivalent to `-diff -merge -text`.

Section 20.3: Prefilled .gitattribute Templates

If you are unsure which rules to list in your `.gitattributes` file, or you just want to add generally accepted attributes to your project, you can shoose or generate a `.gitattributes` file at:

- https://gitattributes.io/
- https://github.com/alexkaratarakis/gitattributes

Section 20.4: Disable Line Ending Normalization

Create a `.gitattributes` file in the project root containing:

```
* -text
```

This is equivalent to setting `core.autocrlf = false`.

Chapter 21: .mailmap file: Associating contributor and email aliases

Section 21.1: Merge contributers by aliases to show commit count in shortlog

When contributors add to a project from different machines or operating systems, it may happen that they use different email addresses or names for this, which will fragment contributor lists and statistics.

Running `git shortlog -sn` to get a list of contributors and the number of commits by them could result in the following output:

```
Patrick Rothfuss 871
Elizabeth Moon 762
E. Moon 184
Rothfuss, Patrick 90
```

This fragmentation/disassociation may be adjusted by providing a plain text file `.mailmap`, containing email mappings.

All names and email addresses listed in one line will be associated to the first named entity respectively.

For the example above, a mapping could look like this:

```
Patrick Rothfuss <fussy@kingkiller.com> Rothfuss, Patrick <fussy@kingkiller.com>
Elizabeth Moon <emoon@marines.mil> E. Moon <emoon@scifi.org>
```

Once this file exists in the project's root, running `git shortlog -sn` again will result in a condensed list:

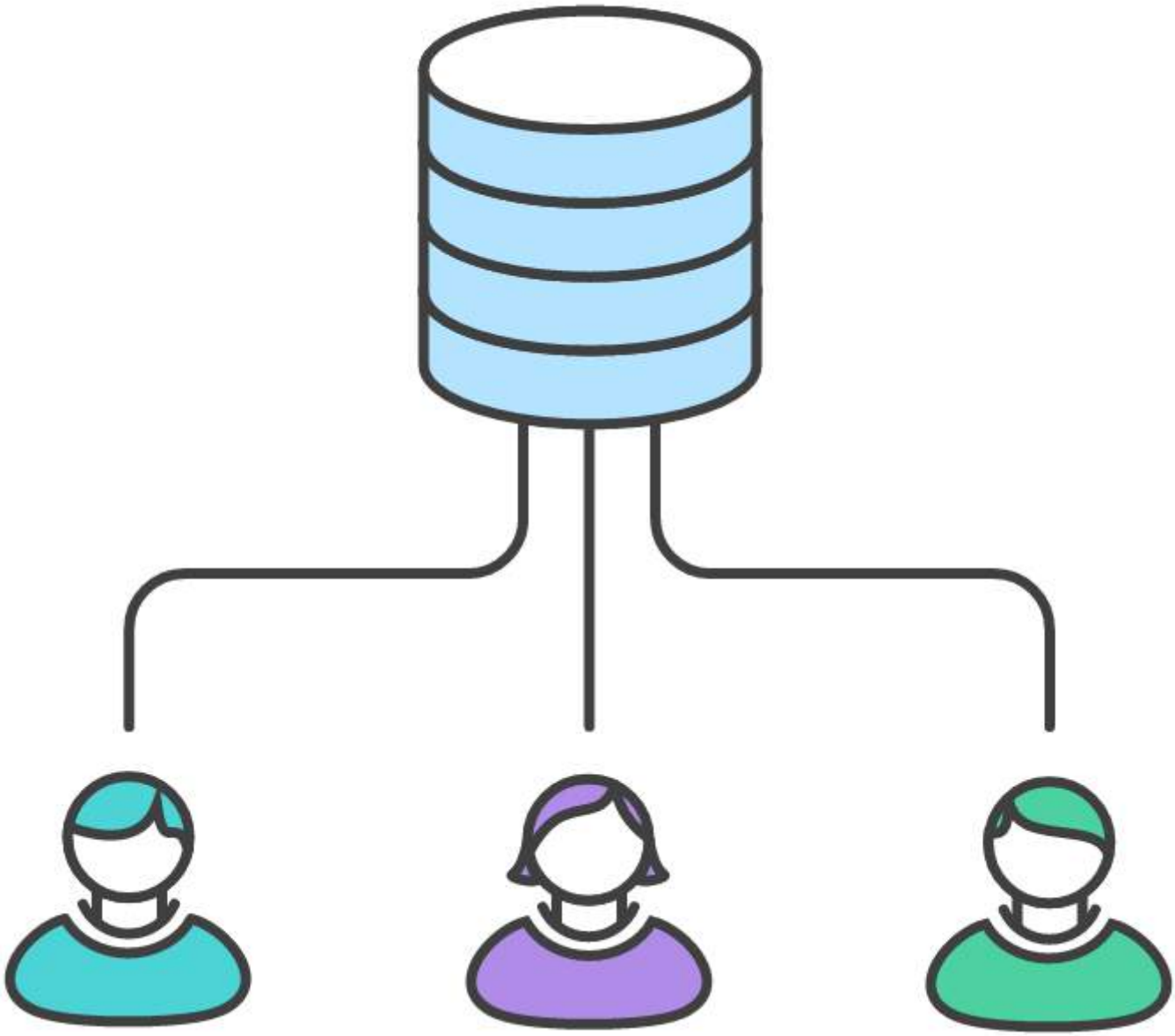
```
Patrick Rothfuss 961
Elizabeth Moon 946
```

Chapter 22: Analyzing types of workflows

Section 22.1: Centralized Workflow

With this fundamental workflow model, a `master` branch contains all active development. Contributors will need to be especially sure they pull the latest changes before continuing development, for this branch will be changing rapidly. Everyone has access to this repo and can commit changes right to the master branch.

Visual representation of this model:



This is the classic version control paradigm, upon which older systems like Subversion and CVS were built. Softwares that work this way are called Centralized Version Control Systems, or CVCS's. While Git is capable of working this way, there are notable disadvantages, such as being required to precede every pull with a merge. It's very possible for a team to work this way, but the constant merge conflict resolution can end up eating a lot of valuable time.

This is why Linus Torvalds created Git not as a CVCS, but rather as a *DVCS*, or *Distributed Version Control System*, similar to Mercurial. The advantage to this new way of doing things is the flexibility demonstrated in the other examples on this page.

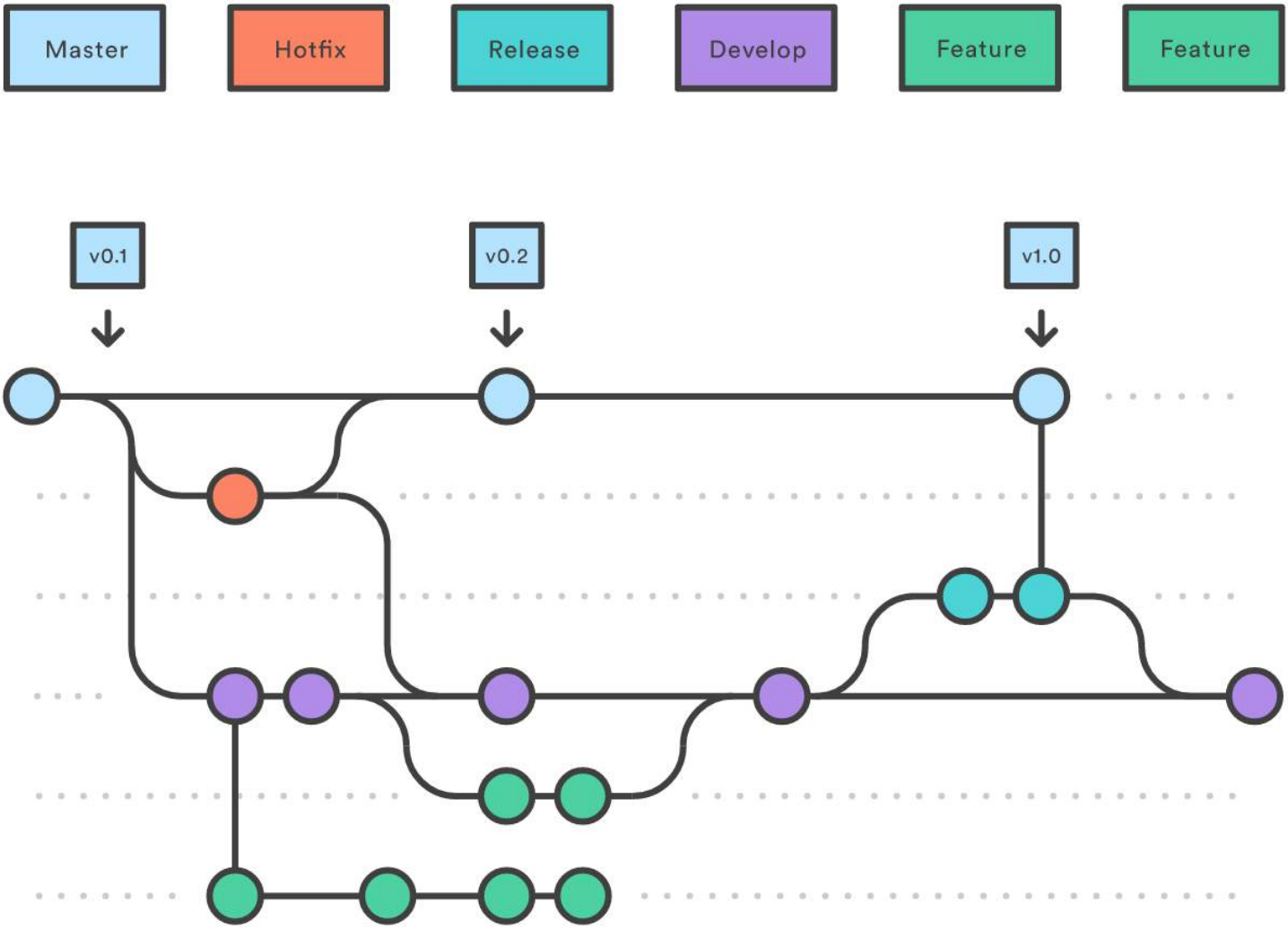
Section 22.2: Gitflow Workflow

Originally proposed by Vincent Driessen, Gitflow is a development workflow using git and several pre-defined branches. This can seen as a special case of the Feature Branch Workflow.

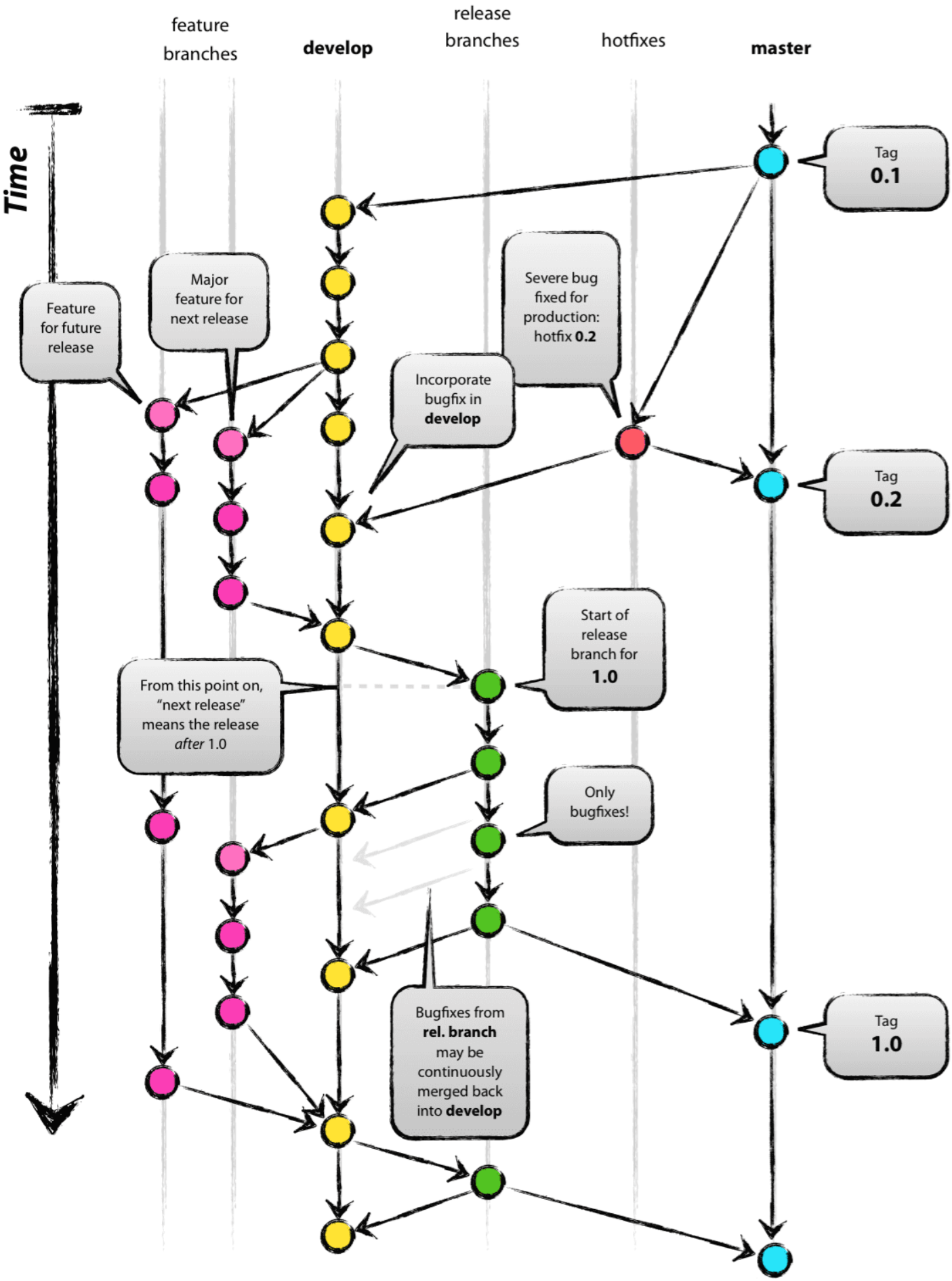
The idea of this one is to have separate branches reserved for specific parts in development:

- `master` branch is always the most recent *production* code. Experimental code does not belong here.
- `develop` branch contains all of the latest *development*. These developmental changes can be pretty much anything, but larger features are reserved for their own branches. Code here is always worked on and merged into `release` before release / deployment.
- `hotfix` branches are for minor bug fixes, which cannot wait until the next release. `hotfix` branches come off of `master` and are merged back into both `master` and `develop`.
- `release` branches are used to release new development from `develop` to `master`. Any last minute changes, such as bumping version numbers, are done in the release branch, and then are merged back into `master` and `develop`. When deploying a new version, `master` should be tagged with the current version number (e.g. using semantic versioning) for future reference and easy rollback.
- `feature` branches are reserved for bigger features. These are specifically developed in designated branches and integrated with `develop` when finished. Dedicated `feature` branches help to separate development and to be able to deploy *done* features independently from each other.

A visual representation of this model:



The original representation of this model:


feature branches
develop
release branches
hotfixes
master
Time
Tag 0.1
Feature for future release
Major feature for next release
Severe bug fixed for production: hotfix 0.2
Incorporate bugfix in develop
Tag 0.2
Start of release branch for 1.0
From this point on, "next release" means the release after 1.0
Only bugfixes!
Bugfixes from rel. branch may be continuously merged back into develop
Tag 1.0

Section 22.3: Feature Branch Workflow

The core idea behind the Feature Branch Workflow is that all feature development should take place in a dedicated branch instead of the `master` branch. This encapsulation makes it easy for multiple developers to work on a particular feature without disturbing the main codebase. It also means the `master` branch will never contain broken code, which is a huge advantage for continuous integration environments.

Encapsulating feature development also makes it possible to leverage pull requests, which are a way to initiate discussions around a branch. They give other developers the opportunity to sign off on a feature before it gets integrated into the official project. Or, if you get stuck in the middle of a feature, you can open a pull request asking for suggestions from your colleagues. The point is, pull requests make it incredibly easy for your team to comment on each other's work.

based on Atlassian Tutorials.

Section 22.4: GitHub Flow

Popular within many open source projects but not only.

Master branch of a specific location (Github, Gitlab, Bitbucket, local server) contains the latest shippable version. For each new feature/bug fix/architectural change each developer creates a branch.

Changes happen on that branch and can be discussed in a pull request, code review, etc. Once accepted they get merged to the master branch.

Full flow by Scott Chacon:

- Anything in the master branch is deployable
- To work on something new, create a descriptively named branch off of master (ie: new-oauth2-scopes)
- Commit to that branch locally and regularly push your work to the same named branch on the server
- When you need feedback or help, or you think the branch is ready for merging, open a pull request
- After someone else has reviewed and signed off on the feature, you can merge it into master
- Once it is merged and pushed to 'master', you can and should deploy immediately

Originally presented on Scott Chacon's personal web site.

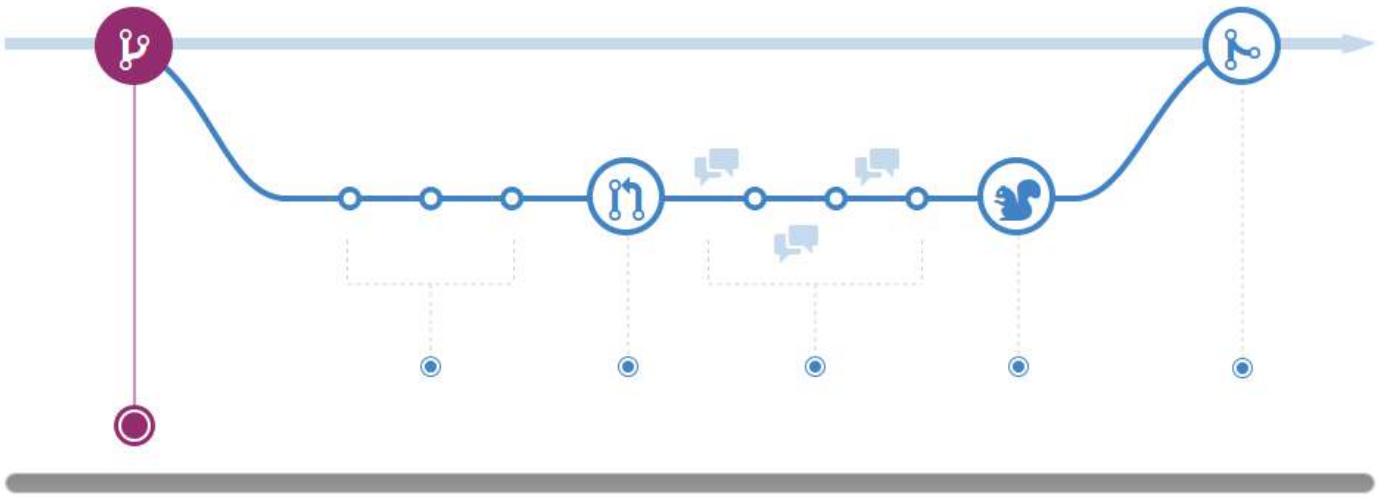
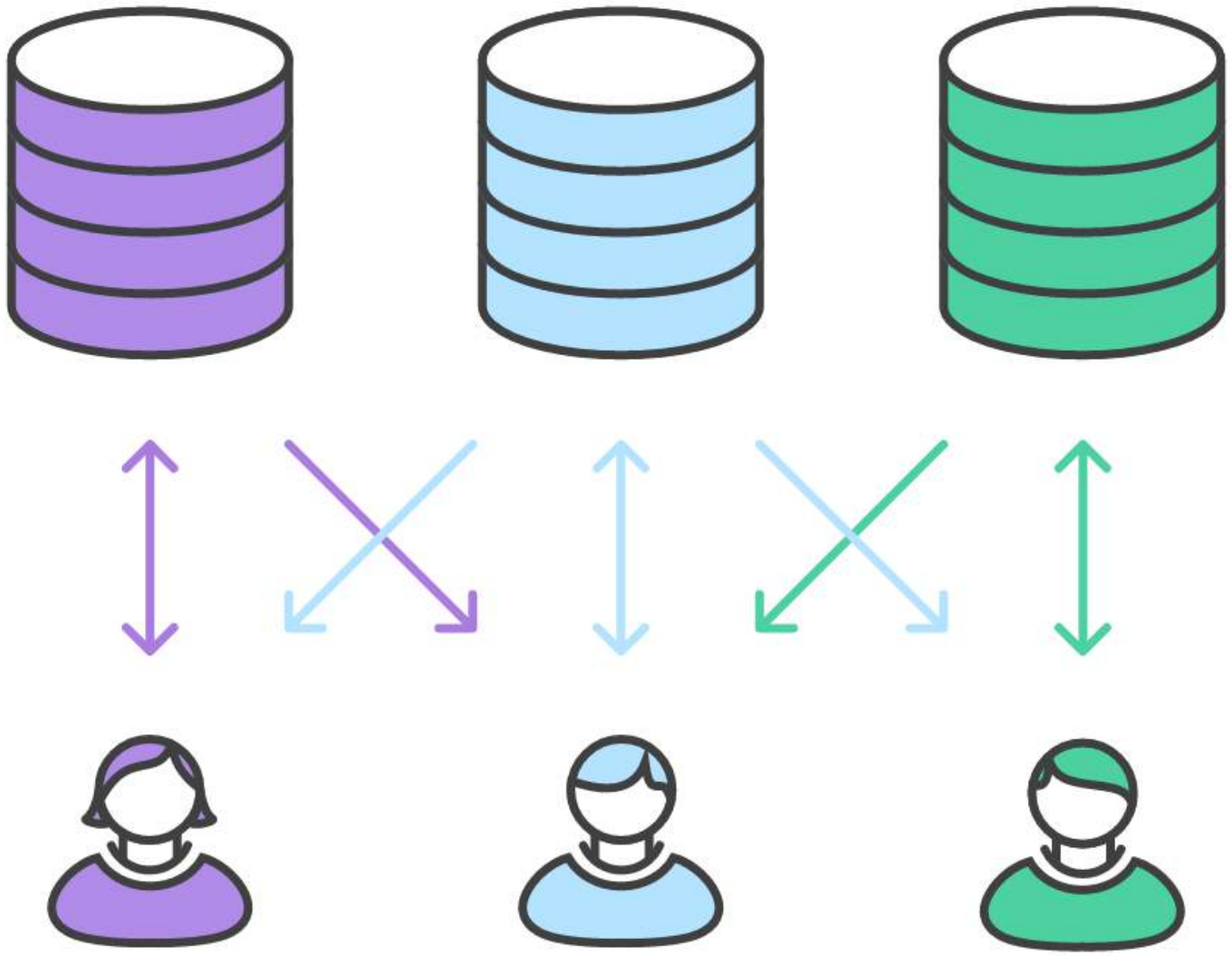


Image courtesy of the GitHub Flow reference

Section 22.5: Forking Workflow

This type of workflow is fundamentally different than the other ones mentioned on this topic. Instead of having one centralized repo that all developers have access to, each developer has his/her *own* repo that is forked from the main repo. The advantage of this is that developers can post to their own repos rather than a shared repo and a maintainer can integrate the changes from the forked repos into the original whenever appropriate.

A visual representation of this workflow is as follows:



Chapter 23: Pulling

Parameters	Details
`--quiet`	No text output
`-q`	shorthand for `--quiet`
`--verbose`	verbose text output. Passed to fetch and merge/rebase commands respectively.
`-v`	shorthand for `--verbose`
`--[no-]recurse-submodules[=yes|on-demand|no]`	Fetch new commits for submodules? (Not that this is not a pull/checkout)

Unlike pushing with Git where your local changes are sent to the central repository's server, pulling with Git takes the current code on the server and 'pulls' it down from the repository's server to your local machine. This topic explains the process of pulling code from a repository using Git as well as the situations one might encounter while pulling different code into the local copy.

Section 23.1: Pulling changes to a local repository

Simple pull

When you are working on a remote repository (say, GitHub) with someone else, you will at some point want to share your changes with them. Once they have pushed their changes to a remote repository, you can retrieve those changes by *pulling* from this repository.

```
git pull
```

Will do it, in the majority of cases.

Pull from a different remote or branch

You can pull changes from a different remote or branch by specifying their names

```
git pull origin feature-A
```

Will pull the branch `feature-A` form `origin` into your local branch. Note that you can directly supply an URL instead of a remote name, and an object name such as a commit SHA instead of a branch name.

Manual pull

To imitate the behavior of a git pull, you can use `git fetch` then `git merge`

```
git fetch origin # retrieve objects and update refs from origin
git merge origin/feature-A # actually perform the merge
```

This can give you more control, and allows you to inspect the remote branch before merging it. Indeed, after fetching, you can see the remote branches with `git branch -a`, and check them out with

```
git checkout -b local-branch-name origin/feature-A # checkout the remote branch
# inspect the branch, make commits, squash, ammend or whatever
git checkout merging-branches # moving to the destination branch
```

```
git merge local-branch-name # performing the merge
```

This can be very handy when processing pull requests.

Section 23.2: Updating with local changes

When local changes are present, the `git pull` command aborts reporting :

> error: Your local changes to the following files would be overwritten by merge

In order to update (like svn update did with subversion), you can run :

```
git stash
git pull --rebase
git stash pop
```

A convenient way could be to define an alias using :

Version < 2.9

```
git config --global alias.up '!git stash && git pull --rebase && git stash pop'
```

Version ≥ 2.9

```
git config --global alias.up 'pull --rebase --autostash'
```

Next you can simply use :

```
git up
```

Section 23.3: Pull, overwrite local

```
git fetch
git reset --hard origin/master
```

Beware: While commits discarded using `reset --hard` can be recovered using `reflog` and `reset`, uncommitted changes are deleted forever.

Change `origin` and `master` to the remote and branch you want to forcibly pull to, respectively, if they are named differently.

Section 23.4: Pull code from remote

```
git pull
```

Section 23.5: Keeping linear history when pulling

Rebasing when pulling

If you are pulling in fresh commits from the remote repository and you have local changes on the current branch then git will automatically merge the remote version and your version. If you would like to reduce the number of merges on your branch you can tell git to rebase your commits on the remote version of the branch.

```
git pull --rebase
```

Making it the default behavior

To make this the default behavior for newly created branches, type the following command:

```
git config branch.autosetuprebase always
```

To change the behavior of an existing branch, use this:

```
git config branch.BRANCH_NAME.rebase true
```

And

```
git pull --no-rebase
```

To perform a normal merging pull.

Check if fast-forwardable

To only allow fast forwarding the local branch, you can use:

```
git pull --ff-only
```

This will display an error when the local branch is not fast-forwardable, and needs to be either rebased or merged with upstream.

Section 23.6: Pull, "permission denied"

Some problems can occur if the `.git` folder has wrong permission. Fixing this problem by setting the owner of the complete `.git` folder. Sometimes it happen that another user pull and change the rights of the `.git` folder or files.

To fix the problem:

```
chown -R youruser:yourgroup .git/
```

Chapter 24: Hooks

Section 24.1: Pre-push

Available in Git 1.8.2 and above.

Version ≥ 1.8

Pre-push hooks can be used to prevent a push from going though. Reasons this is helpful include: blocking accidental manual pushes to specific branches, or blocking pushes if an established check fails (unit tests, syntax).

A pre-push hook is created by simply creating a file named `pre-push` under `.git/hooks/`, and (**gotcha alert**), making sure the file is executable: `chmod +x ./git/hooks/pre-push`.

Here's an example from Hannah Wolfe that blocks a push to master:

```
#!/bin/bash

protected_branch='master'
current_branch=$(git symbolic-ref HEAD | sed -e 's,.*/\(.*\),\1,')

if [ $protected_branch = $current_branch ]
then
    read -p "You're about to push master, is that what you intended? [y|n] " -n 1 -r < /dev/tty
    echo
    if echo $REPLY | grep -E '^[Yy]$' > /dev/null
    then
        exit 0 # push will execute
    fi
    exit 1 # push will not execute
else
    exit 0 # push will execute
fi
```

Here's an example from Volkan Unsal which makes sure RSpec tests pass before allowing the push:

```
#!/usr/bin/env ruby
require 'pty'
html_path = "rspec_results.html"
begin
  PTY.spawn( "rspec spec --format h > rspec_results.html" ) do |stdin, stdout, pid|
  begin
    stdin.each { |line| print line }
  rescue Errno::EIO
  end
end
rescue PTY::ChildExited
  puts "Child process exit!"
end

# find out if there were any errors
html = open(html_path).read
examples = html.match(/(\d+) examples/)[0].to_i rescue 0
errors = html.match(/(\d+) errors/)[0].to_i rescue 0
if errors == 0 then
  errors = html.match(/(\d+) failure/)[0].to_i rescue 0
end
pending = html.match(/(\d+) pending/)[0].to_i rescue 0
```

```
if errors.zero?
  puts "0 failed! #{examples} run, #{pending} pending"
  # HTML Output when tests ran successfully:
  # puts "View spec results at #{File.expand_path(html_path)}"
  sleep 1
  exit 0
else
  puts "\aCOMMIT FAILED!!"
  puts "View your rspec results at #{File.expand_path(html_path)}"
  puts
  puts "#{errors} failed! #{examples} run, #{pending} pending"
  # Open HTML Ooutput when tests failed
  # `open #{html_path}`
  exit 1
end
```

As you can see, there are lots of possibilities, but the core piece is to `exit 0` if good things happened, and `exit 1` if bad things happened. Anytime you `exit 1` the push will be prevented and your code will be in the state it was before running `git push`...

When using client side hooks, keep in mind that users can skip all client side hooks by using the option "--no-verify" on a push. If you're relying on the hook to enforce process, you can get burned.

Documentation: https://git-scm.com/docs/githooks#_pre_push
Official Sample:
https://github.com/git/git/blob/87c86dd14abe8db7d00b0df5661ef8cf147a72a3/templates/hooks--pre-push.sample

Section 24.2: Verify Maven build (or other build system) before committing

`.git/hooks/pre-commit`

```
#!/bin/sh
if [ -s pom.xml ]; then
    echo "Running mvn verify"
    mvn clean verify
    if [ $? -ne 0 ]; then
        echo "Maven build failed"
        exit 1
    fi
fi
```

Section 24.3: Automatically forward certain pushes to other repositories

`post-receive` hooks can be used to automatically forward incoming pushes to another repository.

```
$ cat .git/hooks/post-receive

#!/bin/bash

IFS=' '
while read local_ref local_sha remote_ref remote_sha
do

  echo "$remote_ref" | egrep '^refs\/heads\/[A-Z]+-[0-9]+$' >/dev/null && {
    ref=`echo $remote_ref | sed -e 's/^refs\/heads\///'`
```

```
    echo Forwarding feature branch to other repository: $ref
    git push -q --force other_repos $ref
  }

done
```

In this example, the `egrep` regexp looks for a specific branch format (here: JIRA-12345 as used to name Jira issues). You can leave this part off if you want to forward all branches, of course.

Section 24.4: Commit-msg

This hook is similar to the `prepare-commit-msg` hook, but it's called after the user enters a commit message rather than before. This is usually used to warn developers if their commit message is in an incorrect format.

The only argument passed to this hook is the name of the file that contains the message. If you don't like the message that the user has entered, you can either alter this file in-place (same as `prepare-commit-msg`) or you can abort the commit entirely by exiting with a non-zero status.

The following example is used to check if the word ticket followed by a number is present on the commit message

```
word="ticket [0-9]"
isPresent=$(grep -Eoh "$word" $1)

if [[ -z $isPresent ]]
  then echo "Commit message KO, $word is missing"; exit 1;
  else echo "Commit message OK"; exit 0;
fi
```

Section 24.5: Local hooks

Local hooks affect only the local repositories in which they reside. Each developer can alter their own local hooks, so they can't be used reliably as a way to enforce a commit policy. They are designed to make it easier for developers to adhere to certain guidelines and avoid potential problems down the road.

There are six types of local hooks: pre-commit, prepare-commit-msg, commit-msg, post-commit, post-checkout, and pre-rebase.

The first four hooks relate to commits and allow you to have some control over each part in a commit's life cycle. The final two let you perform some extra actions or safety checks for the git checkout and git rebase commands.

All of the "pre-" hooks let you alter the action that's about to take place, while the "post-" hooks are used primarily for notifications.

Section 24.6: Post-checkout

This hook works similarly to the `post-commit` hook, but it's called whenever you successfully check out a reference with `git checkout`. This could be a useful tool for clearing out your working directory of auto-generated files that would otherwise cause confusion.

This hook accepts three parameters:

1. the ref of the previous HEAD,
2. the ref of the new HEAD, and
3. a flag indicating if it was a branch checkout or a file checkout (`1` or `0`, respectively).

Its exit status has no affect on the `git checkout` command.

Section 24.7: Post-commit

This hook is called immediately after the `commit-msg` hook. It cannot alter the outcome of the `git commit` operation, therefore it's used primarily for notification purposes.

The script takes no parameters, and its exit status does not affect the commit in any way.

Section 24.8: Post-receive

This hook is called after a successful push operation. It is typically used for notification purposes.

The script takes no parameters, but is sent the same information as `pre-receive` via standard input:

```
<old-value> <new-value> <ref-name>
```

Section 24.9: Pre-commit

This hook is executed every time you run `git commit`, to verify what is about to be committed. You can use this hook to inspect the snapshot that is about to be committed.

This type of hook is useful for running automated tests to make sure the incoming commit doesn't break existing functionality of your project. This type of hook may also check for whitespace or EOL errors.

No arguments are passed to the pre-commit script, and exiting with a non-zero status aborts the entire commit.

Section 24.10: Prepare-commit-msg

This hook is called after the `pre-commit` hook to populate the text editor with a commit message. This is typically used to alter the automatically generated commit messages for squashed or merged commits.

One to three arguments are passed to this hook:

- The name of a temporary file that contains the message.
- The type of commit, either
  - message (`-m` or `-F` option),
  - template (`-t` option),
  - merge (if it's a merge commit), or
  - squash (if it's squashing other commits).
- The SHA1 hash of the relevant commit. This is only given if `-c`, `-C`, or `--amend` option was given.

Similar to `pre-commit`, exiting with a non-zero status aborts the commit.

Section 24.11: Pre-rebase

This hook is called before `git rebase` begins to alter code structure. This hook is typically used for making sure a rebase operation is appropriate.

This hook takes 2 parameters:

1. the upstream branch that the series was forked from, and
2. the branch being rebased (empty when rebasing the current branch).

You can abort the rebase operation by exiting with a non-zero status.

Section 24.12: Pre-receive

This hook is executed every time somebody uses `git push` to push commits to the repository. It always resides in the remote repository that is the destination of the push and not in the originating (local) repository.

The hook runs before any references are updated. It is typically used to enforce any kind of development policy.

The script takes no parameters, but each ref that is being pushed is passed to the script on a separate line on standard input in the following format:

```
<old-value> <new-value> <ref-name>
```

Section 24.13: Update

This hook is called after `pre-receive`, and it works the same way. It's called before anything is actually updated, but is called separately for each ref that was pushed rather than all of the refs at once.

This hook accepts the following 3 arguments:

- name of the ref being updated,
- old object name stored in the ref, and
- new object name stored in the ref.

This is the same information passed to `pre-receive`, but since `update` is invoked separately for each ref, you can reject some refs while allowing others.

Chapter 25: Cloning Repositories

Section 25.1: Shallow Clone

Cloning a huge repository (like a project with multiple years of history) might take a long time, or fail because of the amount of data to be transferred. In cases where you don't need to have the full history available, you can do a shallow clone:

```
git clone [repo_url] --depth 1
```

The above command will fetch just the last commit from the remote repository.

Be aware that you may not be able to resolve merges in a shallow repository. It's often a good idea to take at least as many commits are you are going to need to backtrack to resolve merges. For example, to instead get the last 50 commits:

```
git clone [repo_url] --depth 50
```

Later, if required, you can the fetch the rest of the repository:

Version ≥ 1.8.3

```
git fetch --unshallow     # equivalent of git fetch --depth=2147483647
                          # fetches the rest of the repository
```

Version < 1.8.3

```
git fetch --depth=1000    # fetch the last 1000 commits
```

Section 25.2: Regular Clone

To download the entire repository including the full history and all branches, type:

```
git clone <url>
```

The example above will place it in a directory with the same name as the repository name.

To download the repository and save it in a specific directory, type:

```
git clone <url> [directory]
```

For more details, visit Clone a repository.

Section 25.3: Clone a specific branch

To clone a specific branch of a repository, type `--branch <branch name>` before the repository url:

```
git clone --branch <branch name> <url> [directory]
```

To use the shorthand option for `--branch`, type `-b`. This command downloads entire repository and checks out `<branch name>`.

To save disk space you can clone history leading only to single branch with:

```
git clone --branch <branch_name> --single-branch <url> [directory]
```

If `--single-branch` is not added to the command, history of all branches will be cloned into `[directory]`. This can be issue with big repositories.

To later undo `--single-branch` flag and fetch the rest of repository use command:

```
git config remote.origin.fetch "+refs/heads/*:refs/remotes/origin/*"
git fetch origin
```

Section 25.4: Clone recursively

Version ≥ 1.6.5

```
git clone <url> --recursive
```

Clones the repository and also clones all submodules. If the submodules themselves contain additional submodules, Git will also clone those.

Section 25.5: Clone using a proxy

If you need to download files with git under a proxy, setting proxy server system-wide couldn't be enough. You could also try the following:

```
git config --global http.proxy http://<proxy-server>:<port>/
```

Chapter 26: Stashing

Parameter	Details
show	Show the changes recorded in the stash as a diff between the stashed state and its original parent. When no <stash> is given, shows the latest one.
list	List the stashes that you currently have. Each stash is listed with its name (e.g. stash@{0} is the latest stash, stash@{1} is the one before, etc.), the name of the branch that was current when the stash was made, and a short description of the commit the stash was based on.
pop	Remove a single stashed state from the stash list and apply it on top of the current working tree state.
apply	Like `pop`, but do not remove the state from the stash list.
clear	Remove all the stashed states. Note that those states will then be subject to pruning, and may be impossible to recover.
drop	Remove a single stashed state from the stash list. When no <stash> is given, it removes the latest one. i.e. stash@{0}, otherwise <stash> must be a valid stash log reference of the form stash@{<revision>}.
create	Create a stash (which is a regular commit object) and return its object name, without storing it anywhere in the ref namespace. This is intended to be useful for scripts. It is probably not the command you want to use; see "save" above.
store	Store a given stash created via git stash create (which is a dangling merge commit) in the stash ref, updating the stash reflog. This is intended to be useful for scripts. It is probably not the command you want to use; see "save" above.

Section 26.1: What is Stashing?

When working on a project, you might be half-way through a feature branch change when a bug is raised against master. You're not ready to commit your code, but you also don't want to lose your changes. This is where `git stash` comes in handy.

Run `git status` on a branch to show your uncommitted changes:

```
(master) $ git status
On branch master
Your branch is up-to-date with 'origin/master'.
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git checkout -- <file>..." to discard changes in working directory)

    modified:   business/com/test/core/actions/Photo.c

no changes added to commit (use "git add" and/or "git commit -a")
```

Then run `git stash` to save these changes to a stack:

```
(master) $ git stash
Saved working directory and index state WIP on master:
2f2a6e1 Merge pull request #1 from test/test-branch
HEAD is now at 2f2a6e1 Merge pull request #1 from test/test-branch
```

If you have added files to your working directory these can be stashed as well. You just need to stage them first.

```
(master) $ git stash
Saved working directory and index state WIP on master:
(master) $ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
```

```
        NewPhoto.c

nothing added to commit but untracked files present (use "git add" to track)
(master) $ git stage NewPhoto.c
(master) $ git stash
Saved working directory and index state WIP on master:
(master) $ git status
On branch master
nothing to commit, working tree clean
(master) $
```

Your working directory is now clean of any changes you made. You can see this by re-running `git status`:

```
(master) $ git status
On branch master
Your branch is up-to-date with 'origin/master'.
nothing to commit, working directory clean
```

To apply the very last stash, run `git stash apply` (additionally, you can apply *and* remove the last stashed changed with `git stash pop`):

```
(master) $ git stash apply
On branch master
Your branch is up-to-date with 'origin/master'.
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git checkout -- <file>..." to discard changes in working directory)

    modified:   business/com/test/core/actions/Photo.c

no changes added to commit (use "git add" and/or "git commit -a")
```

Note, however, that stashing does not remember the branch you were working on. In the above examples, the user was stashing on **master**. If they switch to the **dev** branch, **dev**, and run `git stash apply` the last stash is put on the **dev** branch.

```
(master) $ git checkout -b dev
Switched to a new branch 'dev'
(dev) $ git stash apply
On branch dev
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git checkout -- <file>..." to discard changes in working directory)

    modified:   business/com/test/core/actions/Photo.c

no changes added to commit (use "git add" and/or "git commit -a")
```

Section 26.2: Create stash

Save the current state of working directory and the index (also known as the staging area) in a stack of stashes.

```
git stash
```

To include all untracked files in the stash use the `--include-untracked` or `-u` flags.

```
git stash --include-untracked
```

To include a message with your stash to make it more easily identifiable later

```
git stash save "<whatever message>"
```

To leave the staging area in current state after stash use the `--keep-index` or `-k` flags.

```
git stash --keep-index
```

Section 26.3: Apply and remove stash

To apply the last stash and remove it from the stack - type:

```
git stash pop
```

To apply specific stash and remove it from the stack - type:

```
git stash pop stash@{n}
```

Section 26.4: Apply stash without removing it

Applies the last stash without removing it from the stack

```
git stash apply
```

Or a specific stash

```
git stash apply stash@{n}
```

Section 26.5: Show stash

Shows the changes saved in the last stash

```
git stash show
```

Or a specific stash

```
git stash show stash@{n}
```

To show content of the changes saved for the specific stash

```
git stash show -p stash@{n}
```

Section 26.6: Partial stash

If you would like to stash only *some* diffs in your working set, you can use a partial stash.

```
git stash -p
```

And then interactively select which hunks to stash.

As of version 2.13.0 you can also avoid the interactive mode and create a partial stash with a pathspec using the new **push** keyword.

```
git stash push -m "My partial stash" -- app.config
```

Section 26.7: List saved stashes

```
git stash list
```

This will list all stashes in the stack in reverse chronological order.
You will get a list that looks something like this:

```
stash@{0}: WIP on master: 67a4e01 Merge tests into develop
stash@{1}: WIP on master: 70f0d95 Add user role to localStorage on user login
```

You can refer to specific stash by its name, for example `stash@{1}`.

Section 26.8: Move your work in progress to another branch

If while working you realize you're on wrong branch and you haven't created any commits yet, you can easily move your work to correct branch using stashing:

```
git stash
git checkout correct-branch
git stash pop
```

Remember `git stash pop` will apply the last stash and delete it from the stash list. To keep the stash in the list and only apply to some branch you can use:

```
git stash apply
```

Section 26.9: Remove stash

Remove all stash

```
git stash clear
```

Removes the last stash

```
git stash drop
```

Or a specific stash

```
git stash drop stash@{n}
```

Section 26.10: Apply part of a stash with checkout

You've made a stash and wish to checkout only some of the files in that stash.

```
git checkout stash@{0} -- myfile.txt
```

Section 26.11: Recovering earlier changes from stash

To get your most recent stash after running git stash, use

```
git stash apply
```

To see a list of your stashes, use

```
git stash list
```

You will get a list that looks something like this

```
stash@{0}: WIP on master: 67a4e01 Merge tests into develop
stash@{1}: WIP on master: 70f0d95 Add user role to localStorage on user login
```

Choose a different git stash to restore with the number that shows up for the stash you want

```
git stash apply stash@{2}
```

Section 26.12: Interactive Stashing

Stashing takes the dirty state of your working directory – that is, your modified tracked files and staged changes – and saves it on a stack of unfinished changes that you can reapply at any time.

Stashing only modified files:

Suppose you don't want to stash the staged files and only stash the modified files so you can use:

```
git stash --keep-index
```

Which will stash only the modified files.

Stashing untracked files:

Stash never saves the untracked files it only stashes the modified and staged files. So suppose if you need to stash the untracked files too then you can use this:

```
git stash -u
```

this will track the untracked, staged and modified files.

Stash some particular changes only:

Suppose you need to stash only some part of code from the file or only some files only from all the modified and stashed files then you can do it like this:

```
git stash --patch
```

Git will not stash everything that is modified but will instead prompt you interactively which of the changes you would like to stash and which you would like to keep in your working directory.

Section 26.13: Recover a dropped stash

If you have only just popped it and the terminal is still open, you will still have the hash value printed by `git stash pop` on screen:

```
$ git stash pop
```

```
[...]
Dropped refs/stash@{0} (2ca03e22256be97f9e40f08e6d6773c7d41dbfd1)
```

(Note that git stash drop also produces the same line.)

Otherwise, you can find it using this:

```
git fsck --no-reflog | awk '/dangling commit/ {print $3}'
```

This will show you all the commits at the tips of your commit graph which are no longer referenced from any branch or tag – every lost commit, including every stash commit you've ever created, will be somewhere in that graph.

The easiest way to find the stash commit you want is probably to pass that list to `gitk`:

```
gitk --all $( git fsck --no-reflog | awk '/dangling commit/ {print $3}' )
```

This will launch a repository browser showing you *every single commit in the repository ever*, regardless of whether it is reachable or not.

You can replace `gitk` there with something like `git log --graph --oneline --decorate` if you prefer a nice graph on the console over a separate GUI app.

To spot stash commits, look for commit messages of this form:

> WIP on *somebranch*: *commithash Some old commit message*

Once you know the hash of the commit you want, you can apply it as a stash:

`git stash apply` sh_hash

Or you can use the context menu in `gitk` to create branches for any unreachable commits you are interested in. After that, you can do whatever you want with them with all the normal tools. When you're done, just blow those branches away again.

Chapter 27: Subtrees

Section 27.1: Create, Pull, and Backport Subtree

Create Subtree

Add a new remote called `plugin` pointing to the plugin's repository:

```
git remote add plugin https://path.to/remotes/plugin.git
```

Then Create a subtree specifying the new folder prefix `plugins/demo`. `plugin` is the remote name, and `master` refers to the master branch on the subtree's repository:

```
git subtree add --prefix=plugins/demo plugin master
```

Pull Subtree Updates

Pull normal commits made in plugin:

```
git subtree pull --prefix=plugins/demo plugin master
```

Backport Subtree Updates

1. Specify commits made in superproject to be backported:

   ```
   git commit -am "new changes to be backported"
   ```

2. Checkout new branch for merging, set to track subtree repository:

   ```
   git checkout -b backport plugin/master
   ```

3. Cherry-pick backports:

   ```
   git cherry-pick -x --strategy=subtree master
   ```

4. Push changes back to plugin source:

   ```
   git push plugin backport:master
   ```

Chapter 28: Renaming

Parameter	Details
`-f` or `--force`	Force renaming or moving of a file even if the target exists

Section 28.1: Rename Folders

To rename a folder from `oldName` to `newName`

```
git mv directoryToFolder/oldName directoryToFolder/newName
```

Followed by `git commit` and/or `git push`

If this error occurs:

> fatal: renaming 'directoryToFolder/oldName' failed: Invalid argument

Use the following command:

```
git mv directoryToFolder/oldName temp && git mv temp directoryToFolder/newName
```

Section 28.2: rename a local and the remote branch

the easiest way is to have the local branch checked out:

```
git checkout old_branch
```

then rename the local branch, delete the old remote and set the new renamed branch as upstream:

```
git branch -m new_branch
git push origin :old_branch
git push --set-upstream origin new_branch
```

Section 28.3: Renaming a local branch

You can rename branch in local repository using this command:

```
git branch -m old_name new_name
```

Chapter 29: Pushing

Parameter	Details
--force	Overwrites the remote ref to match your local ref. *Can cause the remote repository to lose commits, so use with care*.
--verbose	Run verbosely.
<remote>	The remote repository that is destination of the push operation.
<refspec>...	Specify what remote ref to update with what local ref or object.

After changing, staging, and committing code with Git, pushing is required to make your changes available to others and transfers your local changes to the repository server. This topic will cover how to properly push code using Git.

Section 29.1: Push a specific object to a remote branch

General syntax

```
git push <remotename> <object>:<remotebranchname>
```

Example

```
git push origin master:wip-yourname
```

Will push your master branch to the `wip-yourname` branch of origin (most of the time, the repository you cloned from).

Delete remote branch

Deleting the remote branch is the equivalent of pushing an empty object to it.

```
git push <remotename> :<remotebranchname>
```

Example

```
git push origin :wip-yourname
```

Will delete the remote branch `wip-yourname`

Instead of using the colon, you can also use the --delete flag, which is better readable in some cases.

Example

```
git push origin --delete wip-yourname
```

Push a single commit

If you have a single commit in your branch that you want to push to a remote without pushing anything else, you can use the following

```
git push <remotename> <commit SHA>:<remotebranchname>
```

Example

Assuming a git history like this

```
eeb32bc Commit 1 - already pushed
347d700 Commit 2 - want to push
e539af8 Commit 3 - only local
5d339db Commit 4 - only local
```

to push only commit *347d700* to remote *master* use the following command

```
git push origin 347d700:master
```

Section 29.2: Push

```
git push
```

will push your code to your existing upstream. Depending on the push configuration, it will either push code from you current branch (default in Git 2.x) or from all branches (default in Git 1.x).

Specify remote repository

When working with git, it can be handy to have multiple remote repositories. To specify a remote repository to push to, just append its name to the command.

```
git push origin
```

Specify Branch

To push to a specific branch, say `feature_x`:

```
git push origin feature_x
```

Set the remote tracking branch

Unless the branch you are working on originally comes from a remote repository, simply using `git push` won't work the first time. You must perform the following command to tell git to push the current branch to a specific remote/branch combination

```
git push --set-upstream origin master
```

Here, `master` is the branch name on the remote `origin`. You can use `-u` as a shorthand for `--set-upstream`.

Pushing to a new repository

To push to a repository that you haven't made yet, or is empty:

1. Create the repository on GitHub (if applicable)
2. Copy the url given to you, in the form `https://github.com/USERNAME/REPO_NAME.git`
3. Go to your local repository, and execute `git remote add origin URL`
    - To verify it was added, run `git remote -v`
4. Run `git push origin master`

Your code should now be on GitHub

For more information view Adding a remote repository

Explanation

Push code means that git will analyze the differences of your local commits and remote and send them to be written on the upstream. When push succeeds, your local repository and remote repository are synchronized and other users can see your commits.

For more details on the concepts of "upstream" and "downstream", see Remarks.

Section 29.3: Force Pushing

Sometimes, when you have local changes incompatible with remote changes (ie, when you cannot fast-forward the remote branch, or the remote branch is not a direct ancestor of your local branch), the only way to push your changes is a force push.

```
git push -f
```

or

```
git push --force
```

Important notes

This will **overwrite** any remote changes and your remote will match your local.

Attention: Using this command may cause the remote repository to **lose commits**. Moreover, it is strongly advised against doing a force push if you are sharing this remote repository with others, since their history will retain every overwritten commit, thus rending their work out of sync with the remote repository.

As a rule of thumb, only force push when:

- Nobody except you pulled the changes you are trying to overwrite
- You can force everyone to clone a fresh copy after the forced push and make everyone apply their changes to it (people may hate you for this).

Section 29.4: Push tags

```
git push --tags
```

Pushes all of the `git tags` in the local repository that are not in the remote one.

Section 29.5: Changing the default push behavior

Current updates the branch on the remote repository that shares a name with the current working branch.

```
git config push.default current
```

Simple pushes to the upstream branch, but will not work if the upstream branch is called something else.

```
git config push.default simple
```

Upstream pushes to the upstream branch, no matter what it is called.

```
git config push.default upstream
```

Matching pushes all branches that match on the local and the remote git config push.default upstream

After you've set the preferred style, use

```
git push
```

to update the remote repository.

Chapter 30: Internals

Section 30.1: Repo

A `git repository` is an on-disk data structure which stores metadata for a set of files and directories.

It lives in your project's `.git/` folder. Every time you commit data to git, it gets stored here. Inversely, `.git/` contains every single commit.

It's basic structure is like this:

```
.git/
    objects/
    refs/
```

Section 30.2: Objects

`git` is fundamentally a key-value store. When you add data to `git`, it builds an `object` and uses the SHA-1 hash of the `object`'s contents as a key.

Therefore, any content in `git` can be looked up by it's hash:

```
git cat-file -p 4bb6f98
```

There are 4 types of `Object`:

- `blob`
- `tree`
- `commit`
- `tag`

Section 30.3: HEAD ref

`HEAD` is a special `ref`. It always points to the current object.

You can see where it's currently pointing by checking the `.git/HEAD` file.

Normally, `HEAD` points to another `ref`:

```
$cat .git/HEAD
ref: refs/heads/mainline
```

But it can also point directly to an `object`:

```
$ cat .git/HEAD
4bb6f98a223abc9345a0cef9200562333
```

This is what's known as a "detached head" - because `HEAD` is not attached to (pointing at) any `ref`, but rather points directly to an `object`.

Section 30.4: Refs

A `ref` is essentially a pointer. It's a name that points to an `object`. For example,

```
"master" --> 1a410e...
```

They are stored in `.git/refs/heads/ in plain text files.

```
$ cat .git/refs/heads/mainline
4bb6f98a223abc9345a0cef9200562333
```

This is commonly what are called `branches`. However, you'll note that in `git` there is no such thing as a `branch` - only a `ref`.

Now, it's possible to navigate `git` purely by jumping around to different `objects` directly by their hashes. But this would be terribly inconvenient. A `ref` gives you a convenient name to refer to `objects` by. It's much easier to ask `git` to go to a specific place by name rather than by hash.

Section 30.5: Commit Object

A `commit` is probably the `object` type most familiar to `git` users, as it's what they are used to creating with the `git commit` commands.

However, the `commit` does not directly contain any changed files or data. Rather, it contains mostly metadata and pointers to other `objects` which contain the actual contents of the `commit`.

A `commit` contains a few things:

- hash of a `tree`
- hash of a parent `commit`
- author name/email, commiter name/email
- commit message

You can see the contents of any commit like this:

```
$ git cat-file commit 5bac93
tree 04d1daef...
parent b7850ef5...
author Geddy Lee <glee@rush.com>
commiter Neil Peart <npeart@rush.com>

First commit!
```

Tree

A very important note is that the `tree` objects stores EVERY file in your project, and it stores whole files not diffs. This means that each `commit` contains a snapshot of the entire project*.

**Technically, only changed files are stored. But this is more an implementation detail for efficiency. From a design perspective, a `commit` should be considered as containing a complete copy of the project.*

Parent

The `parent` line contains a hash of another `commit` object, and can be thought of as a "parent pointer" that points to the "previous commit". This implicitly forms a graph of commits known as the **commit graph**. Specifically, it's a directed acyclic graph (or DAG).

Section 30.6: Tree Object

A `tree` basically represents a folder in a traditional filesystem: nested containers for files or other folders.

A `tree` contains:

- 0 or more `blob` objects
- 0 or more `tree` objects

Just as you can use `ls` or `dir` to list the contents of a folder, you can list the contents of a `tree` object.

```
$ git cat-file -p 07b1a631
100644 blob b91bba1b   .gitignore
100644 blob cc0956f1   Makefile
040000 tree 92e1ca7e   src
...
```

You can look up the files in a `commit` by first finding the hash of the `tree` in the `commit`, and then looking at that `tree`:

```
$ git cat-file commit 4bb6f93a
tree 07b1a631
parent ...
author ...
commiter ...

$ git cat-file -p 07b1a631
100644 blob b91bba1b   .gitignore
100644 blob cc0956f1   Makefile
040000 tree 92e1ca7e   src
...
```

Section 30.7: Blob Object

A `blob` contains arbitrary binary file contents. Commonly, it will be raw text such as source code or a blog article. But it could just as easily be the bytes of a PNG file or anything else.

If you have the hash of a `blob`, you can look at it's contents.

```
$ git cat-file -p d429810
package com.example.project

class Foo {
 ...
}
...
```

For example, you can browse a `tree` as above, and then look at one of the `blobs` in it.

```
$ git cat-file -p 07b1a631
100644 blob b91bba1b   .gitignore
100644 blob cc0956f1   Makefile
040000 tree 92e1ca7e   src
100644 blob cae391ff   Readme.txt

$ git cat-file -p cae391ff
Welcome to my project! This is the readmefile
```

```
...
```

Section 30.8: Creating new Commits

The `git commit` command does a few things:

1. Create `blobs` and `trees` to represent your project directory - stored in `.git/objects`
2. Creates a new `commit` object with your author information, commit message, and the root `tree` from step 1 - also stored in `.git/objects`
3. Updates the `HEAD` ref in `.git/HEAD` to the hash of the newly-created `commit`

This results in a new snapshot of your project being added to `git` that is connected to the previous state.

Section 30.9: Moving HEAD

When you run `git checkout` on a commit (specified by hash or ref) you're telling `git` to make your working directory look like how it did when the snapshot was taken.

1. Update the files in the working directory to match the `tree` inside the `commit`
2. Update `HEAD` to point to the specified hash or ref

Section 30.10: Moving refs around

Running `git reset --hard` moves refs to the specified hash/ref.

Moving `MyBranch` to `b8dc53`:

```
$ git checkout MyBranch      # moves HEAD to MyBranch
$ git reset --hard b8dc53    # makes MyBranch point to b8dc53
```

Section 30.11: Creating new Refs

Running `git checkout -b <refname>` will create a new ref that points to the current `commit`.

```
$ cat .git/head
1f324a

$ git checkout -b TestBranch

$ cat .git/refs/heads/TestBranch
1f324a
```

Chapter 31: git-tfs

Section 31.1: git-tfs clone

This will create a folder with the same name as the project, i.e. /My.Project.Name

```
$ git tfs clone http://tfs:8080/tfs/DefaultCollection/ $/My.Project.Name
```

Section 31.2: git-tfs clone from bare git repository

Cloning from a git repository is ten times faster than cloning directly from TFVS and works well in a team environment. At least one team member will have to create the bare git repository by doing the regular git-tfs clone first. Then the new repository can be bootstrapped to work with TFVS.

```
$ git clone x:/fileshare/git/My.Project.Name.git
$ cd My.Project.Name
$ git tfs bootstrap
$ git tfs pull
```

Section 31.3: git-tfs install via Chocolatey

The following assumes you will use kdiff3 for file diffing and although not essential it is a good idea.

```
C:\> choco install kdiff3
```

Git can be installed first so you can state any parameters you wish. Here all the Unix tools are also installed and 'NoAutoCrlf' means checkout as is, commit as is.

```
C:\> choco install git -params '"/GitAndUnixToolsOnPath /NoAutoCrlf"'
```

This is all you really need to be able to install git-tfs via chocolatey.

```
C:\> choco install git-tfs
```

Section 31.4: git-tfs Check In

Launch the Check In dialog for TFVS.

```
$ git tfs checkintool
```

This will take all of your local commits and create a single check-in.

Section 31.5: git-tfs push

Push all local commits to the TFVS remote.

```
$ git tfs rcheckin
```

Note: this will fail if Check-in Notes are required. These can be bypassed by adding `git-tfs-force: rcheckin` to the commit message.

Chapter 32: Empty directories in Git

Section 32.1: Git doesn't track directories

Assume you've initialized a project with the following directory structure:

```
/build
app.js
```

Then you add everything so you've created so far and commit:

```
git init
git add .
git commit -m "Initial commit"
```

Git will only track the file app.js.

Assume you added a build step to your application and rely on the "build" directory to be there as the output directory (and you don't want to make it a setup instruction every developer has to follow), a *convention* is to include a ".gitkeep" file inside the directory and let Git track that file.

```
/build
  .gitkeep
app.js
```

Then add this new file:

```
git add build/.gitkeep
git commit -m "Keep the build directory around"
```

Git will now track the file build/.gitkeep file and therefore the build folder will be made available on checkout.

Again, this is just a convention and not a Git feature.

Chapter 33: git-svn

Section 33.1: Cloning the SVN repository

You need to create a new local copy of the repository with the command

```
git svn clone SVN_REPO_ROOT_URL [DEST_FOLDER_PATH] -T TRUNK_REPO_PATH -t TAGS_REPO_PATH -b
BRANCHES_REPO_PATH
```

If your SVN repository follows the standard layout (trunk, branches, tags folders) you can save some typing:

```
git svn clone -s SVN_REPO_ROOT_URL [DEST_FOLDER_PATH]
```

`git svn clone` checks out each SVN revision, one by one, and makes a git commit in your local repository in order to recreate the history. If the SVN repository has a lot of commits this will take a while.

When the command is finished you will have a full fledged git repository with a local branch called master that tracks the trunk branch in the SVN repository.

Section 33.2: Pushing local changes to SVN

The command

```
git svn dcommit
```

will create a SVN revision for each of your local git commits. As with SVN, your local git history must be in sync with the latest changes in the SVN repository, so if the command fails, try performing a `git svn rebase` first.

Section 33.3: Working locally

Just use your local git repository as a normal git repo, with the normal git commands:

- `git add FILE` and `git checkout -- FILE` To stage/unstage a file
- `git commit` To save your changes. Those commits will be local and will not be "pushed" to the SVN repo, just like in a normal git repository
- `git stash` and `git stash pop` Allows using stashes
- `git reset HEAD --hard` Revert all your local changes
- `git log` Access all the history in the repository
- `git rebase -i` so you can rewrite your local history freely
- `git branch` and `git checkout` to create local branches

As the git-svn documentation states "Subversion is a system that is far less sophisticated than Git" so you can't use all the full power of git without messing up the history in the Subversion server. Fortunately the rules are very simple: **Keep the history linear**

This means you can make almost any git operation: creating branches, removing/reordering/squashing commits, move the history around, delete commits, etc. Anything *but merges*. If you need to reintegrate the history of local branches use `git rebase` instead.

When you perform a merge, a merge commit is created. The particular thing about merge commits is that they have two parents, and that makes the history non-linear. Non-linear history will confuse SVN in the case you "push" a merge commit to the repository.

However do not worry: **you won't break anything if you "push" a git merge commit to SVN**. If you do so, when

the git merge commit is sent to the svn server it will contain all the changes of all commits for that merge, so you will lose the history of those commits, but not the changes in your code.

Section 33.4: Getting the latest changes from SVN

The equivalent to `git pull` is the command

```
git svn rebase
```

This retrieves all the changes from the SVN repository and applies them *on top* of your local commits in your current branch.

You can also use the command

```
git svn fetch
```

to retrieve the changes from the SVN repository and bring them to your local machine but without applying them to your local branch.

Section 33.5: Handling empty folders

git does not recognice the concept of folders, it just works with files and their filepaths. This means git does not track empty folders. SVN, however, does. Using git-svn means that, by default, *any change you do involving empty folders with git will not be propagated to SVN*.

Using the `--rmdir` flag when issuing a comment corrects this issue, and removes an empty folder in SVN if you locally delete the last file inside it:

```
git svn dcommit --rmdir
```

Unfortunately **it does not removes existing empty folders**: you need to do it manually.

To avoid adding the flag each time you do a dcommit, or to play it safe if you are using a git GUI tool (like SourceTree) you can set this behaviour as default with the command:

```
git config --global svn.rmdir true
```

This changes your .gitconfig file and adds these lines:

```
[svn]
rmdir = true
```

To remove all untracked files and folders that should be kept empty for SVN use the git command:

```
git clean -fd
```

Please note: the previous command will remove all untracked files and empty folders, even the ones that should be tracked by SVN! If you need to generate againg the empty folders tracked by SVN use the command

```
git svn mkdirs
```

In practices this means that if you want to cleanup your workspace from untracked files and folders you should always use both commands to recreate the empty folders tracked by SVN:

```
git clean -fd && git svn mkdirs
```

Chapter 34: Archive

Parameter	Details
--format=<fmt>	Format of the resulting archive: `tar` or `zip`. If this options is not given and the output file is specified, the format is inferred from the filename if possible. Otherwise, defaults to `tar`.
-l, --list	Show all available formats.
-v, --verbose	Report progress to stderr.
--prefix=<prefix>/	Prepend <prefix>/ to each filename in the archive.
-o <file>, --output=<file>	Write the archive to <file> instead of stdout.
--worktree-attributes	Look for attributes in `.gitattributes` files in the working tree.
<extra>	This can be any options that the archiver backend understands. For `zip` backend, using `-0` will store the files without deflating them, while `-1` through `-9` can be used to adjust compression speed and ratio.
--remote=<repo>	Retrieve a tar archive from a remote repository **`<repo>`** rather than the local repository.
--exec=<git-upload-archive>	Used with `--remote` to specify the path to the **`<git-upload-archive>`** on the remote.
<tree-ish>	The tree or commit to produce an archive for.
<path>	Without an optional parameter, all files and directories in the current working directory are included in the archive. If one or more paths are specified, only these are included.

Section 34.1: Create an archive of git repository

With `git archive` it is possible to create compressed archives of a repository, for example for distributing releases.

Create a tar archive of current `HEAD` revision:

```
git archive --format tar HEAD | cat > archive-HEAD.tar
```

Create a tar archive of current `HEAD` revision with gzip compression:

```
git archive --format tar HEAD | gzip > archive-HEAD.tar.gz
```

This can also be done with (which will use the in-built tar.gz handling):

```
git archive --format tar.gz HEAD > archive-HEAD.tar.gz
```

Create a zip archive of current `HEAD` revision:

```
git archive --format zip HEAD > archive-HEAD.zip
```

Alternatively it is possible to just specify an output file with valid extension and the format and compression type will be inferred from it:

```
git archive --output=archive-HEAD.tar.gz HEAD
```

Section 34.2: Create an archive of git repository with directory prefix

It is considered good practice to use a prefix when creating git archives, so that extraction will place all files inside a

directory. To create an archive of `HEAD` with a directory prefix:

```
git archive --output=archive-HEAD.zip --prefix=src-directory-name HEAD
```

When extracted all the files will be extracted inside a directory named `src-directory-name` in the current directory.

Section 34.3: Create archive of git repository based on specific branch, revision, tag or directory

It is also possible to create archives of other items than `HEAD`, such as branches, commits, tags, and directories.

To create an archive of a local branch `dev`:

```
git archive --output=archive-dev.zip --prefix=src-directory-name dev
```

To create an archive of a remote branch `origin/dev`:

```
git archive --output=archive-dev.zip --prefix=src-directory-name origin/dev
```

To create an archive of a tag `v.01`:

```
git archive --output=archive-v.01.zip --prefix=src-directory-name v.01
```

Create an archive of files inside a specific sub directory (`sub-dir`) of revision `HEAD`:

```
git archive zip --output=archive-sub-dir.zip --prefix=src-directory-name HEAD:sub-dir/
```

Chapter 35: Rewriting history with filter-branch

Section 35.1: Changing the author of commits

You can use an environment filter to change the author of commits. Just modify and export `$GIT_AUTHOR_NAME` in the script to change who authored the commit.

Create a file `filter.sh` with contents like so:

```
if [ "$GIT_AUTHOR_NAME" = "Author to Change From" ]
then
    export GIT_AUTHOR_NAME="Author to Change To"
    export GIT_AUTHOR_EMAIL="email.to.change.to@example.com"
fi
```

Then run `filter-branch` from the command line:

```
chmod +x ./filter.sh
git filter-branch --env-filter ./filter.sh
```

Section 35.2: Setting git committer equal to commit author

This command, given a commit range `commit1..commit2`, rewrites history so that git commit author becomes also git committer:

```
git filter-branch -f --commit-filter \
   'export GIT_COMMITTER_NAME=\"$GIT_AUTHOR_NAME\";
    export GIT_COMMITTER_EMAIL=\"$GIT_AUTHOR_EMAIL\";
    export GIT_COMMITTER_DATE=\"$GIT_AUTHOR_DATE\";
    git commit-tree $@' \
    -- commit1..commit2
```

Chapter 36: Migrating to Git

Section 36.1: SubGit

SubGit may be used to perform a one-time import of an SVN repository to git.

```
$ subgit import --non-interactive --svn-url http://svn.my.co/repos/myproject myproject.git
```

Section 36.2: Migrate from SVN to Git using Atlassian conversion utility

Download the Atlassian conversion utility here. This utility requires Java, so please ensure that you have the Java Runtime Environment JRE installed on the machine you plan to do the conversion.

Use the command `java -jar svn-migration-scripts.jar verify` to check if your machine is missing any of the programs necessary to complete the conversion. Specifically, this command checks for the Git, subversion, and `git-svn` utilities. It also verifies that you are performing the migration on a case-sensitive file system. Migration to Git should be done on a case-sensitive file system to avoid corrupting the repository.

Next, you need to generate an authors file. Subversion tracks changes by the committer's username only. Git, however, uses two pieces of information to distinguish a user: a real name and an email address. The following command will generate a text file mapping the subversion usernames to their Git equivalents:

```
java -jar svn-migration-scripts.jar authors <svn-repo> authors.txt
```

where `<svn-repo>` is the URL of the subversion repository you wish to convert. After running this command, the contributors' identification information will be mapped in `authors.txt`. The email addresses will be of the form `<username>@mycompany.com`. In the authors file, you will need to manually change each person's default name (which by default has become their username) to their actual names. Make sure to also check all of the email addresses for correctness before proceeding.

The following command will clone an svn repo as a Git one:

```
git svn clone --stdlayout --authors-file=authors.txt <svn-repo> <git-repo-name>
```

where `<svn-repo>` is the same repository URL used above and `<git-repo-name>` is the folder name in the current directory to clone the repository into. There are a few considerations before using this command:

- The `--stdlayout` flag from above tells Git that you're using a standard layout with `trunk`, `branches`, and `tags` folders. Subversion repositories with non-standard layouts require you to specify the locations of the `trunk` folder, any/all `branch` folders, and the `tags` folder. This can be done by following this example: `git svn clone --trunk=/trunk --branches=/branches --branches=/bugfixes --tags=/tags --authors-file=authors.txt <svn-repo> <git-repo-name>`.
- This command could take many hours to complete depending on the size of your repo.
- To cut down the conversion time for large repositories, the conversion can be run directly on the server hosting the subversion repository in order to eliminate network overhead.

`git svn clone` imports the subversion branches (and trunk) as remote branches including subversion tags (remote branches prefixed with `tags/`). To convert these to actual branches and tags, run the following commands on a Linux machine in the order they are provided. After running them, `git branch -a` should show the correct branch names, and `git tag -l` should show the repository tags.

```
git for-each-ref refs/remotes/origin/tags | cut -d / -f 5- | grep -v @ | while read tagname; do git
tag $tagname origin/tags/$tagname; git branch -r -d origin/tags/$tagname; done
git for-each-ref refs/remotes | cut -d / -f 4- | grep -v @ | while read branchname; do git branch
"$branchname" "refs/remotes/origin/$branchname"; git branch -r -d "origin/$branchname"; done
```

The conversion from svn to Git is now complete! Simply `push` your local repo to a server and you can continue to contribute using Git as well as having a completely preserved version history from svn.

Section 36.3: Migrating Mercurial to Git

One can use the following methods in order to import a `Mercurial` Repo into `Git`:

1. Using fast export:

```
cd
git clone git://repo.or.cz/fast-export.git
git init git_repo
cd git_repo
~/fast-export/hg-fast-export.sh -r /path/to/old/mercurial_repo
git checkout HEAD
```

2. Using Hg-Git: A very detailed answer here: https://stackoverflow.com/a/31827990/5283213
3. Using GitHub's Importer: Follow the (detailed) instructions at GitHub.

Section 36.4: Migrate from Team Foundation Version Control (TFVC) to Git

You could migrate from team foundation version control to git by using an open source tool called Git-TF. Migration will also transfer your existing history by converting tfs checkins to git commits.

To put your solution into Git by using Git-TF follow these steps:

Download Git-TF

You can download (and install) Git-TF from Codeplex: Git-TF @ Codeplex

Clone your TFVC solution

Launch powershell (win) and type the command

```
git-tf clone http://my.tfs.server.address:port/tfs/mycollection '$/myproject/mybranch/mysolution' -
-deep
```

The --deep switch is the keeyword to note as this tells Git-Tf to copy your checkin-history. You now have a local git repository in the folder from which you called your cloe command from.

Cleanup

- Add a .gitignore file. If you are using Visual Studio the editor can do this for you, otherwise you could do this manually by downloading a complete file from github/gitignore.
- RemoveTFS source control bindings from solution (remove all *.vssscc files). You could also modify your solution file by removing the GlobalSection(TeamFoundationVersionControl).....EndClobalSection

Commit & Push

Complete your conversion by committing and pushing your local repository to your remote.

```
git add .
git commit -a -m "Coverted solution source control from TFVC to Git"

git remote add origin https://my.remote/project/repo.git

git push origin master
```

Section 36.5: Migrate from SVN to Git using svn2git

svn2git is a Ruby wrapper around git's native SVN support through git-svn, helping you with migrating projects from Subversion to Git, keeping history (incl. trunk, tags and branches history).

Examples

To migrate a svn repository with the standard layout (ie. branches, tags and trunk at the root level of the repository):

```
$ svn2git http://svn.example.com/path/to/repo
```

To migrate a svn repository which is not in standard layout:

```
$ svn2git http://svn.example.com/path/to/repo --trunk trunk-dir --tags tags-dir --branches
branches-dir
```

In case you do not want to migrate (or do not have) branches, tags or trunk you can use options `--notrunk`, `--nobranches`, and `--notags`.

For example, `$ svn2git http://svn.example.com/path/to/repo --trunk trunk-dir --notags --nobranches` will migrate only trunk history.

To reduce the space required by your new repository you may want to exclude any directories or files you once added while you should not have (eg. build directory or archives):

```
$ svn2git http://svn.example.com/path/to/repo --exclude build --exclude '.*\.zip$'
```

Post-migration optimization

If you already have a few thousand of commits (or more) in your newly created git repository, you may want to reduce space used before pushing your repository on a remote. This can be done using the following command:

```
$ git gc --aggressive
```

Note: The previous command can take up to several hours on large repositories (tens of thousand of commits and/or hundreds of megabytes of history).

Chapter 37: Show

Section 37.1: Overview

`git show` shows various Git objects.

For commits:

Shows the commit message and a diff of the changes introduced.

Command	Description
`git show`	shows the previous commit
`git show @~3`	shows the 3rd-from-last commit

For trees and blobs:

Shows the tree or blob.

Command	Description
`git show @~3:`	shows the project root directory as it was 3 commits ago (a tree)
`git show @~3:src/program.js`	shows `src/program.js` as it was 3 commits ago (a blob)
`git show @:a.txt @:b.txt`	shows `a.txt` concatenated with `b.txt` from current commit

For tags:

Shows the tag message and the referenced object.

Chapter 38: Resolving merge conflicts

Section 38.1: Manual Resolution

While performing a `git merge` you may find that git reports a "merge conflict" error. It will report to you which files have conflicts, and you will need to resolve the conflicts.

A `git status` at any point will help you see what still needs editing with a helpful message like

```
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")

Unmerged paths:
  (use "git add <file>..." to mark resolution)

    both modified:      index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

Git leaves markers in the files to tell you where the conflict arose:

```
<<<<<<<<< HEAD: index.html #indicates the state of your current branch
<div id="footer">contact : email@somedomain.com</div>
========= #indicates break between conflicts
<div id="footer">
please contact us at email@somedomain.com
</div>
>>>>>>>>> iss2: index.html #indicates the state of the other branch (iss2)
```

In order to resolve the conflicts, you must edit the area between the <<<<<< and >>>>>>> markers appropriately, remove the status lines (the <<<<<<<, >>>>>>>, and ======== lines) completely. Then `git add index.html` to mark it resolved and `git commit` to finish the merge.

Chapter 39: Bundles

Section 39.1: Creating a git bundle on the local machine and using it on another

Sometimes you may want maintain versions of a git repository on machines that have no network connection. Bundles allow you to package git objects and references in a repository on one machine and import those into a repository on another.

```
git tag 2016_07_24
git bundle create changes_between_tags.bundle [some_previous_tag]..2016_07_24
```

Somehow transfer the **changes_between_tags.bundle** file to the remote machine; e.g., via thumb drive. Once you have it there:

```
git bundle verify changes_between_tags.bundle  # make sure bundle arrived intact
git checkout [some branch]       # in the repo on the remote machine
git bundle list-heads changes_between_tags.bundle # list the references in the bundle
git pull changes_between_tags.bundle [reference from the bundle, e.g. last field from the previous output]
```

The reverse is also possible. Once you've made changes on the remote repository you can bundle up the deltas; put the changes on, e.g., a thumb drive, and merge them back into the local repository so the two can stay in sync without requiring direct `git`, `ssh`, `rsync`, or `http` protocol access between the machines.

Chapter 40: Display commit history graphically with Gitk

Section 40.1: Display commit history for one file

```
gitk path/to/myfile
```

Section 40.2: Display all commits between two commits

Let's say you have two commits `d9e1db9` and `5651067` and want to see what happened between them. `d9e1db9` is the oldest ancestor and `5651067` is the final descendant in the chain of commits.

```
gitk --ancestry-path d9e1db9 5651067
```

Section 40.3: Display commits since version tag

If you have the version tag `v2.3` you can display all commits since that tag.

```
gitk v2.3..
```

Chapter 41: Bisecting/Finding faulty commits

Section 41.1: Binary search (git bisect)

`git bisect` allows you to find which commit introduced a bug using a binary search.

Start by bisecting a session by providing two commit references: a good commit before the bug, and a bad commit after the bug. Generally, the bad commit is `HEAD`.

```
# start the git bisect session
$ git bisect start

# give a commit where the bug doesn't exist
$ git bisect good 49c747d

# give a commit where the bug exist
$ git bisect bad HEAD
```

`git` starts a binary search: It splits the revision in half and switches the repository to the intermediate revision. Inspect the code to determine if the revision is good or bad:

```
# tell git the revision is good,
# which means it doesn't contain the bug
$ git bisect good

# if the revision contains the bug,
# then tell git it's bad
$ git bisect bad
```

`git` will continue to run the binary search on each remaining subset of bad revisions depending on your instructions. `git` will present a single revision that, unless your flags were incorrect, will represent exactly the revision where the bug was introduced.

Afterwards remember to run `git bisect reset` to end the bisect session and return to HEAD.

```
$ git bisect reset
```

If you have a script that can check for the bug, you can automate the process with:

```
$ git bisect run [script] [arguments]
```

Where `[script]` is the path to your script and `[arguments]` is any arguments that should be passed to your script.

Running this command will automatically run through the binary search, executing `git bisect good` or `git bisect bad` at each step depending on the exit code of your script. Exiting with 0 indicates `good`, while exiting with 1-124, 126, or 127 indicates bad. 125 indicates that the script cannot test that revision (which will trigger a `git bisect skip`).

Section 41.2: Semi-automatically find a faulty commit

Imagine you are on the `master` branch and something is not working as expected (a regression was introduced), but you don't know where. All you know is, that is was working in the last release (which was e.g., tagged or you know the commit hash, lets take `old-rel` here).

Git has help for you, finding the faulty commit which introduced the regression with a very low number of steps (binary search).

First of all start bisecting:

```
git bisect start master old-rel
```

This will tell git that `master` is a broken revision (or the first broken version) and `old-rel` is the last known version.

Git will now check out a detached head in the middle of both commits. Now, you can do your testing. Depending on whether it works or not issue

```
git bisect good
```

or

```
git bisect bad
```

. In case this commit cannot be tested, you can easily `git reset` and test that one, git willl take care of this.

After a few steps git will output the faulty commit hash.

In order to abort the bisect process just issue

```
git bisect reset
```

and git will restore the previous state.

Chapter 42: Blaming

Parameter	Details
filename	Name of the file for which details need to be checked
-f	Show the file name in the origin commit
-e	Show the author email instead of author name
-w	Ignore white spaces while making a comparison between child and parent's version
-L start,end	Show only the given line range Example: `git blame -L 1,2 [filename]`
--show-stats	Shows additional statistics at end of blame output
-l	Show long rev (Default: off)
-t	Show raw timestamp (Default: off)
-reverse	Walk history forward instead of backward
-p, --porcelain	Output for machine consumption
-M	Detect moved or copied lines within a file
-C	In addition to -M, detect lines moved or copied from other files that were modified in the same commit
-h	Show the help message
-c	Use the same output mode as git-annotate (Default: off)
-n	Show the line number in the original commit (Default: off)

Section 42.1: Only show certain lines

Output can be restricted by specifying line ranges as

```
git blame -L <start>,<end>
```

Where `<start>` and `<end>` can be:

- line number

  ```
  git blame -L 10,30
  ```

- /regex/

  ```
  git blame -L /void main/, git blame -L 46,/void foo/
  ```

- +offset, -offset (only for `<end>`)

  ```
  git blame -L 108,+30, git blame -L 215,-15
  ```

Multiple line ranges can be specified, and overlapping ranges are allowed.

```
git blame -L 10,30 -L 12,80 -L 120,+10 -L ^/void main/,+40
```

Section 42.2: To find out who changed a file

```
// Shows the author and commit per line of specified file
git blame test.c
```

```
// Shows the author email and commit per line of specified
git blame -e test.c file

// Limits the selection of lines by specified range
git blame -L 1,10 test.c
```

Section 42.3: Show the commit that last modified a line

`git blame <file>`

will show the file with each line annotated with the commit that last modified it.

Section 42.4: Ignore whitespace-only changes

Sometimes repos will have commits that only adjust whitespace, for example fixing indentation or switching between tabs and spaces. This makes it difficult to find the commit where the code was actually written.

`git blame -w`

will ignore whitespace-only changes to find where the line really came from.

Chapter 43: Git revisions syntax

Section 43.1: Specifying revision by object name

```
$ git show dae86e1950b1277e545cee180551750029cfe735
$ git show dae86e19
```

You can specify revision (or in truth any object: tag, tree i.e. directory contents, blob i.e. file contents) using SHA-1 object name, either full 40-byte hexadecimal string, or a substring that is unique to the repository.

Section 43.2: Symbolic ref names: branches, tags, remote-tracking branches

```
$ git log master    # specify branch
$ git show v1.0     # specify tag
$ git show HEAD     # specify current branch
$ git show origin   # specify default remote-tracking branch for remote 'origin'
```

You can specify revision using a symbolic ref name, which includes branches (for example 'master', 'next', 'maint'), tags (for example 'v1.0', 'v0.6.3-rc2'), remote-tracking branches (for example 'origin', 'origin/master'), and special refs such as 'HEAD' for current branch.

If the symbolic ref name is ambiguous, for example if you have both branch and tag named 'fix' (having branch and tag with the same name is not recommended), you need to specify the kind of ref you want to use:

```
$ git show heads/fix      # or 'refs/heads/fix', to specify branch
$ git show tags/fix       # or 'refs/tags/fix', to specify tag
```

Section 43.3: The default revision: HEAD

```
$ git show         # equivalent to 'git show HEAD'
```

'HEAD' names the commit on which you based the changes in the working tree, and is usually the symbolic name for the current branch. Many (but not all) commands that take revision parameter defaults to 'HEAD' if it is missing.

Section 43.4: Reflog references: <refname>@{<n>}

```
$ git show @{1}            # uses reflog for current branch
$ git show master@{1}      # uses reflog for branch 'master'
$ git show HEAD@{1}        # uses 'HEAD' reflog
```

A ref, usually a branch or HEAD, followed by the suffix `@` with an ordinal specification enclosed in a brace pair (e.g. `{1}`, `{15}`) specifies the n-th prior value of that ref *in your **local** repository*. You can check recent reflog entries with `git reflog` command, or `--walk-reflogs` / `-g` option to `git log`.

```
$ git reflog
08bb350 HEAD@{0}: reset: moving to HEAD^
4ebf58d HEAD@{1}: commit: gitweb(1): Document query parameters
08bb350 HEAD@{2}: pull: Fast-forward
f34be46 HEAD@{3}: checkout: moving from af40944bda352190f05d22b7cb8fe88beb17f3a7 to master
af40944 HEAD@{4}: checkout: moving from master to v2.6.3

$ git reflog gitweb-docs
```

```
4ebf58d gitweb-docs@{0}: branch: Created from master
```

Note: using reflogs practically replaced older mechanism of utilizing `ORIG_HEAD` ref (roughly equivalent to `HEAD@{1}`).

Section 43.5: Reflog references: <refname>@{<date>}

```
$ git show master@{yesterday}
$ git show HEAD@{5 minutes ago}   # or HEAD@{5.minutes.ago}
```

A ref followed by the suffix `@` with a date specification enclosed in a brace pair (e.g. `{yesterday}`, `{1 month 2 weeks 3 days 1 hour 1 second ago}` or `{1979-02-26 18:30:00}`) specifies the value of the ref at a prior point in time (or closest point to it). Note that this looks up the state of your **local** ref at a given time; e.g., what was in your local *'master'* branch last week.

You can use `git reflog` with a date specifier to look up exact time where you did something to given ref in the local repository.

```
$ git reflog HEAD@{now}
08bb350 HEAD@{Sat Jul 23 19:48:13 2016 +0200}: reset: moving to HEAD^
4ebf58d HEAD@{Sat Jul 23 19:39:20 2016 +0200}: commit: gitweb(1): Document query parameters
08bb350 HEAD@{Sat Jul 23 19:26:43 2016 +0200}: pull: Fast-forward
```

Section 43.6: Tracked / upstream branch: <branchname>@{upstream}

```
$ git log @{upstream}..       # what was done locally and not yet published, current branch
$ git show master@{upstream}  # show upstream of branch 'master'
```

The suffix `@{upstream}` appended to a branchname (short form `<branchname>@{u}`) refers to the branch that the branch specified by branchname is set to build on top of (configured with `branch.<name>.remote` and `branch.<name>.merge`, or with `git branch --set-upstream-to=<branch>`). A missing branchname defaults to the current one.

Together with syntax for revision ranges it is very useful to see the commits your branch is ahead of upstream (commits in your local repository not yet present upstream), and what commits you are behind (commits in upstream not merged into local branch), or both:

```
$ git log --oneline @{u}..
$ git log --oneline ..@{u}
$ git log --oneline --left-right @{u}...  # same as ...@{u}
```

Section 43.7: Commit ancestry chain: <rev>^, <rev>~<n>, etc

```
$ git reset --hard HEAD^            # discard last commit
$ git rebase --interactive HEAD~5   # rebase last 4 commits
```

A suffix `^` to a revision parameter means the first parent of that commit object. `^<n>` means the <n>-th parent (i.e. `<rev>^` is equivalent to `<rev>^1`).

A suffix `~<n>` to a revision parameter means the commit object that is the <n>-th generation ancestor of the named commit object, following only the first parents. This means that for example `<rev>~3` is equivalent to `<rev>^^^`. As a shortcut, `<rev>~` means `<rev>~1`, and is equivalent to `<rev>^1`, or `<rev>^` in short.

This syntax is composable.

To find such symbolic names you can use the `git name-rev` command:

```
$ git name-rev 33db5f4d9027a10e477ccf054b2c1ab94f74c85a
33db5f4d9027a10e477ccf054b2c1ab94f74c85a tags/v0.99~940
```

Note that `--pretty=oneline` and not `--oneline` must be used in the following example

```
$ git log --pretty=oneline | git name-rev --stdin --name-only
master Sixth batch of topics for 2.10
master~1 Merge branch 'ls/p4-tmp-refs'
master~2 Merge branch 'js/am-call-theirs-theirs-in-fallback-3way'
[...]
master~14^2 sideband.c: small optimization of strbuf usage
master~16^2 connect: read $GIT_SSH_COMMAND from config file
[...]
master~22^2~1 t7810-grep.sh: fix a whitespace inconsistency
master~22^2~2 t7810-grep.sh: fix duplicated test name
```

Section 43.8: Dereferencing branches and tags: <rev>^0, <rev>^{<type>}

In some cases the behavior of a command depends on whether it is given branch name, tag name, or an arbitrary revision. You can use "de-referencing" syntax if you need the latter.

A suffix `^` followed by an object type name (`tag`, `commit`, `tree`, `blob`) enclosed in brace pair (for example `v0.99.8^{commit}`) means dereference the object at `<rev>` recursively until an object of type `<type>` is found or the object cannot be dereferenced anymore. `<rev>^0` is a short-hand for `<rev>^{commit}`.

```
$ git checkout HEAD^0    # equivalent to 'git checkout --detach' in modern Git
```

A suffix `^` followed by an empty brace pair (for example `v0.99.8^{}`) means to dereference the tag recursively until a non-tag object is found.

Compare

```
$ git show v1.0
$ git cat-file -p v1.0
$ git replace --edit v1.0
```

with

```
$ git show v1.0^{}
$ git cat-file -p v1.0^{}
$ git replace --edit v1.0^{}
```

Section 43.9: Youngest matching commit: <rev>^{/<text>}, :/<text>

```
$ git show HEAD^{/fix nasty bug}   # find starting from HEAD
$ git show ':/fix nasty bug'       # find starting from any branch
```

A colon (`':'`), followed by a slash (`'/'`), followed by a text, names a commit whose commit message matches the specified regular expression. This name returns the youngest matching commit which is reachable from *any* ref.

The regular expression can match any part of the commit message. To match messages starting with a string, one can use e.g. `:/^foo`. The special sequence `:/!` is reserved for modifiers to what is matched. `:/!-foo` performs a negative match, while `:/!!foo` matches a literal ! character, followed by `foo`.

A suffix `^` to a revision parameter, followed by a brace pair that contains a text led by a slash, is the same as the `:/<text>` syntax below that it returns the youngest matching commit which is reachable from the `<rev>` before `^`.

Chapter 44: Worktrees

Parameter	Details
-f --force	By default, add refuses to create a new working tree when `<branch>` is already checked out by another working tree. This option overrides that safeguard.
-b `<new-branch>` -B `<new-branch>`	With add, create a new branch named `<new-branch>` starting at `<branch>`, and check out `<new-branch>` into the new working tree. If `<branch>` is omitted, it defaults to `HEAD`. By default, `-b` refuses to create a new branch if it already exists. `-B` overrides this safeguard, resetting `<new-branch>` to `<branch>`.
--detach	With add, detach `HEAD` in the new working tree.
--[no-] checkout	By default, add checks out `<branch>`, however, `--no-checkout` can be used to suppress checkout in order to make customizations, such as configuring sparse-checkout.
-n --dry-run	With prune, do not remove anything; just report what it would remove.
--porcelain	With list, output in an easy-to-parse format for scripts. This format will remain stable across Git versions and regardless of user configuration.
-v --verbose	With prune, report all removals.
--expire `<time>`	With prune, only expire unused working trees older than `<time>`.

Section 44.1: Using a worktree

You are right in the middle of working on a new feature, and your boss comes in demanding that you fix something immediately. You may typically want use `git stash` to store your changes away temporarily. However, at this point your working tree is in a state of disarray (with new, moved, and removed files, and other bits and pieces strewn around) and you don't want to disturb your progress.

By adding a worktree, you create a temporary linked working tree to make the emergency fix, remove it when done, and then resume your earlier coding session:

```
$ git worktree add -b emergency-fix ../temp master
$ pushd ../temp
# ... work work work ...
$ git commit -a -m 'emergency fix for boss'
$ popd
$ rm -rf ../temp
$ git worktree prune
```

NOTE: In this example, the fix still is in the emergency-fix branch. At this point you probably want to `git merge` or `git format-patch` and afterwards remove the emergency-fix branch.

Section 44.2: Moving a worktree

Currently (as of version 2.11.0) there is no built-in functionality to move an already existing worktree. This is listed as an official bug (see https://git-scm.com/docs/git-worktree#_bugs).

To get around this limitation it is possible to perform manual operations directly in the `.git` reference files.

In this example, the main copy of the repo is living at `/home/user/project-main` and the secondary worktree is located at `/home/user/project-1` and we want to move it to `/home/user/project-2`.

Don't perform any git command in between these steps, otherwise the garbage collector might be triggered and the references to the secondary tree can be lost. Perform these steps from the start until the end without interruption:

1. Change the worktree's `.git` file to point to the new location inside the main tree. The file `/home/user/project-1/.git` should now contain the following:

```
gitdir: /home/user/project-main/.git/worktrees/project-2
```

2. Rename the worktree inside the `.git` directory of the main project by moving the worktree's directory that exists in there:

```
$ mv /home/user/project-main/.git/worktrees/project-1 /home/user/project-main/.git/worktrees/project-2
```

3. Change the reference inside `/home/user/project-main/.git/worktrees/project-2/gitdir` to point to the new location. In this example, the file would have the following contents:

```
/home/user/project-2/.git
```

4. Finally, move your worktree to the new location:

```
$ mv /home/user/project-1 /home/user/project-2
```

If you have done everything correctly, listing the existing worktrees should refer to the new location:

```
$ git worktree list
/home/user/project-main  23f78ad [master]
/home/user/project-2     78ac3f3 [branch-name]
```

It should now also be safe to run `git worktree prune`.

Chapter 45: Git Remote

Parameter	Details
-v, --verbose	Run verbosely.
-m <master>	Sets head to remote's <master> branch
--mirror=fetch	Refs will not be stored in refs/remotes namespace, but instead will be mirrored in the local repo
--mirror=push	`git push` will behave as if --mirror was passed
--no-tags	`git fetch <name>` does not import tags from the remote repo
-t <branch>	Specifies the remote to track *only* <branch>
-f	`git fetch <name>` is run immediately after remote is set up
--tags	`git fetch <name>` imports every tag from the remote repo
-a, --auto	The symbolic-ref's HEAD is set to the same branch as the remote's HEAD
-d, --delete	All listed refs are deleted from the remote repository
--add	Adds <name> to list of currently tracked branches (set-branches)
--add	Instead of changing some URL, new URL is added (set-url)
--all	Push all branches.
--delete	All urls matching <url> are deleted. (set-url)
--push	Push URLS are manipulated instead of fetch URLS
-n	The remote heads are not queried first with `git ls-remote <name>`, cached information is used instead
--dry-run	report what branches will be pruned, but do not actually prune them
--prune	Remove remote branches that don't have a local counterpart

Section 45.1: Display Remote Repositories

To list all configured remote repositories, use `git remote`.

It shows the short name (aliases) of each remote handle that you have configured.

```
$ git remote
premium
premiumPro
origin
```

To show more detailed information, the `--verbose` or `-v` flag can be used. The output will include the URL and the type of the remote (`push` or `pull`):

```
$ git remote -v
premiumPro    https://github.com/user/CatClickerPro.git (fetch)
premiumPro    https://github.com/user/CatClickerPro.git (push)
premium    https://github.com/user/CatClicker.git (fetch)
premium    https://github.com/user/CatClicker.git (push)
origin    https://github.com/ud/starter.git (fetch)
origin    https://github.com/ud/starter.git (push)
```

Section 45.2: Change remote url of your Git repository

You may want to do this if the remote repository is migrated. The command for changing the remote url is:

```
git remote set-url
```

It takes 2 arguments: an existing remote name (origin, upstream) and the url.

Check your current remote url:

```
git remote -v
origin    https://bitbucket.com/develop/myrepo.git (fetch)
origin    https://bitbucket.com/develop/myrepo.git (push)
```

Change your remote url:

```
git remote set-url origin https://localserver/develop/myrepo.git
```

Check again your remote url:

```
git remote -v
origin    https://localserver/develop/myrepo.git (fetch)
origin    https://localserver/develop/myrepo.git (push)
```

Section 45.3: Remove a Remote Repository

Remove the remote named `<name>`. All remote-tracking branches and configuration settings for the remote are removed.

To remove a remote repository `dev`:

```
git remote rm dev
```

Section 45.4: Add a Remote Repository

To add a remote, use `git remote add` in the root of your local repository.

For adding a remote Git repository <url> as an easy short name <name> use

```
git remote add <name> <url>
```

The command `git fetch <name>` can then be used to create and update remote-tracking branches `<name>/<branch>`.

Section 45.5: Show more information about remote repository

You can view more information about a remote repository by `git remote show <remote repository alias>`

```
git remote show origin
```

result:

```
remote origin
Fetch URL:  https://localserver/develop/myrepo.git
Push  URL:  https://localserver/develop/myrepo.git
HEAD branch: master
Remote branches:
  master      tracked
Local branches configured for 'git pull':
  master      merges with remote master
```

```
Local refs configured for 'git push':
  master      pushes to master      (up to date)
```

Section 45.6: Rename a Remote Repository

Rename the remote named `<old>` to `<new>`. All remote-tracking branches and configuration settings for the remote are updated.

To rename a remote branch name `dev` to `dev1` :

```
git remote rename dev dev1
```

Chapter 46: Git Large File Storage (LFS)

Section 46.1: Declare certain file types to store externally

A common workflow for using Git LFS is to declare which files are intercepted through a rules-based system, just like `.gitignore` files.

Much of time, wildcards are used to pick certain file-types to blanket track.

e.g. `git lfs track "*.psd"`

When a file matching the above pattern is added them committed, when it is then pushed to the remote, it will be uploaded separately, with a pointer replacing the file in the remote repository.

After a file has been tracked with lfs, your `.gitattributes` file will be updated accordingly. Github recommends committing your local `.gitattributes` file, rather than working with a global `.gitattributes` file, to help ensure you don't have any issues when working with different projects.

Section 46.2: Set LFS config for all clones

To set LFS options that apply to all clones, create and commit a file named `.lfsconfig` at the repository root. This file can specify LFS options the same way as allowed in `.git/config`.

For example, to exclude a certain file from LFS fetches be default, create and commit `.lfsconfig` with the following contents:

```
[lfs]
    fetchexclude = ReallyBigFile.wav
```

Section 46.3: Install LFS

Download and install, either via Homebrew, or from website.

For Brew,

```
brew install git-lfs
git lfs install
```

Often you will also need to do some setup on the service that hosts your remote to allow it to work with lfs. This will be different for each host, but will likely just be checking a box saying you want to use git lfs.

Chapter 47: Git Patch

Parameter	Details
(<mbox>|<Maildir>)...	The list of mailbox files to read patches from. If you do not supply this argument, the command reads from the standard input. If you supply directories, they will be treated as Maildirs.
-s, --signoff	Add a Signed-off-by: line to the commit message, using the committer identity of yourself.
-q, --quiet	Be quiet. Only print error messages.
-u, --utf8	Pass `-u` flag to `git mailinfo`. The proposed commit log message taken from the e-mail is re-coded into UTF-8 encoding (configuration variable `i18n.commitencoding` can be used to specify project's preferred encoding if it is not UTF-8). You can use `--no-utf8` to override this.
--no-utf8	Pass -n flag to git mailinfo.
-3, --3way	When the patch does not apply cleanly, fall back on 3-way merge if the patch records the identity of blobs it is supposed to apply to and we have those blobs available locally.
--ignore-date, --ignore-space-change, --ignore-whitespace, --whitespace=<option>, -C<n>, -p<n>, --directory=<dir>, --exclude=<path>, --include=<path>, --reject	These flags are passed to the git apply program that applies the patch.
--patch-format	By default the command will try to detect the patch format automatically. This option allows the user to bypass the automatic detection and specify the patch format that the patch(es) should be interpreted as. Valid formats are `mbox`, `stgit`, `stgit-series`, and `hg`.
-i, --interactive	Run interactively.
--committer-date-is-author-date	By default the command records the date from the e-mail message as the commit author date, and uses the time of commit creation as the committer date. This allows the user to lie about the committer date by using the same value as the author date.
--ignore-date	By default the command records the date from the e-mail message as the commit author date, and uses the time of commit creation as the committer date. This allows the user to lie about the author date by using the same value as the committer date.
--skip	Skip the current patch. This is only meaningful when restarting an aborted patch.
-S[<keyid>], --gpg-sign[=<keyid>]	GPG-sign commits.
--continue, -r, --resolved	After a patch failure (e.g. attempting to apply conflicting patch), the user has applied it by hand and the index file stores the result of the application. Make a commit using the authorship and commit log extracted from the e-mail message and the current index file, and continue.
--resolvemsg=<msg>	When a patch failure occurs, `<msg>` will be printed to the screen before exiting. This overrides the standard message informing you to use `--continue` or `--skip` to handle the failure. This is solely for internal use between `git rebase` and `git am`.
--abort	Restore the original branch and abort the patching operation.

Section 47.1: Creating a patch

To create a patch, there are two steps.

1. Make your changes and commit them.

2. Run `git format-patch <commit-reference>` to convert all commits since the commit <commit-reference> (not including it) into patch files.

For example, if patches should be generated from the latest two commits:

```
git format-patch HEAD~~
```

This will create 2 files, one for each commit since `HEAD~~`, like this:

```
0001-hello_world.patch
0002-beginning.patch
```

Section 47.2: Applying patches

We can use `git apply some.patch` to have the changes from the `.patch` file applied to your current working directory. They will be unstaged and need to be committed.

To apply a patch as a commit (with its commit message), use

```
git am some.patch
```

To apply all patch files to the tree:

```
git am *.patch
```

Chapter 48: Git statistics

Parameter	Details
`-n`, `--numbered`	Sort output according to the number of commits per author instead of alphabetic order
`-s`, `--summary`	Only provide a commit count summary
`-e`, `--email`	Show the email address of each author
`--format`[=<format>]	Instead of the commit subject, use some other information to describe each commit. <format> can be any string accepted by the `--format` option of `git log`.
`-w`[<width>[,<indent1>[,<indent2>]]]	Linewrap the output by wrapping each line at `width`. The first line of each entry is indented by `indent1` number of spaces, and subsequent lines are indented by `indent2` spaces.
<revision range>	Show only commits in the specified revision range. Default to the whole history until the current commit.
[`--`] <path>	Show only commits that explain how the files matching `path` came to be. Paths may need to be prefixed with "-- " to separate them from options or the revision range.

Section 48.1: Lines of code per developer

```
git ls-tree -r HEAD | sed -Ee 's/^.{53}//' | \
while read filename; do file "$filename"; done | \
grep -E ': .*text' | sed -E -e 's/: .*//' | \
while read filename; do git blame --line-porcelain "$filename"; done | \
sed -n 's/^author //p' | \
sort | uniq -c | sort -rn
```

Section 48.2: Listing each branch and its last revision's date

```
for k in `git branch -a | sed s/^../`; do echo -e `git log -1 --pretty=format:"%Cgreen%ci
%Cblue%cr%Creset" $k --`\\t"$k";done | sort
```

Section 48.3: Commits per developer

Git `shortlog` is used to summarize the git log outputs and group the commits by author.

By default, all commit messages are shown but argument `--summary` or `-s` skips the messages and gives a list of authors with their total number of commits.

`--numbered` or `-n` changes the ordering from alphabetical (by author ascending) to number of commits descending.

```
git shortlog -sn        #Names and Number of commits

git shortlog -sne       #Names along with their email ids and the Number of commits
```

or

```
git log --pretty=format:%ae \
| gawk -- '{ ++c[$0]; } END { for(cc in c) printf "%5d %s\n",c[cc],cc; }'
```

Note: Commits by the same person may not be grouped together where their name and/or email address has been spelled differently. For example `John Doe` and `Johnny Doe` will appear separately in the list. To resolve this,

refer to the `.mailmap` feature.

Section 48.4: Commits per date

```
git log --pretty=format:"%ai" | awk '{print " : "$1}' | sort -r | uniq -c
```

Section 48.5: Total number of commits in a branch

```
git log  --pretty=oneline |wc -l
```

Section 48.6: List all commits in pretty format

```
git log --pretty=format:"%Cgreen%ci %Cblue%cn  %Cgreen%cr%Creset %s"
```

This will give a nice overview of all commits (1 per line) with date, user and commit message.

The `--pretty` option has many placeholders, each starting with `%`. All options can be found here

Section 48.7: Find All Local Git Repositories on Computer

To list all the git repository locations on your you can run the following

```
find $HOME -type d -name ".git"
```

Assuming you have `locate`, this should be much faster:

```
locate .git |grep git$
```

If you have `gnu locate` or `mlocate`, this will select only the git dirs:

```
locate -ber \\.git$
```

Section 48.8: Show the total number of commits per author

In order to get the total number of commits that each developer or contributor has made on a repository, you can simply use the `git shortlog`:

```
git shortlog -s
```

which provides the author names and number of commits by each one.

Additionally, if you want to have the results calculated on all branches, add `--all` flag to the command:

```
git shortlog -s --all
```

Chapter 49: git send-email

Section 49.1: Use git send-email with Gmail

Background: if you work on a project like the Linux kernel, rather than make a pull request you will need to submit your commits to a listserv for review. This entry details how to use git-send email with Gmail.

Add the following to your .gitconfig file:

```
[sendemail]
    smtpserver = smtp.googlemail.com
    smtpencryption = tls
    smtpserverport = 587
    smtpuser = name@gmail.com
```

Then on the web: Go to Google -> My Account -> Connected Apps & Sites -> Allow less secure apps -> Switch ON

To create a patch set:

```
git format-patch HEAD~~~~ --subject-prefix="PATCH <project-name>"
```

Then send the patches to a listserv:

```
git send-email --annotate --to project-developers-list@listserve.example.com 00*.patch
```

To create and send updated version (version 2 in this example) of the patch:

```
git format-patch -v 2 HEAD~~~~  ......
git send-email --to project-developers-list@listserve.example.com v2-00*.patch
```

Section 49.2: Composing

```
--from                    * Email From:
--[no-]to                 * Email To:
--[no-]cc                 * Email Cc:
--[no-]bcc                * Email Bcc:
--subject                 * Email "Subject:"
--in-reply-to             * Email "In-Reply-To:"
--[no-]xmailer                * Add "X-Mailer:" header (default).
--[no-]annotate               * Review each patch that will be sent in an editor.
--compose                     * Open an editor for introduction.
--compose-encoding        * Encoding to assume for introduction.
--8bit-encoding           * Encoding to assume 8bit mails if undeclared
--transfer-encoding       * Transfer encoding to use (quoted-printable, 8bit, base64)
```

Section 49.3: Sending patches by mail

Suppose you've got a lot of commit against a project (here ulogd2, official branch is git-svn) and that you wan to send your patchset to the Mailling list devel@netfilter.org. To do so, just open a shell at the root of the git directory and use:

```
git format-patch --stat -p --raw --signoff  --subject-prefix="ULOGD PATCH" -o /tmp/ulogd2/ -n git-
svn
```

```
git send-email --compose --no-chain-reply-to --to devel@netfilter.org /tmp/ulogd2/
```

First command will create a serie of mail from patches in /tmp/ulogd2/ with statistic report and second will start your editor to compose an introduction mail to the patchset. To avoid awful threaded mail series, one can use :

```
git config sendemail.chainreplyto false
```

source

Chapter 50: Git GUI Clients

Section 50.1: gitk and git-gui

When you install Git, you also get its visual tools, gitk and git-gui.

`gitk` is a graphical history viewer. Think of it like a powerful GUI shell over git log and git grep. This is the tool to use when you're trying to find something that happened in the past, or visualize your project's history.

Gitk is easiest to invoke from the command-line. Just cd into a Git repository, and type:

```
$ gitk [git log options]
```

Gitk accepts many command-line options, most of which are passed through to the underlying git log action. Probably one of the most useful is the `--all` flag, which tells gitk to show commits reachable from any ref, not just HEAD. Gitk's interface looks like this:

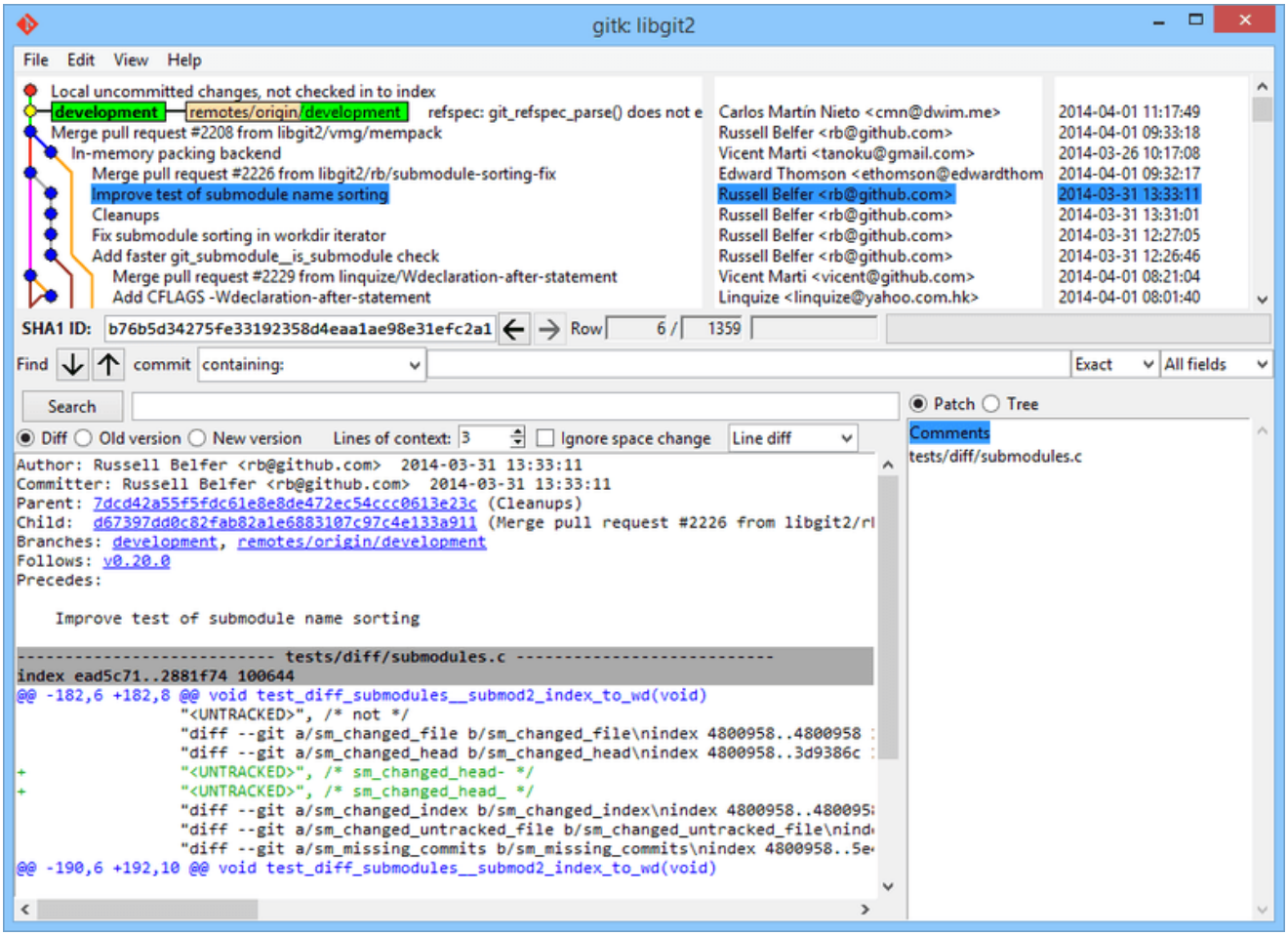


Figure 1-1. The gitk history viewer.

On the top is something that looks a bit like the output of git log --graph; each dot represents a commit, the lines represent parent relationships, and refs are shown as colored boxes. The yellow dot represents HEAD, and the red dot represents changes that are yet to become a commit. At the bottom is a view of the selected commit; the comments and patch on the left, and a summary view on the right. In between is a collection of controls used for searching history.

You can access many git related functions via right-click on a branch name or a commit message. For example checking out a different branch or cherry pick a commit is easily done with one click.

`git-gui`, on the other hand, is primarily a tool for crafting commits. It, too, is easiest to invoke from the command line:

```
$ git gui
```

And it looks something like this:

The `git-gui` commit tool.

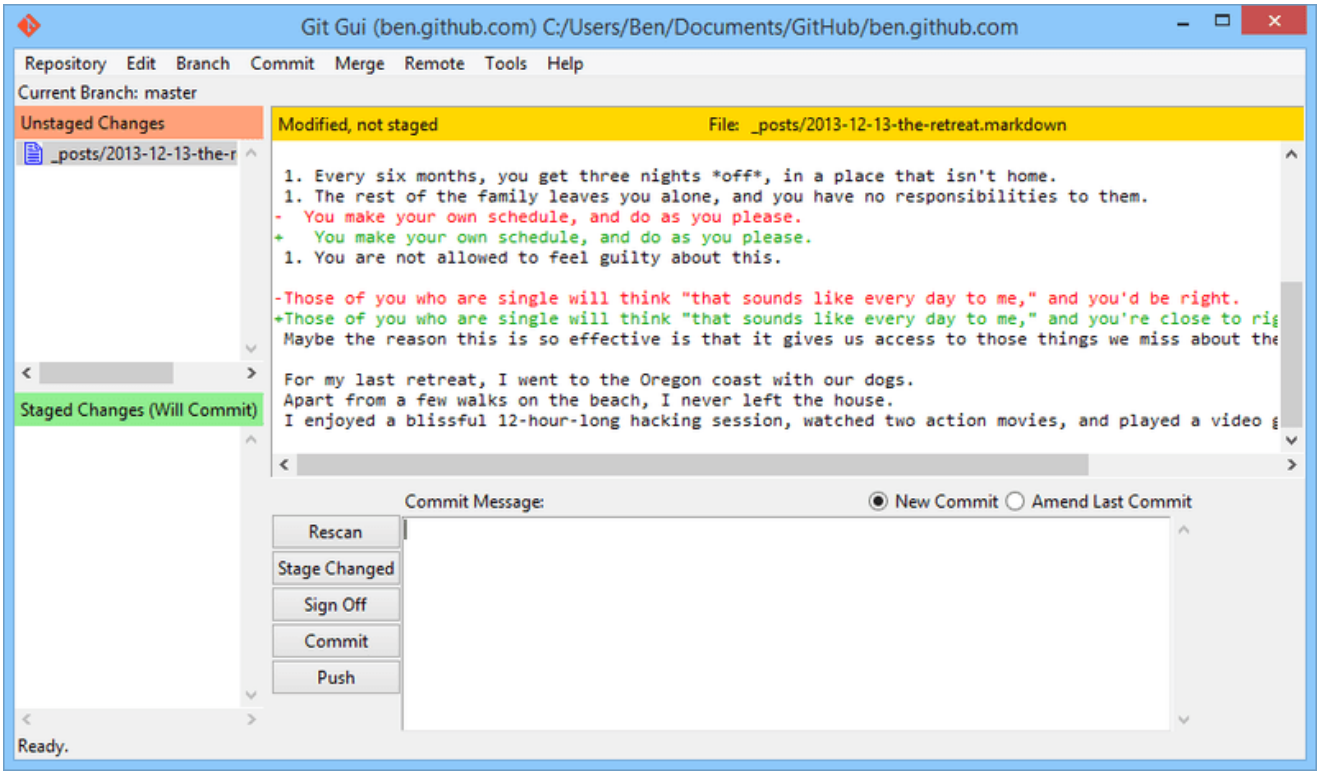


Figure 1-2. The git-gui commit tool.

On the left is the index; unstaged changes are on top, staged changes on the bottom. You can move entire files between the two states by clicking on their icons, or you can select a file for viewing by clicking on its name.

At top right is the diff view, which shows the changes for the currently-selected file. You can stage individual hunks (or individual lines) by right-clicking in this area.

At the bottom right is the message and action area. Type your message into the text box and click "Commit" to do something similar to git commit. You can also choose to amend the last commit by choosing the "Amend" radio button, which will update the "Staged Changes" area with the contents of the last commit. Then you can simply stage or unstage some changes, alter the commit message, and click "Commit" again to replace the old commit with a new one.

gitk and git-gui are examples of task-oriented tools. Each of them is tailored for a specific purpose (viewing history and creating commits, respectively), and omit the features not necessary for that task.

Source: https://git-scm.com/book/en/v2/Git-in-Other-Environments-Graphical-Interfaces

Section 50.2: GitHub Desktop

Website: https://desktop.github.com
Price: free

Platforms: OS X and Windows
Developed by: GitHub

Section 50.3: Git Kraken

Website:https://www.gitkraken.com
Price: $60/years (free for For open source, education, non-profit, startups or personal use)
Platforms: Linux, OS X, Windows
Developed by: Axosoft

Section 50.4: SourceTree

Website: https://www.sourcetreeapp.com
Price: free (account is necessary)
Platforms: OS X and Windows
Developer: Atlassian

Section 50.5: Git Extensions

Website: https://gitextensions.github.io
Price: free
Platform: Windows

Section 50.6: SmartGit

Website: http://www.syntevo.com/smartgit/
Price: Free for non-commercial use only. A perpetual license costs 99 USD
Platforms: Linux, OS X, Windows
Developed by: syntevo

Chapter 51: Reflog - Restoring commits not shown in git log

Section 51.1: Recovering from a bad rebase

Suppose that you had started an interactive rebase:

```
git rebase --interactive HEAD~20
```

and by mistake, you squashed or dropped some commits that you didn't want to lose, but then completed the rebase. To recover, do `git reflog`, and you might see some output like this:

```
aaaaaaa HEAD@{0} rebase -i (finish): returning to refs/head/master
bbbbbbb HEAD@{1} rebase -i (squash): Fix parse error
...
ccccccc HEAD@{n} rebase -i (start): checkout HEAD~20
ddddddd HEAD@{n+1} ...
...
```

In this case, the last commit, `ddddddd` (or `HEAD@{n+1}`) is the tip of your *pre-rebase* branch. Thus, to recover that commit (and all parent commits, including those accidentally squashed or dropped), do:

```
$ git checkout HEAD@{n+1}
```

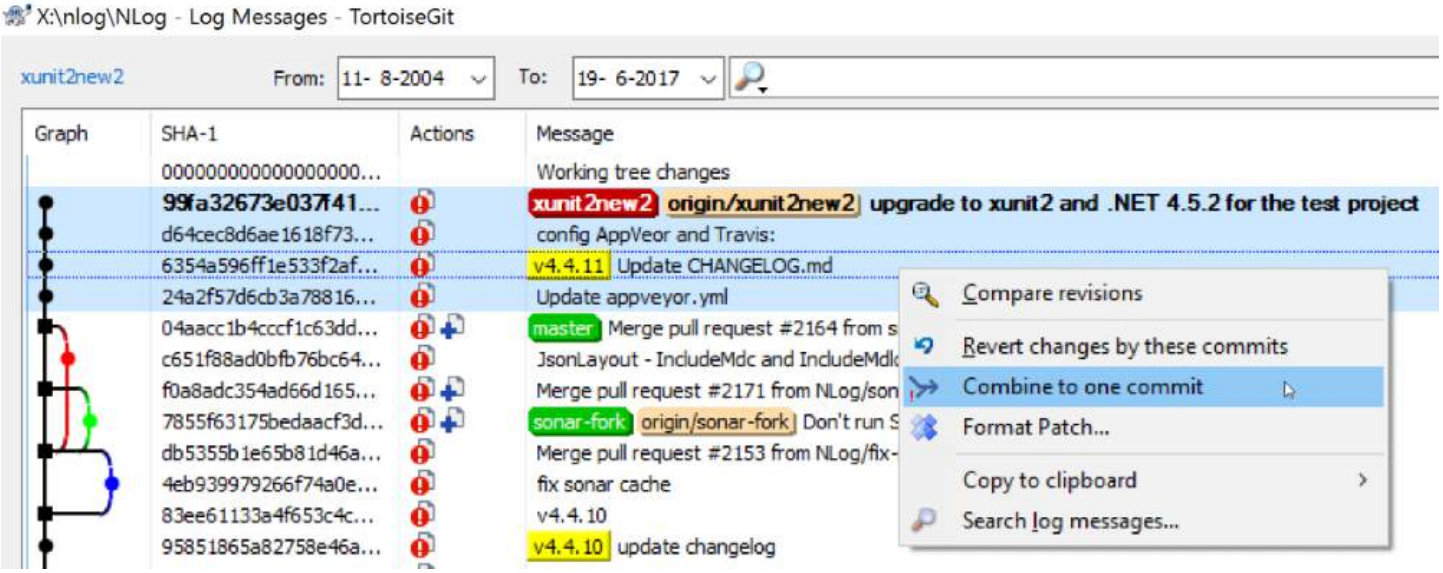
You can then create a new branch at that commit with `git checkout -b [branch]`. See Branching for more information.

Chapter 52: TortoiseGit

Section 52.1: Squash commits

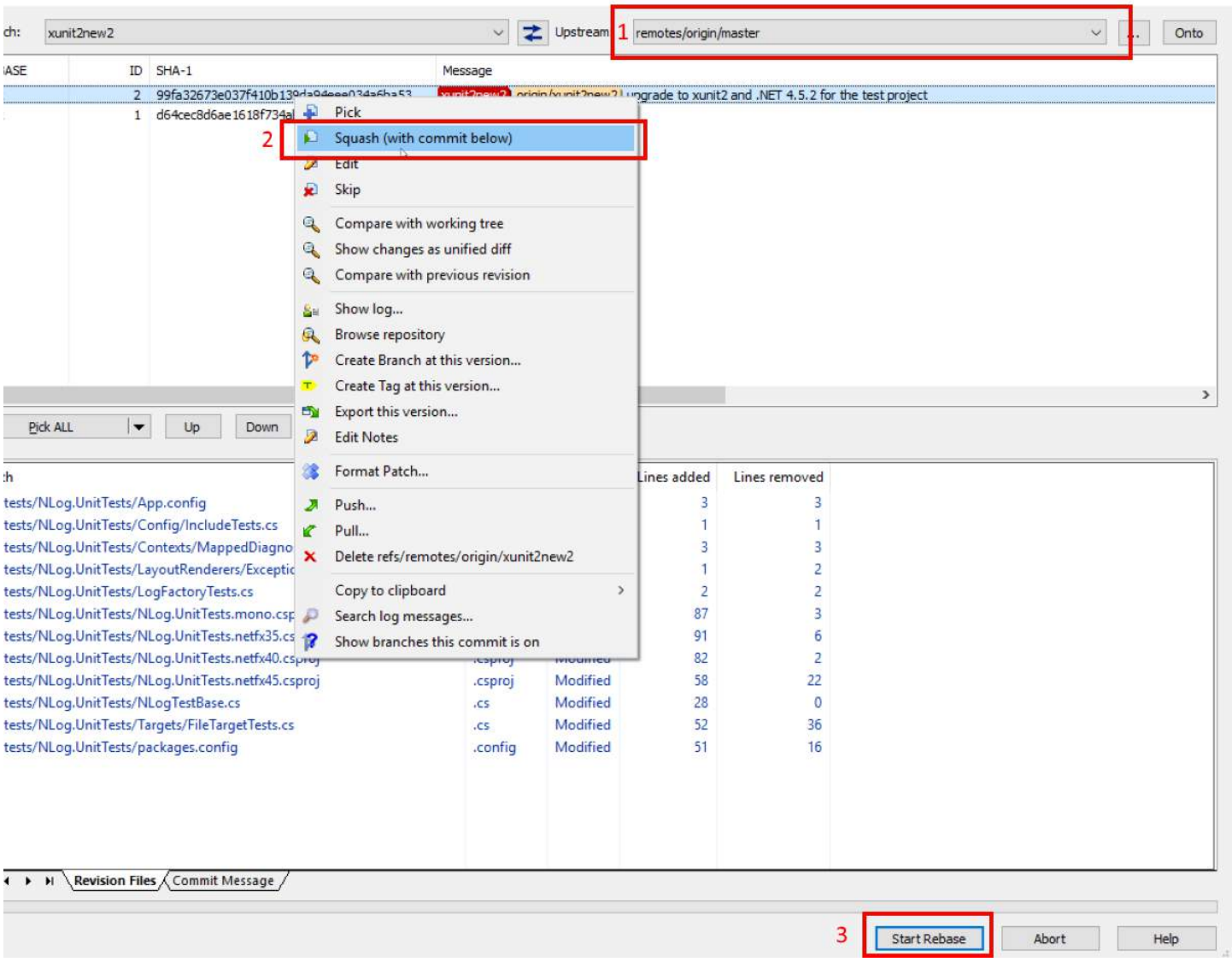
The easy way

This won't work if there are merge commits in your selection



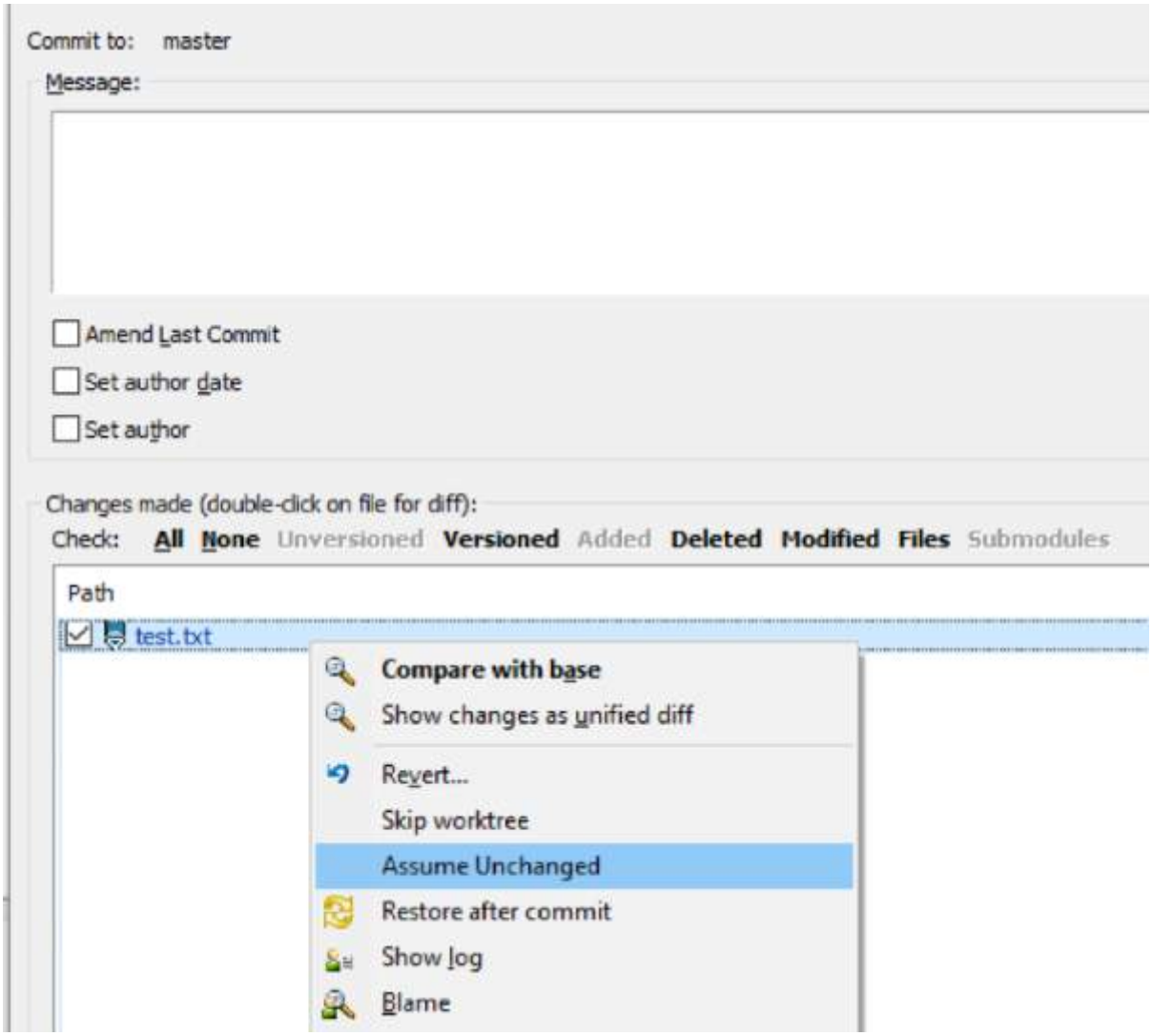
The advanced way

Start the rebase dialog:



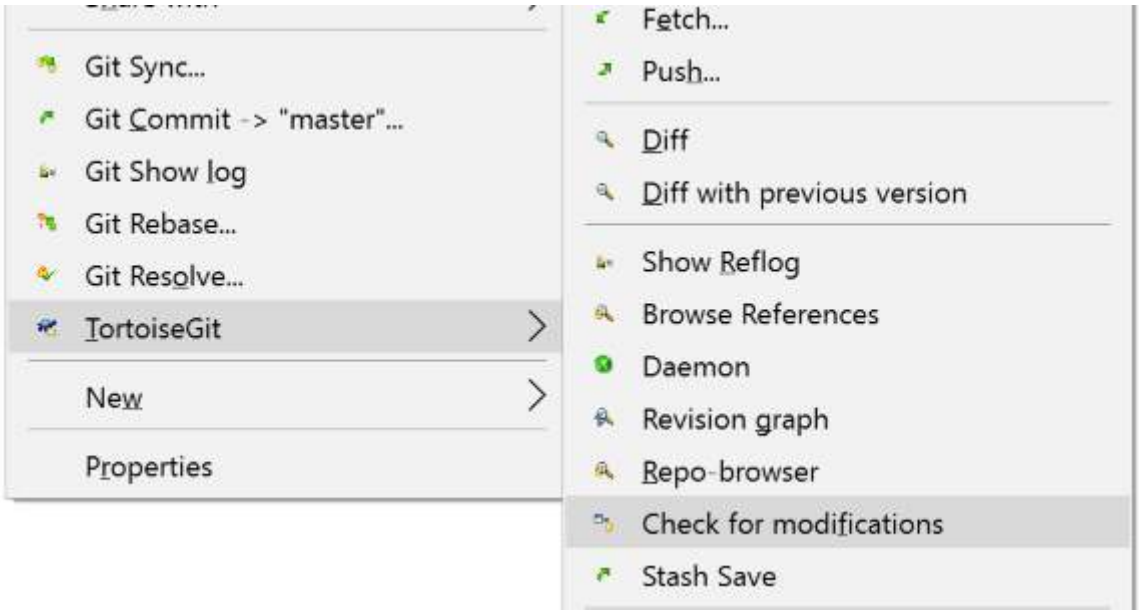
Section 52.2: Assume unchanged

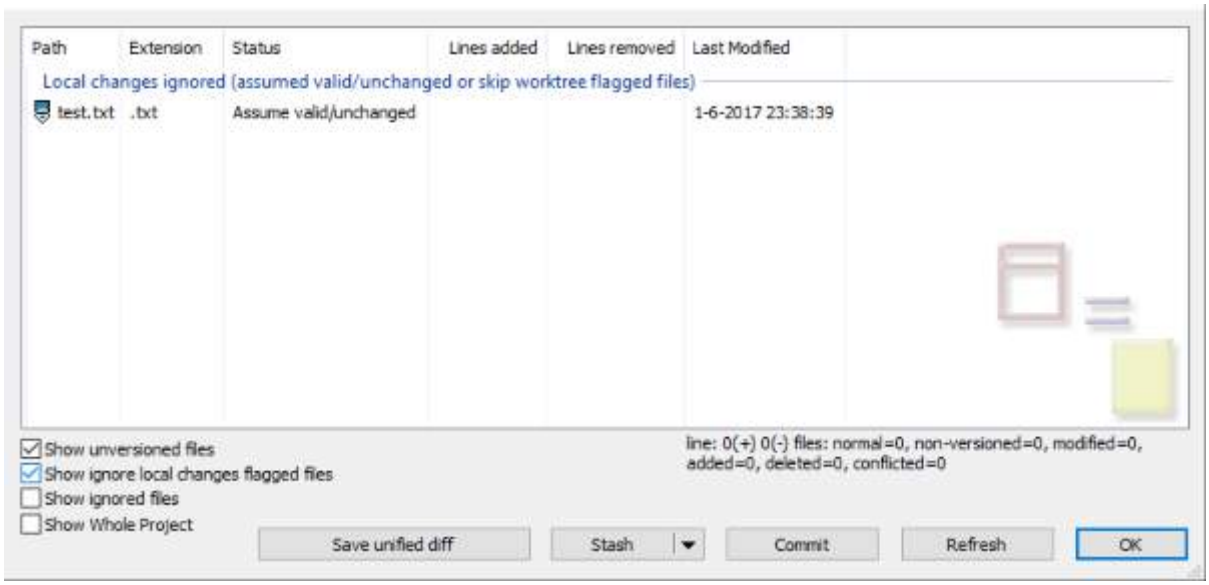
If a file is changed, but you don't like to commit it, set the file as "Assume unchanged"

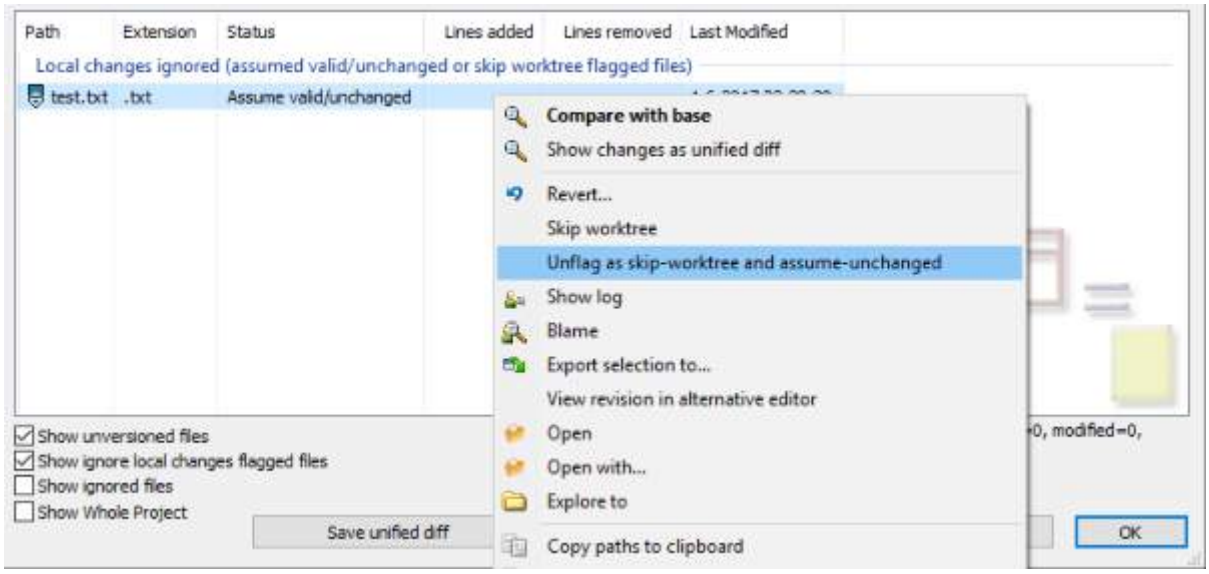


Revert "Assume unchanged"

Need some steps:

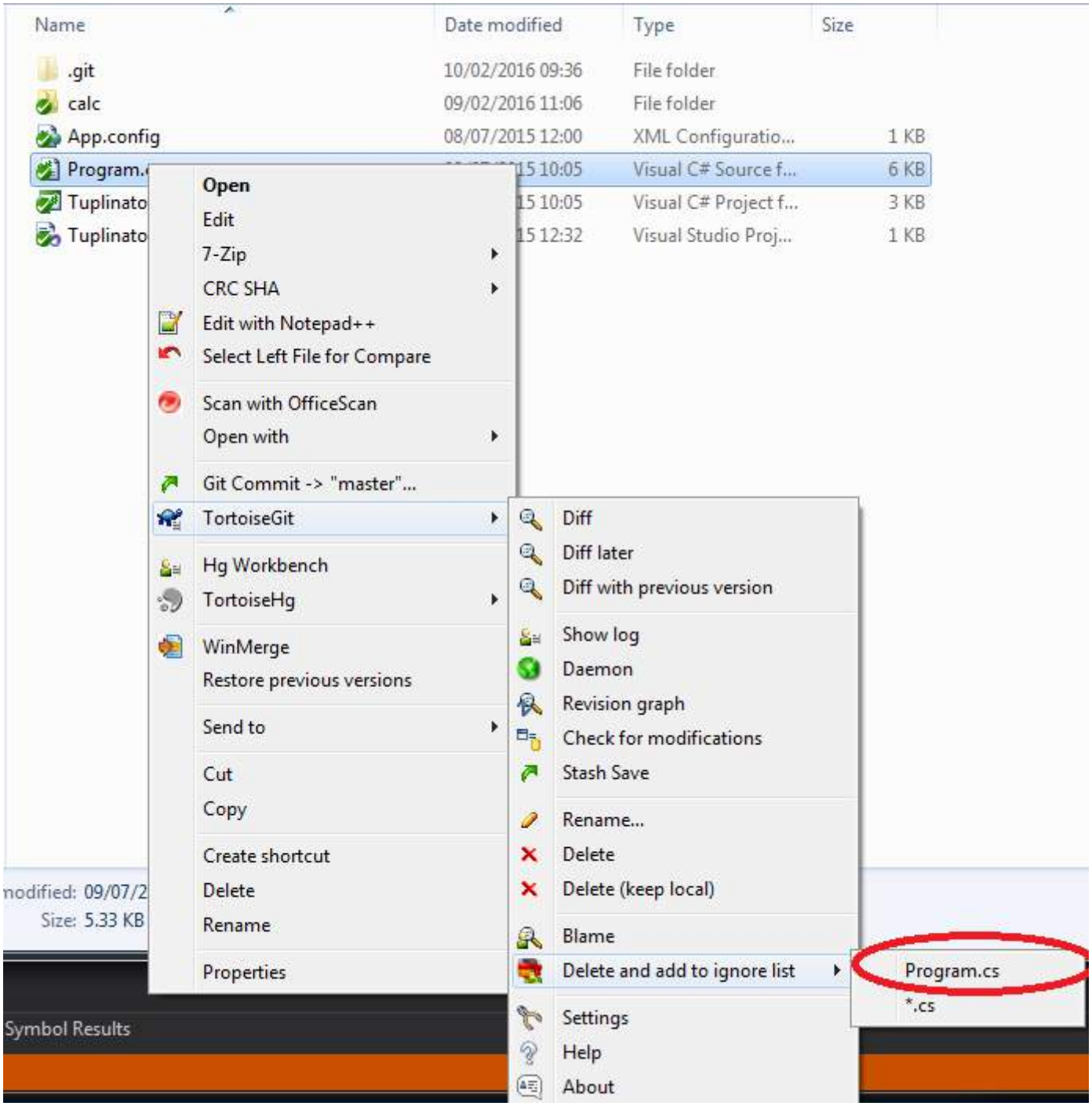






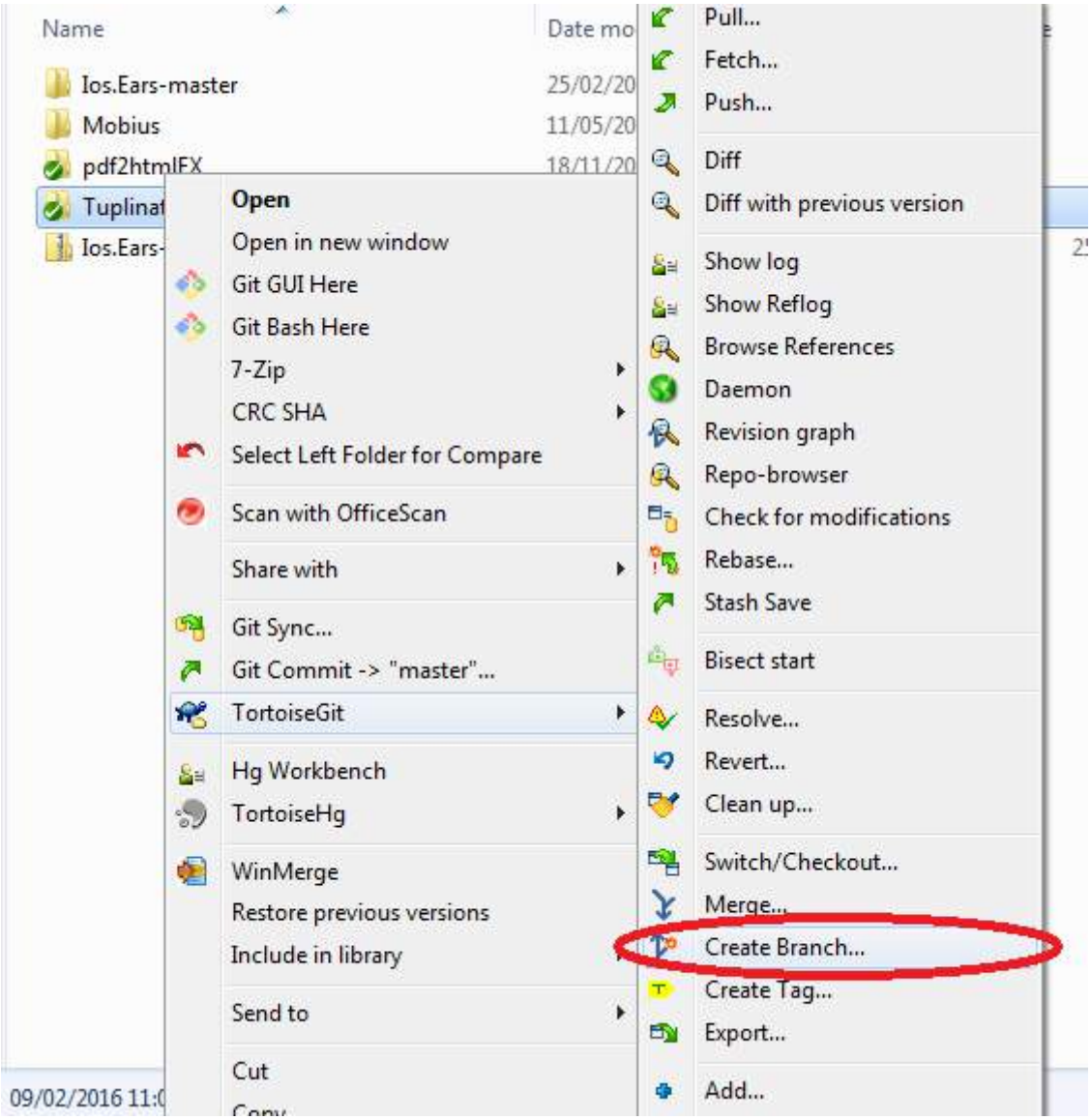
Section 52.3: Ignoring Files and Folders

Those that are using TortioseGit UI click `Right Mouse` on the file (or folder) you want to ignore -> `TortoiseGit` -> `Delete and add to ignore list`, here you can choose to ignore all files of that type or this specific file -> dialog will pop out Click `Ok` and you should be done.

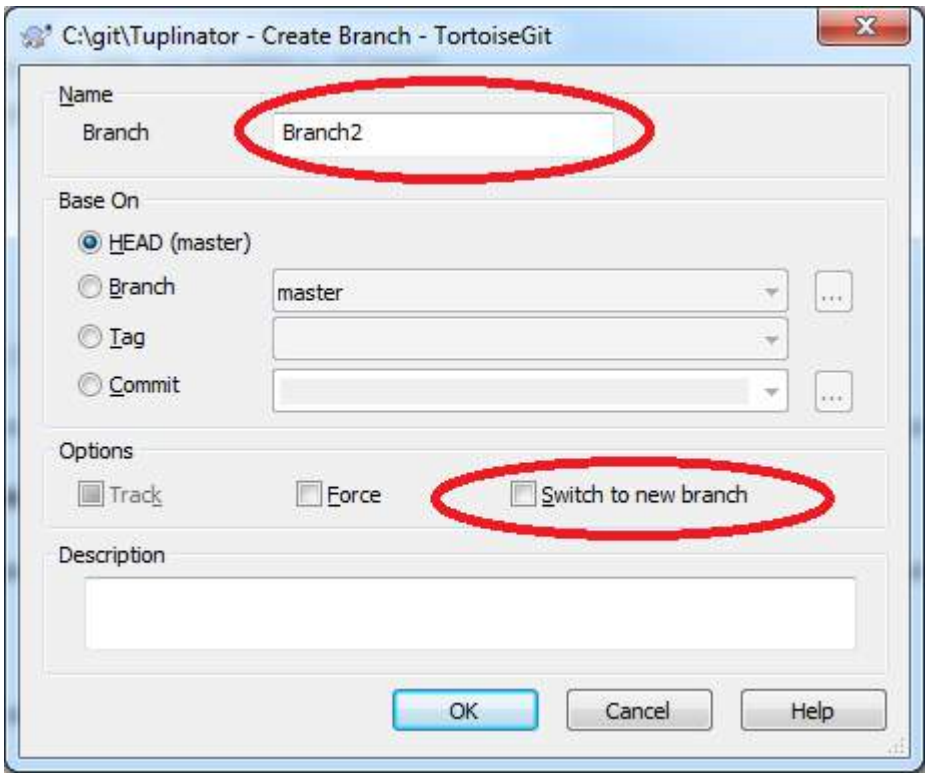


Section 52.4: Branching

For those that are using UI to branch click `Right Mouse` on repository then `Tortoise Git` -> `Create Branch...`



New window will open -> `Give branch a name` -> Tick the box `Switch to new branch` (Chances are you want to start working with it after branching). -> Click `OK` and you should be done.



Chapter 53: External merge and difftools

Section 53.1: Setting up KDiff3 as merge tool

The following should be added to your global `.gitconfig` file

```
[merge]
    tool = kdiff3
[mergetool "kdiff3"]
    path = D:/Program Files (x86)/KDiff3/kdiff3.exe
    keepBackup = false
    keepbackup = false
    trustExitCode = false
```

Remember to set the `path` property to point to the directory where you have installed KDiff3

Section 53.2: Setting up KDiff3 as diff tool

```
[diff]
    tool = kdiff3
    guitool = kdiff3
[difftool "kdiff3"]
    path = D:/Program Files (x86)/KDiff3/kdiff3.exe
    cmd = \"D:/Program Files (x86)/KDiff3/kdiff3.exe\" \"$LOCAL\" \"$REMOTE\"
```

Section 53.3: Setting up an IntelliJ IDE as merge tool (Windows)

```
[merge]
    tool = intellij
[mergetool "intellij"]
    cmd = cmd \"/C D:\\workspace\\tools\\symlink\\idea\\bin\\idea.bat merge $(cd $(dirname
"$LOCAL") && pwd)/$(basename "$LOCAL") $(cd $(dirname "$REMOTE") && pwd)/$(basename "$REMOTE") $(cd
$(dirname "$BASE") && pwd)/$(basename "$BASE") $(cd $(dirname "$MERGED") && pwd)/$(basename
"$MERGED")\"
    keepBackup = false
    keepbackup = false
    trustExitCode = true
```

The one gotcha here is that this `cmd` property does not accept any weird characters in the path. If your IDE's install location has weird characters in it (e.g. it's installed in `Program Files (x86)`, you'll have to create a symlink

Section 53.4: Setting up an IntelliJ IDE as diff tool (Windows)

```
[diff]
    tool = intellij
    guitool = intellij
[difftool "intellij"]
    path = D:/Program Files (x86)/JetBrains/IntelliJ IDEA 2016.2/bin/idea.bat
    cmd = cmd \"/C D:\\workspace\\tools\\symlink\\idea\\bin\\idea.bat diff $(cd $(dirname "$LOCAL")
&& pwd)/$(basename "$LOCAL") $(cd $(dirname "$REMOTE") && pwd)/$(basename "$REMOTE")\"
```

The one gotcha here is that this `cmd` property does not accept any weird characters in the path. If your IDE's install location has weird characters in it (e.g. it's installed in `Program Files (x86)`, you'll have to create a symlink

Section 53.5: Setting up Beyond Compare

You can set the path to `bcomp.exe`

```
git config --global difftool.bc3.path 'c:\Program Files (x86)\Beyond Compare 3\bcomp.exe'
```

and configure `bc3` as default

```
git config --global diff.tool bc3
```

Chapter 54: Update Object Name in Reference

Section 54.1: Update Object Name in Reference

Use

Update the object name which is stored in reference

SYNOPSIS

```
git update-ref [-m <reason>] (-d <ref> [<oldvalue>] | [--no-deref] [--create-reflog] <ref>
<newvalue> [<oldvalue>] | --stdin [-z])
```

General Syntax

1. Dereferencing the symbolic refs, update the current branch head to the new object.

   ```
   git update-ref HEAD <newvalue>
   ```

2. Stores the `newvalue` in `ref`, after verify that the current value of the `ref` matches `oldvalue`.

   ```
   git update-ref refs/head/master <newvalue> <oldvalue>
   ```

   above syntax updates the master branch head to `newvalue` only if its current value is `oldvalue`.

Use `-d` flag to deletes the named **`<ref>`** after verifying it still contains **`<oldvalue>`**.

Use `--create-reflog`, update-ref will create a reflog for each ref even if one would not ordinarily be created.

Use `-z` flag to specify in NUL-terminated format, which has values like update, create, delete, verify.

Update

Set **`<ref>`** to **`<newvalue>`** after verifying **`<oldvalue>`**, if given. Specify a zero **`<newvalue>`** to ensure the ref does not exist after the update and/or a zero **`<oldvalue>`** to make sure the ref does not exist before the update.

Create

Create **`<ref>`** with **`<newvalue>`** after verifying it does not exist. The given **`<newvalue>`** may not be zero.

Delete

Delete **`<ref>`** after verifying it exists with **`<oldvalue>`**, if given. If given, **`<oldvalue>`** may not be zero.

Verify

Verify **`<ref>`** against **`<oldvalue>`** but do not change it. If **`<oldvalue>`** zero or missing, the ref must not exist.

Chapter 55: Git Branch Name on Bash Ubuntu

This documentation deals with the **branch name** of the git on the **bash** terminal. We developers need to find the git branch name very frequently. We can add the branch name along with the path to the current directory.

Section 55.1: Branch Name in terminal

What is PS1

PS1 denotes Prompt String 1. It is the one of the prompt available in Linux/UNIX shell. When you open your terminal, it will display the content defined in PS1 variable in your bash prompt. In order to add branch name to bash prompt we have to edit the PS1 variable (set value of PS1 in ~/.bash_profile).

Display git branch name

Add following lines to your ~/.bash_profile

```
git_branch() {
 git branch 2> /dev/null | sed -e '/^[^*]/d' -e 's/* \(.*\)/ (\1)/'
}
export PS1="\u@\h \[\033[32m\]\w\[\033[33m\]\$(git_branch)\[\033[00m\] $ "
```

This git_branch function will find the branch name we are on. Once we are done with this changes we can navigate to the git repo on the terminal and will be able to see the branch name.

Chapter 56: Git Client-Side Hooks

Like many other Version Control Systems, Git has a way to fire off custom scripts when certain important actions occur. There are two groups of these hooks: client-side and server-side. Client-side hooks are triggered by operations such as committing and merging, while server-side hooks run on network operations such as receiving pushed commits. You can use these hooks for all sorts of reasons.

Section 56.1: Git pre-push hook

pre-push script is called by `git push` after it has checked the remote status, but before anything has been pushed. If this script exits with a non-zero status nothing will be pushed.

This hook is called with the following parameters:

```
 $1 -- Name of the remote to which the push is being done (Ex: origin)
$2 -- URL to which the push is being done (Ex: https://://.git)
```

Information about the commits which are being pushed is supplied as lines to the standard input in the form:

```
<local_ref> <local_sha1> <remote_ref> <remote_sha1>
```

Sample values:

```
local_ref = refs/heads/master
local_sha1 = 68a07ee4f6af8271dc40caae6cc23f283122ed11
remote_ref = refs/heads/master
remote_sha1 = efd4d512f34b11e3cf5c12433bbedd4b1532716f
```

Below example pre-push script was taken from default pre-push.sample which was automatically created when a new repository is initialized with `git init`

```
# This sample shows how to prevent push of commits where the log message starts
# with "WIP" (work in progress).

remote="$1"
url="$2"

z40=0000000000000000000000000000000000000000

while read local_ref local_sha remote_ref remote_sha
do
    if [ "$local_sha" = $z40 ]
    then
        # Handle delete
        :
    else
        if [ "$remote_sha" = $z40 ]
        then
            # New branch, examine all commits
            range="$local_sha"
        else
            # Update to existing branch, examine new commits
            range="$remote_sha..$local_sha"
        fi
```

```
        # Check for WIP commit
        commit=`git rev-list -n 1 --grep '^WIP' "$range"`
        if [ -n "$commit" ]
        then
            echo >&2 "Found WIP commit in $local_ref, not pushing"
            exit 1
        fi
    fi
done

exit 0
```

Section 56.2: Installing a Hook

The hooks are all stored in the `hooks` sub directory of the Git directory. In most projects, that's `.git/hooks`.

To enable a hook script, put a file in the `hooks` subdirectory of your `.git` directory that is named appropriately (without any extension) and is executable.

Chapter 57: Git rerere

`rerere` (reuse recorded resolution) allows you to tell git to remember how you resolved a hunk conflict. This allows it to be automatically resolved the next time that git encounters the same conflict.

Section 57.1: Enabling rerere

To enable `rerere` run the following command:

```
$ git config --global rerere.enabled true
```

This can be done in a specific repository as well as globally.

Chapter 58: Change git repository name

If you change repository name on the remote side, such as your github or bitbucket, when you push your exisiting code, you will see error: Fatal error, repository not found**.

Section 58.1: Change local setting

Go to terminal,

```
cd projectFolder
git remote -v (it will show previous git url)
git remote set-url origin https://username@bitbucket.org/username/newName.git
git remote -v (double check, it will show new git url)
git push (do whatever you want.)
```

Chapter 59: Git Tagging

Like most Version Control Systems (VCSs), `Git` has the ability to `tag` specific points in history as being important. Typically people use this functionality to mark release points (`v1.0`, and so on).

Section 59.1: Listing all available tags

Using the command `git tag` lists out all available tags:

```
$ git tag
<output follows>
v0.1
v1.3
```

> **Note**: the `tags` are output in an **alphabetical** order.

One may also `search` for available `tags`:

```
$ git tag -l "v1.8.5*"
<output follows>
v1.8.5
v1.8.5-rc0
v1.8.5-rc1
v1.8.5-rc2
v1.8.5-rc3
v1.8.5.1
v1.8.5.2
v1.8.5.3
v1.8.5.4
v1.8.5.5
```

Section 59.2: Create and push tag(s) in GIT

Create a tag:

- To create a tag on your current branch:

  ```
  git tag < tagname >
  ```

  This will create a local `tag` with the current state of the branch you are on.

- To create a tag with some commit:

  ```
  git tag tag-name commit-identifier
  ```

  This will create a local `tag` with the commit-identifier of the branch you are on.

Push a commit in GIT:

- Push an individual tag:

```
git push origin tag-name
```

- Push all the tags at once

```
git push origin --tags
```

Chapter 60: Tidying up your local and remote repository

Section 60.1: Delete local branches that have been deleted on the remote

To remote tracking between local and deleted remote branches use

```
git fetch -p
```

you can then use

```
git branch -vv
```

to see which branches are no longer being tracked.

Branches that are no longer being tracked will be in the form below, containing 'gone'

```
 branch               12345e6 [origin/branch: gone] Fixed bug
```

you can then use a combination of the above commands, looking for where 'git branch -vv' returns 'gone' then using '-d' to delete the branches

```
git fetch -p && git branch -vv | awk '/: gone]/{print $1}' | xargs git branch -d
```

Chapter 61: diff-tree

Compares the content and mode of blobs found via two tree objects.

Section 61.1: See the files changed in a specific commit

git diff-tree --no-commit-id --name-only -r COMMIT_ID

Section 61.2: Usage

```
git diff-tree [--stdin] [-m] [-c] [--cc] [-s] [-v] [--pretty] [-t] [-r] [--root] [<common-diff-options>] <tree-ish> [<tree-ish>] [<path>...]
```

Option	Explanation
-r	diff recursively
--root	include the initial commit as diff against /dev/null

Section 61.3: Common diff options

Option	Explanation
-z	output diff-raw with lines terminated with NUL.
-p	output patch format.
-u	synonym for -p.
--patch-with-raw	output both a patch and the diff-raw format.
--stat	show diffstat instead of patch.
--numstat	show numeric diffstat instead of patch.
--patch-with-stat	output a patch and prepend its diffstat.
--name-only	show only names of changed files.
--name-status	show names and status of changed files.
--full-index	show full object name on index lines.
--abbrev=<n>	abbreviate object names in diff-tree header and diff-raw.
-R	swap input file pairs.
-B	detect complete rewrites.
-M	detect renames.
-C	detect copies.
--find-copies-harder	try unchanged files as candidate for copy detection.
-l<n>	limit rename attempts up to paths.
-O	reorder diffs according to the .
-S	find filepair whose only one side contains the string.
--pickaxe-all	show all files diff when -S is used and hit is found.
-a --text	treat all files as text.

Credits

Thank you greatly to all the people from Stack Overflow Documentation who helped provide this content, more changes can be sent to web@petercv.com for new content to be published or updated

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Deep	Chapters 10 and 26
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Jonathan	Chapter 14
Jonathan Lam	Chapter 1
Jordan Knott	Chapter 10
Joseph Dasenbrock	Chapters 1 and 14
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joshng	Chapter 5
jpkrohling	Chapter 16
jready	Chapter 2
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kirrmann	Chapter 14
kisanme	Chapters 14 and 17
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Nemanja Trifunovic	Chapter 50
nepda	Chapter 14
Neui	Chapters 1 and 5
nighthawk454	Chapters 30 and 42
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Noah	Chapters 2, 8 and 14
Noushad PP	Chapter 14
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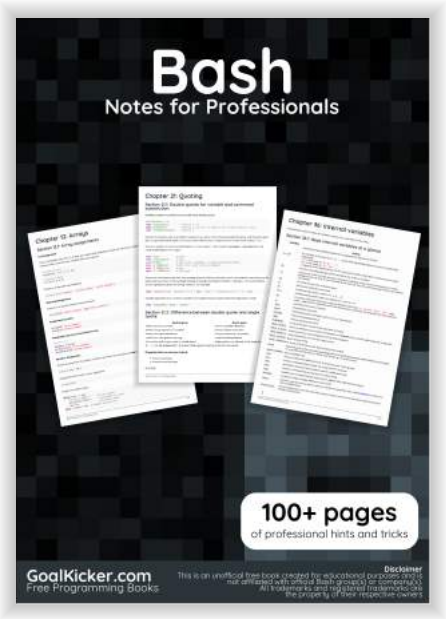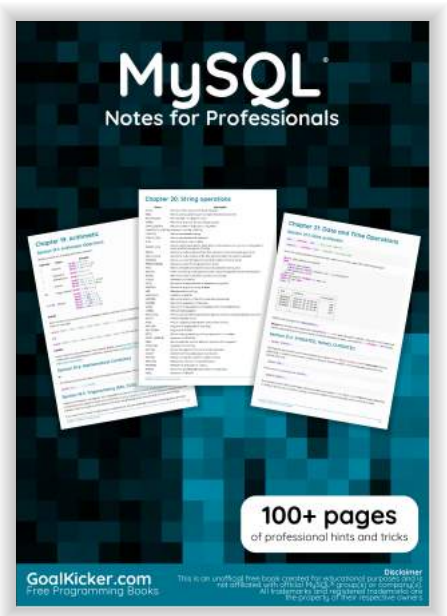
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